Potential modularity of some atypical abelian fourfolds

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1 Introduction

Let A/F be a g-dimensional geometrically simple abelian variety. The geometric endomorphism algebra of A, denoted $D := \mathrm{End}^0_{\overline{F}}(A)$, is a division ring. We say that A is "atypical" if $D \neq \mathbb{Q}$. In this article, we have two main tools. The first is a structure theorem (Theorem 1.3) for the compatible systems associated to atypical abelian varieties. The second is the construction (Theorem 5.1) of twisted PEL Shimura varieties parameterizing abelian varieties with a prescribed Galois action on the endomorphism algebra and the level structure. We apply our structure theorem and these twisted PEL Shimura varieties to prove potential automorphy theorems for abelian fourfolds (Main Theorem 1), construct even four-dimensional symplectic geometric Galois representations with big image (Main Theorem 2), and give new proofs of some results on the fields of definition of endomorphisms of abelian varieties.

Throughout this paper, we write $V_\ell(A) := H^1_{\text{ét}}(A, \mathbb{Q}_\ell)^\vee$, $\rho_{A,\ell}$ for the Galois representation on $V_\ell(A)$, and $\mathcal{V}(A)$ for the compatible system of ℓ-adic representations $\{\rho_{A,\ell}\}_\ell$.

1.1 A special case – abelian surfaces with geometric quaternionic multiplication

We first state a very special case of Theorem 1.3, and use it to deduce a potential modularity result. To our knowledge, this theorem first appeared in work of N. Taylor [Tay20, §3.3].

Theorem 1.1. *Let $A/\mathbb{Q}$ be an abelian surface with $D := \mathrm{End}^0_{\mathbb{Q}}(A)$ a quaternion algebra. For any ℓ, there exist a strongly irreducible two-dimensional $G_\mathbb{Q}$ representation $\xi := \xi_\ell$ and a two-dimensional Artin representation $\varrho := \varrho_\ell$ such that $\rho_{A,\ell} \otimes \overline{\mathbb{Q}_\ell} \cong \xi \otimes \varrho$.*

For abelian surfaces with geometric quaternionic multiplication by D, $H^1(A_\mathbb{C}, \mathbb{Q})$ must be a free module over D, and so the center E of D must be $\mathbb{Q}$. If M/F is the field extension fixed by $\ker \varrho$, $\mathrm{Res}_{G_M} \rho_A \otimes \overline{\mathbb{Q}_\ell} \cong (\mathrm{Res}_{G_M} \xi)^{\oplus 2}$. It follows that ξ is unramified at all but finitely many primes and is potentially de Rham (and hence de Rham) at ℓ. In particular, ξ is a 2-dimensional geometric representation of $G_\mathbb{Q}$ with Hodge–Tate weights $(0, 1)$ and suitably big image (otherwise, by Faltings' theorem, A would have more geometric endomorphisms). By choosing ℓ appropriately, it follows that ξ is modular by [Cal12, Theorem 1.1]. An important point in this case is that 2-dimensional geometric Galois representations with big image are odd.

One can show (Lemma 3.5) that the (symplectic) Weil pairing on $\rho_{A,\ell}$ is the product of pairings on ξ and ϱ. Because ξ is strongly irreducible, Schur's lemma implies that ξ preserves a unique pairing up to scaling. Because ξ is two-dimensional, the pairing on ξ is symplectic, so the pairing on ϱ needs to be orthogonal. Therefore, ϱ is either a direct sum of characters of $G_\mathbb{Q}$ or the induction of a character of G_K for some quadratic extension $K/\mathbb{Q}$.

In the first case, $\rho_{A,\ell}$ is reducible and $\mathrm{End}^0_{\mathbb{Q}}(A)$ is a quadratic field. Because $\mathrm{Lie}(A)$ is a two-dimensional $\mathbb{Q}$-vector space that is a free module over $\mathrm{End}^0_{\mathbb{Q}}(A)$, we must have $\mathrm{End}^0_{\mathbb{Q}}(A) \subsetneq$

$\mathrm{End}^0_{\overline{\mathbb{Q}}}(A) = D$. In particular, ϱ is not a character (by which we mean that it is not of the form $\eta \otimes \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}$), meaning that A is of GL_2-type, and in this case, modularity already follows from e.g. [Rib04, Theorem 4.4]). In the second case, there is a character η of G_K such that

$$\rho_{A,\ell} \cong \xi \otimes \mathrm{Ind}_{G_K}^{G_{\mathbb{Q}}} \eta \cong \mathrm{Ind}_{G_K}^{G_{\mathbb{Q}}}(\mathrm{Res}_{G_K} \xi \otimes \eta).$$

Then, $\rho_{A,\ell}$ has the same L-function as the automorphic induction of a twisted base change of π_f, $\mathrm{Ind}_{K/\mathbb{Q}}(\mathrm{BC}_{K/\mathbb{Q}}(\pi_f) \boxtimes \eta)$. In particular, A is automorphic in this case as well. Automorphy in the second case was first proved by [FP24, Theorem 5.3], who associate a Siegel modular form to A. Using modern potential automorphy theorems for two-dimensional Galois representations ([Sno09, ACC+23], see Theorem 2.3), we can work with abelian surfaces over more general fields:

Theorem 1.2. *Let F be a totally real or CM number field and A/F be an abelian surface with $\mathrm{End}^0_{\overline{\mathbb{Q}}}(A) = D$, where D is a quaternion algebra. Then, A is potentially automorphic, and is automorphic if ξ_ℓ is automorphic for any ℓ.*

When F is totally real, Theorem 1.2 is vastly superseded by e.g. [BCGP21, Theorem 10.2.1], who show potential modularity of all abelian surfaces over F. Even otherwise, this is not an especially interesting potential automorphy result. Nevertheless, it is a useful example to keep in mind for what will follow.

1.2 A structure theorem for atypical abelian varieties

In stating Theorem 1.3, we use the notations CRes and Des. These will be defined properly in Section 2.4, but for now, here is a high level explanation. Let E' be a subfield of $\overline{\mathbb{Q}}$ and E'' a subfield of E'. For a compatible system $\mathcal{R}'$ with coefficients in E', the "coefficient restriction" $\mathrm{CRes}_{E'/E''}(\mathcal{R})$ is a compatible system that, when E'/E'' is Galois, can be thought of as the direct sum of the coefficient-wise $\mathrm{Gal}(E'/E'')$-conjugates of $\mathcal{R}'$. In particular, if $\mathcal{R}'$ is an n-dimensional compatible system with coefficients in E', $\mathrm{CRes}_{E'/E''}(\mathcal{R}')$ is an $([E' : E'']n)$-dimensional compatible system with coefficients in E''. We say that $\mathcal{R}'$ coefficient-wise descends to E'' if the Frobenius characteristic polynomials associated to $\mathcal{R}'$ are actually valued in $E''[X]$, in which case we write $\mathrm{Des}_{E'}\mathcal{R}'$ for the compatible system $\mathcal{R}'$ viewed as having coefficients in E''. Let E be the center of $D := \mathrm{End}^0_{\overline{F}}(A)$ and $E_0 := \mathrm{End}^0_F(A) \cap E$. Let r be the index of D and $e := [E : \mathbb{Q}]$.

Theorem 1.3. *Let A be a geometrically simple abelian variety of dimension g, let L/F be the unique minimal extension over which all the geometric endomorphisms of A are defined, and let K/F be the minimal extension over which $E \hookrightarrow \mathrm{End}^0_K(A)$. Then, there exist a compatible system $\mathcal{R}$ of strongly irreducible $\frac{2g}{er}$-dimensional G_K-representations and a compatible system $\mathcal{A}$ of r-dimensional Artin G_K-representations, both well-defined up to twist by a finite-order character, such that*

1. *The projective kernel of $\mathcal{A}$ is G_L;*
2. *$\mathcal{R} \otimes \mathcal{A}$ is a Schur unobstructed (see Section 2.4) compatible system that coefficient-wise descends to E;*
3. *$\mathrm{Ind}_K^F \mathrm{Des}_E(\mathcal{R} \otimes \mathcal{A})$ is a Schur unobstructed compatible system that coefficient-wise descends to E_0; and*
4. $$\mathcal{V}(A) \cong \mathrm{CRes}_{E_0/\mathbb{Q}}\left(\mathrm{Des}_{E_0}\left(\mathrm{Ind}_K^F \mathrm{Des}_E(\mathcal{R} \otimes \mathcal{A})\right)\right).$$

Furthermore, $\mathcal{R}$ is of symplectic type if A is geometrically (Albert) type I or II, orthogonal type if A is geometrically type III, and unitary type if A is geometrically type IV.

Specializing Theorem 1.3 to the case of a single ℓ, we obtain the following corollary. Let $\Sigma_{E_0} := \mathrm{Hom}_{\mathbb{Q}}(E_0, \overline{\mathbb{Q}})$.

Corollary 1.4. *With the notation of Theorem 1.3, for any rational prime ℓ, there are representations $\{\xi_{\iota_0}\}_{\iota_0 \in \Sigma_{E_0}}$ and $\{\varrho_{\iota_0}\}_{\iota_0 \in \Sigma_{E_0}}$ such that*

$$\rho_{A,\ell} \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{\iota_0 \in \Sigma_{E_0}} \mathrm{Ind}_{G_K}^{G_F}(\xi_{\iota_0} \otimes \varrho_{\iota_0}).$$

As we were circulating a draft of this paper, we learned that Corollary 1.4 was previously obtained by Fité–Guitart [FG22, Theorem 1.1], and proved in essentially the same way. When A is of geometric type I, II, or III and $\frac{2g}{re} = 2$, they also prove the decomposition into compatible systems given by Theorem 1.3. (Their version is also stated for A which are not geometrically simple, though this is not the hard part of this story.)

The fact that $\mathcal{R}$ is symplectic (resp. orthogonal, unitary) type when A is geometrically type I or II (resp. III, IV) is well known. While our language is different, it essentially follows from Schur's lemma combined with the work of Chi, Banaszak, Gajda, and Krasón (see [Chi90, BGK06, BGK10], and also [FFG24, Theorem 7.5]) when A is geometrically type I, II, or III and work of Chi, Banaszak, and Kaim-Garnek [Chi91, BKG21] when A is geometrically type IV.

Because we do need the decomposition in terms of compatible systems in order to apply our modularity results later, we have left our proofs in their original form, but refer the reader to [FG22] for the original appearance of these ideas in the literature.

Theorem 1.3 is not especially exciting when D is a field. When D is a noncommutative division ring however, being able to pull out the dependence on r into an Artin representation enables us to understand $\mathcal{V}$ in terms of a lower-dimensional system $\mathcal{R}$. Furthermore, the Artin representation captures the Galois action on $\mathrm{End}_F^0(A)$ in a precise way (Lemma 3.8). Because $D \otimes \overline{\mathbb{Q}_\ell} \cong M_r(\overline{\mathbb{Q}_\ell})^e$, it is not hard to see how one can decompose $\rho_{A,\ell}$ into $\frac{2g}{re}$-dimensional direct summands after restricting to the absolute Galois group of a field L/F over which all the geometric endomorphisms of A are defined. The point here is that $\mathcal{R}$ is defined over a field K/F which can be much smaller than L. In particular, this K/F is the minimal subextension of L/F over which A has endomorphisms by E. In many cases treated in this paper, we have $K = F$ even when we have little control over L/F.

1.3 Twisted PEL Shimura varieties

Let $\mathcal{O}$ be an order in a semisimple $\mathbb{Q}$-algebra with positive involution, and let n be a positive integer. After specifying some additional data, one can define a PEL Shimura variety Y associated to $\mathcal{O}$ and n [Mil05, Chapter 8]. An F'-point on Y corresponds to a quadruple $(A, \lambda, \iota, \eta)$ up to isomorphism, where A is an abelian variety over F', λ is a polarization on A, ι is an embedding of $\mathcal{O}$ into $\mathrm{End}_{F'}(A)$, and η defines a full level n-structure on $A[n]$ over F'. In particular, the endomorphisms in $\iota(D)$ and the points in $A[n]$ must all be defined over F' in order for $(A, \lambda, \iota, \eta)$ to correspond to an F'-point on Y – the $G_{F'}$-action on the endomorphisms and level structure is trivial.

An important tool in this paper is the construction – via Galois descent from PEL Shimura varieties – of moduli spaces parameterizing polarized abelian varieties with a prescribed Galois action on the endomorphism algebra and level structure. The idea of constructing twists of Shimura varieties parameterizing abelian varieties with a Galois action on the level structure is not a new one – to our knowledge, it first appears Wiles' proof of Fermat's last theorem [Wil95, Proof of Theorem

5.2], and plays a key role in most applications of potential modularity [Tay02]. However, twists with a Galois action on both the endomorphism algebra and the level structure have been less well explored.

In this paper, we use these moduli schemes in conjunction with the Moret–Bailly theorem to construct abelian varieties over number fields with interesting Galois actions on their endomorphism algebras. However, we expect these objects to be of independent interest. For example, most of the abelian surfaces over $\mathbb{Q}$ (on the LMFDB, say) with geometric quaternionic multiplications appear to be of GL_2-type. It is an interesting open question to use these moduli spaces to prove bounds

1.4 Abelian fourfolds and potential automorphy

Our main application of Theorem 1.3 and Theorem 5.1 is to the potential automorphy of abelian fourfolds with nontrivial geometric endomorphisms. Abelian fourfolds are interesting because they can exhibit all four Albert types, and more generally because the simplest examples of many different algebraic cycle phenomena appear amidst abelian fourfolds. In this paper, we prove potential automorphy for many different families of atypical abelian fourfolds. Following the classification of Moonen–Zarhin [MZ95, Table 1], we describe our results in Table 1.

We say that an abelian variety A/F is geometrically type $\mathrm{X}(e)$, $X \in \{\mathrm{I}, \mathrm{II}, \mathrm{III}\}$ and $e \in \mathbb{Z}_{>0}$, if A is geometrically Albert type X and the center of $\mathrm{End}^0_{\overline{F}}(A)$ has degree e. We say that A/F is geometrically type $\mathrm{IV}(r, e_+)$ if it is geometrically Albert type IV and $\mathrm{End}^0_{\overline{F}}(A)$ is a division algebra with index r whose center is a CM field whose totally real subfield has degree e_+.

Suppose that F is totally real. We say that A/F is "odd type" if every member r_λ of the compatible system $\mathcal{R}$ associated to A by Theorem 1.3 is totally odd – this means that the similitude character of r_λ is -1 on every complex conjugation of F. Let K/F be the field associated to A by Theorem 1.3. Inspired by [BCGP21, p. 254], we say that an abelian fourfold A/F is "challenging" if A has geometric type I(2) and $\mathcal{V}(A) \cong \mathrm{Des}_{\mathbb{Q}} \mathrm{Ind}_K^F \mathcal{R}$ for K a number field which is not totally real, A has geometric type I(4) and $\mathcal{V}(A) \cong \mathrm{Des}_{\mathbb{Q}} \mathrm{Ind}_K^F \mathcal{R}$ for K a number field which is neither totally real nor CM, or if A has geometric type II(2) and $\mathcal{V}(A) \cong \mathrm{Des}_{\mathbb{Q}} \mathrm{Ind}_K^F (\mathcal{R} \otimes \mathcal{A})$ for K a number field which is neither totally real nor CM. When $F = \mathbb{Q}$, all abelian fourfolds of geometric type I(4) or II(2) are non-challenging (Corollary 6.9).

Main Theorem 1. *Let $F/\mathbb{Q}$ be a field and A/F an abelian fourfold such that either:*

1. *F is totally real and A/F is odd type of geometric type II(1);*
2. *F is totally real, A/F is non-challenging of geometric type I(2), and $K^{\mathrm{gal}} \not\supseteq \mathrm{End}^0_{\overline{F}}(A)$, where K is as in Theorem 1.3;*
3. *A is non-challenging of geometric type I(4) or II(2); or*
4. *A is of geometric type IV(1,4).*

Then, A is potentially automorphic, and is automorphic if any representation in the compatible system $\mathcal{R}$ associated to A by Theorem 1.3 is automorphic. These results are summarized in Table 1.

Type	Rep'n of $\mathfrak{hg}_{\mathbb{C}}$ on $H^1(A(\mathbb{C}),\mathbb{C})$	Potentially automorphic over totally real fields		Notes
I(1)	$\mathfrak{sp}_8$	Open		Generic abelian fourfolds
	$\mathfrak{sl}_2^{\boxtimes 3}$	Open		Mumford type fourfolds
I(2)	$\mathfrak{sp}_4^{\boxplus 2}$	Non-challenging	Yes (Theorem 6.6)	
		Challenging or $K^{\mathrm{gal}} \supseteq \mathrm{End}_F^0(A)$	Open	Induced from a 4-dim'l symplectic rep'n over a CM field
I(4)	$\mathfrak{sl}_2^{\boxplus 4}$	Non-challenging	Yes ‡ (Lemma 6.8)	
		Challenging	Open	Induced from a 2-dim'l symplectic rep'n over a non-CM non-totally real field
II(1)	$\mathfrak{sp}_4$	Odd type	Yes (Theorem 1.5)	$\rho_{A,\ell} \cong \xi_\ell \otimes \varrho_\ell$, ξ odd symplectic 4-dim'l
		Not odd type	Open	$\rho_{A,\ell} \cong \xi_\ell \otimes \varrho_\ell$, ξ symplectic 4-dim'l, not odd
II(2)	$\mathfrak{sl}_2^{\boxplus 2}$	Non-challenging	Yes ‡ (Lemma 6.8)	
		Challenging	Open	Induced from a 2-dim'l symplectic rep'n over a non-CM non-totally real field
III	$\mathfrak{so}_4$	Open		$\rho_{A,\ell} \cong \xi_\ell \otimes \varrho_\ell$, ξ orthogonal 4-dim'l
IV(1,1)	$\mathfrak{gl}_4$	Open		
	$\mathfrak{sl}_4$	Open		Weil type abelian fourfolds
IV(1,2)	$\mathfrak{gl}_2$	Open		Involves 2-dim'l rep'n with irregular Hodge–Tate weights
IV(1,4)	$\mathbb{C}^{\boxplus 4}$	Yes ‡ (Corollary 3.3)		
	$\mathbb{C}^{\boxplus 3}$	Yes ‡ (Corollary 3.3)		

Table 1: A summary of the potential automorphy results in this paper for abelian fourfolds over totally real fields. The icon ‡ indicates that for those fourfolds, potential modularity also holds over CM fields. Over $\mathbb{Q}$, fourfolds of geometric type I(4) or II(2) are always non-challenging.

The most conceptually interesting case of Main Theorem 1 is that of odd type geometrically type II(1) abelian fourfolds, and this is what we now discuss. Let $F/\mathbb{Q}$ be a totally real field and A/F be an abelian fourfold with $D := \mathrm{End}_F^0(A)$ an indefinite quaternion algebra with center $E = \mathbb{Q}$. Corollary 1.4 implies that for any rational prime ℓ, there exist a 4-dimensional strongly irreducible symplectic representation ξ and a 2-dimensional Artin representation ϱ such that

$$V_\ell(A) \otimes \overline{\mathbb{Q}_\ell} \cong \xi \otimes \varrho. \tag{1}$$

Assume additionally that A is odd type, so ξ is totally odd. Note that, even though ξ is only well-defined up to twist, being odd does not depend on the choice of twist. Boxer–Calegari–Gee–Pilloni [BCGP21, Theorem 8.4.1] prove an automorphy lifting theorem applying to totally odd geometric Galois representations $\xi\colon G_F \to \mathrm{GSp}_4(\overline{\mathbb{Q}_\ell})$ with "big" residual image and Hodge–Tate weights $(0,0,1,1)$ at every $v|\ell$. In [BCGP21, 9.2], they prove potential automorphy for abelian

surfaces using a Moret–Bailly argument. They work with a variety $Y/\mathbb{Q}$ whose F'-points parameterize polarized abelian surfaces B'/F' along with $G_{F'}$-module isomorphisms $B'[p] \xrightarrow{\sim} A[p]$ and $B'[q] \xrightarrow{\sim} \overline{\rho}_q$ for some $\overline{\rho}_q$ known to be automorphic. Because appropriate subsets of $Y(\mathbb{R})$, $Y(\mathbb{Q}_p)$, and $Y(\mathbb{Q}_q)$ are nonempty, the Moret–Bailly theorem implies that Y has a point over a totally real field F', corresponding to an abelian variety B/F'. In particular, because F is totally real, $A_{\mathbb{R}}$ itself is a point in $Y(\mathbb{R})$, showing that it is nonempty. As usual for potential automorphy arguments, two applications of the automorphy lifting theorem to B then imply that A/F' is automorphic.

When A/F is an odd type abelian fourfold with geometric endomorphisms by an indefinite quaternion algebra D, we might hope to proceed in the same way, with Y replaced with the moduli of abelian fourfolds with endomorphism algebra containing D and again the appropriate Galois module structure on the pq-torsion. However, there is a problem – even though F is totally real, the abelian variety A does not define a real point on Y unless the geometric endomorphisms of A are also defined over a totally real field, and this will not typically be the case. To remedy this, we instead work with a variety Y' representing a moduli functor which – in addition to prescribing a Galois module structure on the pq-torsion, also prescribes a Galois action on the geometric endomorphism algebra. These are exactly the varieties which we discussed earlier in Section 1.3. Once we have Y', the rest of the Moret–Bailly argument goes through with only minor modifications, and we deduce the following.

Theorem 1.5. *Let F be a totally real field and A/F be an odd type abelian fourfold with $\mathrm{End}^0_{\overline{F}}(A) = D$, where D is an indefinite quaternion algebra with center $\mathbb{Q}$. Then, there exists a compatible system $\mathcal{R}$ of four-dimensional strongly irreducible symplectic (odd) representations and a compatible system $\mathcal{A}$ of two-dimensional Artin representations with cyclic or dihedral projective image such that*

1. *$\mathcal{R} \otimes \mathcal{A}$ descends to $\mathbb{Q}$;*
2. *$\mathcal{V}(A) \cong \mathrm{Des}_{\mathbb{Q}}(\mathcal{R} \otimes \mathcal{A})$; and*
3. *A is potentially automorphic, and is automorphic whenever any r_λ in $\mathcal{R}$ is automorphic.*

The case of geometric type I(2) ($\mathrm{End}^0_{\overline{F}}(A) = E$, E real quadratic field) is similar. In this case, $\mathcal{R}$ is again a four-dimensional compatible system of symplectic type, but in this case it is guaranteed to be odd. The issue of finding sufficiently many ordinary primes is slightly more subtle in this setting – it is analogous to showing that a weight 2 modular form with coefficients in a quadratic field has infinitely many ordinary primes – but once this is resolved we again apply Moret–Bailly to an appropriately twisted moduli space.

Separately from the potential automorphy results, we also resolve a question posed in [BCGP21, 10.4.1]: we prove that an abelian fourfold (a so-called "fake abelian surface") associated to a cuspidal automorphic representation of GL_4 of symplectic type will necessarily be type II, i.e. have endomorphism algebra an indefinite quaternion division algebra.

1.5 Even type geometric Galois representations

Because Theorem 1.5 only applies to odd type abelian fourfolds, we may ask if geometrically type II abelian fourfolds which are *not* of odd type even exist. We call a fourfold "even type" if the associated compatible system $\mathcal{R}$ has totally even similitude character, i.e. if $\mathcal{R}$ is a four-dimensional even symplectic compatible system of geometric Galois representations with big image. Under the usual conjectures, an automorphic form on GSp_4 associated to $\mathcal{R}$ for such an abelian fourfold would not be a Siegel modular form but rather a Siegel Maass form. Unfortunately, we do not even

know how to associate Galois representations to algebraic Maass forms, and certainly do not have automorphy theorems in this setting.

However, Calegari [Cal12] showed that any two-dimensional geometric Galois representation with big image in GL_2 and big residual image is odd. Because an even type geometrically type II abelian fourfold would imply the existence of a compatible system $\mathcal{R}$ of even geometric Galois representations with big image in GSp_4 and (for a density 1 subset of primes) big residual image, one might conjecture[1] that this is impossible like in the two-dimensional case. More generally, one might conjecture that $\mathcal{R}$ is symplectic (resp. orthogonal) if and only if it odd (resp. even). Such a conjecture would also be concordant with the results of [BLGGT14, Theorem A], who prove potential automorphy theorems (for geometric Galois representations with regular Hodge–Tate weights) exactly in the odd symplectic and even orthogonal cases.

However, this conjecture is not true! Applying Theorem 1.3, Lemma 4.4, and an argument of Frank Calegari (Lemma 7.1), we are able to show that such "even type" abelian fourfolds exist. We are extremely grateful to Calegari for allowing us to include his argument in this article.

Main Theorem 2. *Let $F/\mathbb{Q}$ be a totally real number field. There exist type II abelian fourfolds A/F for which the four-dimensional compatible system $\mathcal{R}$ given by Theorem 1.3 is even type and symplectic type. In particular, there exists a representation $\xi\colon G_F \to \mathrm{GSp}_4(\overline{\mathbb{Q}_\ell})$ such that ξ is geometric, has Zariski dense image, and is totally even.*

1.6 Disclosure of AI use

Most of this paper was completed before the recent advances in capabilities of LLMs. GPT 6 was used to proofread this paper, playing a role analogous to that of a (rather persnickety) reviewer, and we are grateful that it helped identify several mistakes in earlier drafts. The ensuing fixes were devised and implemented by the author with limited additional AI use.

1.7 Acknowledgements

I am indebted to my Ph.D. advisor, Frank Calegari, for numerous illuminating conversations about this material over the years, and especially for proving a result in the setting of geometrically type III abelian fourfolds that motivated Theorem 1.3 and for allowing me to use his construction in Lemma 7.1. I am also grateful to Chengyang Bao for helpful discussions about the material surrounding Lemma 2.9 and Example 2.10, and to Mateo Attanasio for helpful discussions on Section 5. Part of this work was completed while I was supported by an NSF Graduate Research Fellowship.

2 Preliminaries

2.1 Notation and nomenclature

Given $n \in \mathbb{Z}_{\geq 1}$, we write $[n] := \{1, \dots, n\}$. We write G_F for the absolute Galois group of a field F and $|F|$ for the set of places in F. Let G be a group and V a vector space over a field k. Given a representation $\xi\colon G \to \mathrm{GL}(V)$ and a subgroup $H \subseteq G$, we write $\mathrm{Res}_H \xi$ for the restriction of ξ to H. We say that ξ is strongly irreducible if $\mathrm{Res}_H \xi$ is irreducible for every finite-index subgroup $H \subseteq G$. Given finite-dimensional semisimple representations ξ and ξ' a group G, we write $\langle \xi, \xi' \rangle := \dim \mathrm{Hom}_{k[G]}(\xi, \xi') = \dim \mathrm{Hom}_{k[G]}(\xi', \xi)$. Given an automorphism $\tau \in \mathrm{Aut}(k)$, we write ${}^{\tau}\xi := \tau \circ \xi$. We call ${}^{\tau}\xi$ a coefficient-wise or "external" Galois conjugate of ξ. We write $\mathrm{Proj}\, \xi$ for the

[1] As I myself did for several months!

projective representation $G \xrightarrow{\xi} \mathrm{GL}(V) \to \mathrm{PGL}(V)$ associated to ξ. Let H be a normal subgroup of G, ρ be a representation of H, and σ be an element of G. We write ρ_σ for the representation of H defined by $\rho_\sigma(h) := \rho(\sigma h \sigma^{-1})$ for all $h \in H$. We call ρ_σ an "internal" Galois conjugate of ρ. Observe that $({}^\tau\rho)_\sigma = {}^\tau(\rho_\sigma)$ for $\sigma \in G$ and $\tau \in G_k$.

We say that ξ is an extension of ρ (to G) if $\mathrm{Res}_H \xi \cong \rho$. We say that a projective representation $\xi' \colon G \to \mathrm{PGL}(V)$ is a projective extension of ρ (to G) if $\mathrm{Res}_H \xi' \cong \mathrm{Proj}\,\rho$. Given a projective representation ξ', we will sometimes abuse notation and also write ξ' for an arbitrary lift (as a function) $\xi' \colon G \to \mathrm{GL}(V)$ – in general, this ξ' need not be a homomorphism.

When F is totally real, we say that a representation ξ of G_F that is self-dual up to similitude character ν is odd (resp. even) if ν is odd (resp. even), i.e. if $\nu(c) = -1$ (resp. $\nu(c) = +1$) for every complex conjugation $c \in G_F$. We write ε for the cyclotomic character.

Given an abelian variety A/F, we write $\mathrm{End}_F^0(A) := \mathrm{End}_F(A) \otimes_{\mathbb{Z}} \mathbb{Q}$. We call $\mathrm{End}_{\overline{F}}^0(A)$ the geometric endomorphism algebra of A.

2.2 Embeddings

Throughout this paper, we fix embeddings $\jmath_\ell \colon \overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}_\ell}$ for all rational primes ℓ. This gives rise to an embedding $G_{\mathbb{Q}_\ell} \hookrightarrow G_{\mathbb{Q}}$ for every ℓ. For a number field E, we write $\Sigma_E := \mathrm{Hom}_{\mathbb{Q}}(E, \overline{\mathbb{Q}})$. There is a transitive action of $G_{\mathbb{Q}}$ on Σ_E by postcomposition. For each rational prime ℓ, $\jmath_\ell$ gives an embedding $G_{\mathbb{Q}_\ell} \hookrightarrow G_{\mathbb{Q}}$, and the orbits of Σ_E under the resulting $G_{\mathbb{Q}_\ell}$-action correspond to primes λ of E dividing ℓ. An embedded number field is a tuple (E, ι_0) consisting of a number field E and an embedding $\iota_0 \in \Sigma_E$. In the context of an embedded number field, for each $\iota \in \Sigma_E$, we write $\tilde{\iota} \colon \overline{\mathbb{Q}} \to \overline{\mathbb{Q}}$ for an element of $G_{\mathbb{Q}}$ such that $\tilde{\iota} \circ \iota_0 = \iota \in \Sigma_E$. For any $\tau \in G_{\mathbb{Q}}$, ${}^\tau\iota := \tau \circ \iota \in \Sigma_E$. The embedded number field (E', ι_0') is an extension of (E, ι_0) if $E' \supset E$ and ι_0' extends ι_0, and we call $(E', \iota_0')/(E, \iota_0)$ an extension of embedded number fields.

2.3 Automorphy

Recall that an ℓ-adic representation ξ of G_K is said to be an automorphic Galois representation if there is an automorphic form over K whose L-function is the same as that of ξ. We say that ξ is a potentially automorphic Galois representation if there exists a finite extension K'/K such that $\mathrm{Res}_{G_{K'}} \xi$ is automorphic.

An abelian variety is (potentially) automorphic if any of its ℓ-adic representations are (potentially) automorphic. Given a (potentially) automorphic Galois representation, it is useful to understand operations which preserve (potential) automorphy.

Lemma 2.1. *Let $\xi \colon G_K \to \mathrm{GL}_n(\overline{\mathbb{Q}_\ell})$ be a (potentially) automorphic Galois representation.*

1. *For any $r \in \mathbb{Z}_{\geq 1}$, $\xi^{\oplus r}$ is (potentially) automorphic;*
2. *Any twist of ξ by a character is (potentially) automorphic;*
3. *If K/K'' is a Galois extension, then for any $\sigma \in \mathrm{Gal}(K/K'')$, any conjugate ξ_σ of ξ is (potentially) automorphic;*

Proof. Let L be an extension of K such that $\mathrm{Res}_{G_L} \xi$ is automorphic.

1. The direct sum of automorphic Galois representations is automorphic (via the isobaric sum). The analogue for potential automorphy then follows because $\mathrm{Res}_{G_L}(\xi^{\oplus r}) \cong (\mathrm{Res}_{G_L} \xi)^{\oplus r}$;
2. This follows from class field theory;

3. Let π be the automorphic representation associated to $\mathrm{Res}_{G_L}\xi$. Then, $\pi\circ\sigma$ is an automorphic representation of $\mathrm{GL}_n(\mathbb{A}_{\sigma^{-1}(L)})$ whose Satake parameters at unramified primes are exactly the Frobenius characteristic polynomials of $\mathrm{Res}_{G_{\sigma^{-1}(L)}}\xi_\sigma$.

□

It does not follow from Lemma 2.1 that the direct sum of two different potentially automorphic Galois representations is itself potentially automorphic. Let ξ_1 and ξ_2 be representations of G_{K_1} and G_{K_2} such that $\mathrm{Res}_{L_1}\xi_1$ and $\mathrm{Res}_{L_2}\xi_2$ are automorphic. Unless L_1L_2/K_1 and L_1L_2/K_2 both happen to be solvable, we cannot apply base change to deduce that $\mathrm{Res}_{L_1L_2}\xi_1$ and $\mathrm{Res}_{L_1L_2}\xi_2$ are automorphic.

In practice, however, potential automorphy of a G_K-representation ξ does not arise in a vacuum. We typically have two ingredients. The first is a automorphy lifting theorem which – for any finite extension K'/K satisfying some conditions which are preserved under compositum and Galois closure – given a $G_{K'}$-representation ρ' with big image satisfying some p-adic Hodge theory conditions whose residual representation is automorphic and also has big image, implies that ρ' itself is automorphic. The second is a residual representation $\overline{\rho}\colon G_K\to\mathrm{GL}_m(\overline{\mathbb{F}_\ell})$ that is is strongly automorphic (i.e. $\mathrm{Res}_{G_{K'}}\overline{\rho}$ is automorphic for every K'/K), and satisfies the aforementioned big image conditions for K'/K linearly disjoint from some fixed extension K^{avoid}/K. Then, an argument due to Taylor [Tay02] is used to show that the existence of an L-point on an appropriate moduli space – for any L satisfying all the above conditions – implies that $\mathrm{Res}_{G_L}\xi$ is automorphic.

The Moret–Bailly theorem is then used to find such an extension L and an L-point on an appropriate moduli space Y. As noted, any such point implies the automorphy of $\mathrm{Res}_{G_L}\xi$. If Y has an L-point, it certainly has a $\tilde{L}$-point for any $\tilde{L}\supset L$. However, such a $\tilde{L}$ need not satisfy the hypotheses of our automorphy lifting theorem and our strongly automorphic residual representation with big image.

If potential automorphy of ξ_i ($i\in\{1,2\}$) is proved by finding a point on Y_i with associated data S_i, K_i^{avoid}/K, and $\Omega_{v,i}$ as above, then we can prove potential automorphy of ξ_1 and ξ_2 over a common extension L/K by applying the Moret–Bailly theorem to $Y_1\times Y_2$, taking $S:=S_1\cup S_2$, $K^{\mathrm{avoid}}:=K_1^{\mathrm{avoid}}K_2^{\mathrm{avoid}}$, and $\Omega_v:=\Omega_{v,1}\times\Omega_{v,2}$ for $v\in S$.

Lemma 2.2. *Let $\xi,\xi'\colon G_K\to\mathrm{GL}_n(\overline{\mathbb{Q}_\ell})$ be Galois representations whose potential automorphy follows from Theorem 2.3 or from a Moret–Bailly argument involving Theorem 2.4. Then,*

1. *The direct sum $\xi\oplus\xi'$ is potentially automorphic;*
2. *If K/K'' is a Galois extension, $\mathrm{Ind}_{G_K}^{G_{K''}}\xi$ is potentially automorphic;*
3. *If K'/K is a solvable Galois extension, $\mathrm{Res}_{G_{K'}}\xi$ is potentially automorphic.*

Furthermore, any sequence of these operations is also potentially automorphic.

Sketch. 1. This follows from applying the Moret–Bailly theorem to $Y:=Y_1\times Y_2$, as discussed above the lemma statement.

2. By the Moret–Bailly theorem, we may assume that there is an extension L/K such that L/K'' is Galois and $\mathrm{Res}_{G_L}\xi$ is automorphic. Therefore, for any $\sigma\in\mathrm{Gal}(L/K)$, $(\mathrm{Res}_{G_L}\xi)_\sigma$ is still a representation of G_L. We have

$$\mathrm{Res}_{G_L}\left(\mathrm{Res}_{G_K}\mathrm{Ind}_{G_K}^{G_{K''}}\xi\right)=\mathrm{Res}_{G_L}\left(\bigoplus_{\sigma\in\mathrm{Gal}(K/K'')}\xi_\sigma\right)=\bigoplus_{\sigma\in\mathrm{Gal}(K/K'')}(\mathrm{Res}_{G_L}\xi)_\sigma,$$

so the result follows from Lemma 2.1.

3. By the Moret–Bailly theorem, we may assume that there is an extension L/K linearly disjoint from K'/K such that $\mathrm{Res}_{G_L} \xi$ is automorphic. Then, LK'/L is a solvable Galois extension, and by solvable base change [AC89], $\mathrm{Res}_{G_{LK'}} \mathrm{Res}_{G_L} \xi = \mathrm{Res}_{G_{LK'}} \mathrm{Res}_{G_{K'}} \xi$ is automorphic.

If we have a sequence of such operations, we impose all the relevant conditions when applying the Moret–Bailly theorem at the start, so that the field L over which we have automorphy has all the properties we want. □

Remark. We have deliberately chosen to keep the lemma statement of Lemma 2.2 somewhat imprecise. Because the exact automorphy lifting theorems used in Theorem 2.3 and Theorem 2.4 have their own technical conditions and subtleties, any precise statement applying to both theorems would require a substantial amount of notation and definition that we felt would hinder the exposition.

The following two theorems are our main tools for proving potential automorphy results. The first is a potential automorphy theorem for two-dimensional representations over totally real or CM fields. The second is a automorphy lifting theorem for representations valued in $\mathrm{GSp}_4(\overline{\mathbb{Q}_p})$

Theorem 2.3 ([ACC+23, Sno09, Cal12])**.** *Let $K/\mathbb{Q}$ be a totally real or CM extension, $p \geq 7$ be a prime split in $K/\mathbb{Q}$, and $\xi \colon G_K \to \mathrm{GL}_2(\overline{\mathbb{Q}_p})$ a continuous irreducible representation such that*

1. *ξ is unramified at all but finitely many primes of K;*
2. *$\xi|_{G_{K_v}}$ is de Rham with Hodge–Tate weights $(0,1)$ for every $v|p$*
3. *There exists a finite field $k/\mathbb{F}_p$ such that the image of the residual representation of ξ contains $\mathrm{SL}_2(k)$.*

Then, ξ is potentially automorphic.

Theorem 2.4 ([BCGP21, Theorem 8.4.1])**.** *Let K be a totally real number field, p a rational prime totally split in $K/\mathbb{Q}$, and $\xi \colon G_K \to \mathrm{GSp}_4(\overline{\mathbb{Q}_p})$ a continuous Galois representation such that*

- *$p \geq 3$;*
- *ξ has cyclotomic similitude character;*
- *For all $v|p$, $\xi|_{G_{K_v}}$ is p-distinguished weight 2 ordinary;*
- *$\overline{\xi}$ is vast and tidy in the sense of [BCGP21, Definitions 7.5.6 and 7.5.11];*
- *$\overline{\xi}$ is automorphic, and the associated automorphic form π is parallel weight 2, ordinary at $v|p$, has central character $|\cdot|^2$, and has an associated (characteristic 0) Galois representation ρ_π;*
- *ξ and ρ_π are pure at all finite places of F';*
- *The restriction $\xi|_{G_{K_v}}$ is potentially crystalline and has Hodge–Tate weights $(0,0,1,1)$ for every $v|p$*

Then, ξ is automorphic.

Note that $\overline{\xi}$ is vast and tidy if its image contains $\mathrm{Sp}_4(k)$ for some finite extension $k/\mathbb{F}_p$.

2.4 Compatible systems of Galois representations

We refer to Section 2.2 for notation pertaining to embeddings of number fields.

Definition 2.5. Let K be a number field, (E, ι_0) an embedded number field, and S a finite set of primes of K. A weakly compatible system of m-dimensional semisimple continuous Galois representations over a field K unramified away from S with coefficients in (E, ι_0) consists of, for each rational prime ℓ and each $\iota \in \Sigma_E$, a representation $r_{\ell,\iota} \colon G_K \to \mathrm{GL}_m(\overline{\mathbb{Q}_\ell})$ such that:

1. Away from S and the primes of K dividing ℓ, $r_{\ell,\iota}$ is unramified;
2. If ι and ι' are in the same $G_{\mathbb{Q}_\ell}$-orbit, then $r_{\iota,\ell}$ and $r_{\iota',\ell}$ are externally Galois conjugate by an automorphism of $G_{\mathbb{Q}_\ell}$;
3. For all $v \notin S$, there is a polynomial $Q_v(X) \in \mathrm{im}(\iota_0)[X] \subset \overline{\mathbb{Q}}[X]$ (independent of ℓ) such that for all ℓ not dividing the residue characteristic of v, $r_{\iota,\ell}$ is unramified at v and the characteristic polynomial of $r_{\ell,\iota}(\mathrm{Frob}_v)$ is $(\jmath_\ell \circ \tilde{\iota})(Q_v)$;
4. Each $r_{\ell,\iota}$ is de Rham at places $v|\ell$ of K, and crystalline when $v \notin S$.

We denote these data by the 5-tuple $\mathcal{R} = (K; (E, \iota_0), S, \{Q_v(X)\}_v, \{r_{\iota,\ell}\}_\lambda)$.

By the second property in Definition 2.5, we can specify $\mathcal{R}$ by specifying, for each ℓ, one $r_{\ell,\iota}$ in each $G_{\mathbb{Q}_\ell}$-orbit. Because the $G_{\mathbb{Q}_\ell}$-orbits in Σ_E correspond to prime ideals of E above ℓ, this is equivalent to specifying representations $\{r_\lambda\}_\lambda$ for each $\lambda|\ell$. Typical treatments of compatible systems (e.g. [BLGGT14, 5.1]) implicitly fix once and for all, for each ℓ, a collection of orbit representatives $\{\iota_\lambda\}_{\lambda|\ell}$, and define compatible systems by specifying representations $\{r_\lambda\}_\lambda$ where λ ranges over the primes of E. In this subsection, explicitly tracking all $[E : \mathbb{Q}]$ representations for each ℓ will ameliorate the definition of certain operations involving Galois conjugation, but does not actually change the underlying objects which we are working with. Later in the paper, we also fall back to the more standard $\{r_\lambda\}_\lambda$ notation and suppress the marked embedding ι_0 when the added granularity is not necessary.

Remark. We deviate from the usual notation for compatible systems (e.g. [BLGGT14, 5.1]), by specifying the base field K and by not specifying the Hodge–Tate weights $\{H_\tau\}_{\tau \colon K \hookrightarrow \overline{E}}$. We specify K because we work with operations on compatible systems that that change K (e.g. induction). While the Hodge–Tate weights are important, the majority of the compatible systems appearing in this article are self-dual up to similitude and are associated to abelian varieties, and the Hodge–Tate weights are always half zeroes and half ones in these cases. If $\mathcal{R}$ is unitary type, then this need not hold, but in this article we do not work substantially with unitary type compatible systems.

We now define several operations on weakly compatible systems.

Definition 2.6. Let $\mathcal{R} := (K; (E, \iota_0), S, \{Q_v(X)\}_v, \{r_{\ell,\iota}\}_{\ell,\iota})$ be a weakly compatible system.

1. For any extension of embedded fields $(E', \iota_0')/(E, \iota_0)$, we can define the "coefficient-wise extension"
$$\mathcal{R}_{(E',\iota_0')} := \left(K; (E', \iota_0'), S, \{Q_v(X)\}_v,\ \{r_{\ell,\iota'} := r_{\ell,\iota'|_E}\}_{\ell,\iota}\right).$$
2. Given another compatible system $\mathcal{R}' = (K'; E', S', \{Q_v'(X)\}_v, \{r'_{\ell,\iota'}\}_{\ell,\iota'})$ with $(E, \iota_0) = (E', \iota_0')$, we can define
$$\mathcal{R} \oplus \mathcal{R}' := \left(K; (E, \iota_0), S \cup S', \{Q_v Q_v'\}_v, \{r_{\ell,\iota} \oplus r'_{\ell,\iota}\}_{\ell,\iota}\right)$$
If $E \neq E'$ and $\iota|_{E\cap E'} = \iota'|_{E\cap E'}$, we define $\mathcal{R} \oplus \mathcal{R}' := \mathcal{R}_{(EE', \iota_0\iota_0')} \oplus \mathcal{R}'_{EE', \iota_0\iota_0'}$, where $\iota_0\iota_0'$ is the embedding of EE' extending ι_0 and ι_0'.

3. The tensor product of compatible systems $\mathcal{R}$ and $\mathcal{R}'$ is defined analogously to the direct sum, with $\otimes$ replacing $\oplus$ everywhere and with an appropriate definition of Frobenius characteristic polynomials.

4. For any $\sigma \in G_{\mathbb{Q}}$, the "internal" Galois conjugate (in the sense of Section 2.1) of $\mathcal{R}$ by σ is
$$\mathcal{R}_\sigma := \left(\sigma^{-1}(K); (E, \iota_0), \sigma^{-1}(S), \{Q_{\sigma(v)}(X)\}_v, \{(r_{\ell,\iota})_\sigma\}_{\ell,\iota}\right).$$

5. For any $\tau \in G_{\mathbb{Q}}$, the "external" or "coefficient-wise" Galois conjugate is
$${}^\tau\mathcal{R} := (K; (E, \iota_0), S, \{{}^\tau Q_v(X)\}_v, \{r_{\ell,\tau\circ\iota}\}_{\ell,\iota}).$$

6. For any (E'', ι_0'') contained in (E, ι_0), suppose that $Q_v(X) \in \mathrm{im}(\iota_0'')[X]$ for every prime $v \notin S$ of K. We define
$$\mathrm{Des}_{E''}(\mathcal{R}) := (K; (E'', \iota_0''), S, \{Q_v(X)\}_v, \{r_{\ell,\iota''} := r_{\ell,\iota} \text{ for any } \iota \in \Sigma_E \text{ extending } \iota''\}_{\ell,\iota''}).$$
Because we are working with characteristic 0 representations over an algebraically closed field of a compact group, the Brauer–Nesbitt theorem and Chebotarev density ensure that this operation is well-defined (independent of the choice of ι extending each ι''). If the hypothesis for applying $\mathrm{Des}_{E''}$ holds, we say that $\mathcal{R}$ coefficient-wise descends to E''.

7. For any $(E'', \iota_0'') \subset (E, \iota_0)$,
$$\mathrm{CRes}_{E/E''}(\mathcal{R}) := \mathrm{Des}_{E''}\left(\bigoplus_{\tau\in\mathcal{T}} \mathcal{R}^\tau\right),$$
where $\mathcal{T}$ is any collection of $[E : E'']$ elements of $G_{\mathbb{Q}}$ for which $\{\tau \circ \iota_0\}_{\tau\in\mathcal{T}} \subset \Sigma_E$ is the set of embeddings of E extending ι_0''. In particular, if $E'' = \mathbb{Q}$, then in the notation of Section 2.2, we can take $\mathcal{T} := \{\tilde{\iota} \colon \iota \in \Sigma_E\}$.

8. For any subfield $F \subset K$,
$$\mathrm{Ind}_K^F(\mathcal{R}) := \left(F; E, S' := \{v|_F \colon v \in S\} \cup \{v|\mathrm{disc}(K/F)\}, \{Q'_{v'}(X)\}_{v'\in|F|\setminus S'}, \{\mathrm{Ind}_{G_K}^{G_F} r_\lambda\}_\lambda\right)$$
where each $Q'_v(X)$ is an appropriate polynomial which one can check has coefficients in $E[X]$ (or see e.g. [Neu99, VII §10]).

9. For any extension K'/K, we define
$$\mathrm{Res}_{K'}(\mathcal{R}) := \left(K'; E, S' := \{w \colon w|v \text{ for some } v \in S\}, \{Q_w(X)\}_{w\in|K'|\setminus S'}, \{\mathrm{Res}_{G_{K'}} r_\lambda\}_\lambda\right),$$
where, writing f_w for the residue field degree at w, each $Q_w(X) \in E[X]$ is the polynomial whose roots are the f_w^{th} powers of the roots of $Q_v(X)$.

Because the hypothesis for applying $\mathrm{Des}_{E''}$ is a condition on Frobenius characteristic polynomials, and the Frobenius characteristic polynomials in a compatible system are "independent of ι and ℓ" in an appropriate sense (the third condition of Definition 2.5), we can verify this hypothesis on a single $r_{\iota,\ell}$. We record this as a lemma for future use.

Lemma 2.7. *To prove coefficient-wise descent of $\mathcal{R}$ to (E'', ι_0''), it suffices to fix a rational prime ℓ and an $\iota \in \Sigma_E$ and show that for every $v \notin S$ for which ℓ does not divide the residue characteristic of v, $r_{\iota,\ell}(\mathrm{Frob}_v)$ has characteristic polynomial in $(\jmath_\ell\circ\tilde{\iota}\circ\iota_0'')(E'') = (\jmath_\ell\circ\iota'')(E'')$, where $\iota'' := \tilde{\iota}\circ\iota_0'' \in \Sigma_{E''}$.*

Note that this is a substantially stronger condition than simply showing that the characteristic polynomial of every $r_{\ell,\iota}$ lies in $E''_{\lambda''}$ for some $\lambda''|\lambda$ (e.g. the latter is automatic when λ'' splits completely in E/E'').

The main motivation to define compatible systems using embedded coefficient fields and, for each ℓ, to track $\{r_{\ell,\iota}\}_{\iota\in\Sigma_E}$ instead of simply the $G_{\mathbb{Q}_\ell}$-orbit representatives $\{r_\lambda\}_{\lambda|\ell}$, is in order to define the operation $\mathrm{CRes}_{E/E''}$ above. The following example exhibits its utility.

Example 2.8. Let A be the GL_2-type abelian variety A_f associated to a weight 2 modular newform f with coefficients in $E/\mathbb{Q}$. Let $\mathcal{R}_f$ be the 2-dimensional compatible system of $G_{\mathbb{Q}}$-representations with coefficients in E attached to f. Then,

$$\mathcal{V}(A) \cong \mathrm{CRes}_{E/\mathbb{Q}}\,\mathcal{R}_f.$$

The point of our machinery is that this is a stronger result than we would be able to state while working with a single ℓ-adic representation. For simplicity, assume that $E/\mathbb{Q}$ is Galois. For any ℓ, we have

$$\rho_{A,\ell}\otimes\overline{\mathbb{Q}_\ell} \cong \bigoplus_{\tau\in\mathrm{Gal}(E/\mathbb{Q})} \rho_{{}^\tau f}, \tag{2}$$

where each $\rho_{{}^\tau f}\colon G_{\mathbb{Q}}\to \mathrm{GL}_2(\overline{\mathbb{Q}_\ell})$ is the 2-dimensional Galois representation associated to the newform ([DS05, Theorem 6.5.4]) ${}^\tau f$. We would like to think of $\rho_{A,\ell}$ as the direct sum of the "Galois conjugates" of ρ_f. If ℓ is totally inert in E, this is literally correct, as $\mathrm{Gal}(E/\mathbb{Q})\cong\mathrm{Gal}(E_\lambda/\mathbb{Q}_\ell)$ and we have $\rho_{{}^\tau f}\cong {}^\tau\rho_f$ and $\rho_{A,\ell}$ is the direct sum of external Galois conjugates of ρ_f. However, if ℓ splits completely in E, the representation ρ_f are valued in $\mathbb{Q}_\ell$ and has no nontrivial Galois conjugates. In this case, the Galois representations associated to ${}^\tau f$ cannot be obtained from ρ_f, and must be logged separately, and in order to reconstruct Equation (2), we need the full data of the compatible system.

If $\mathcal{R}$ and $\mathcal{R}'$ are absolutely irreducible, then $\mathrm{CRes}_{E/E''}\mathcal{R}\cong\mathrm{CRes}_{E/E''}\mathcal{R}'$ if and only if $\mathcal{R}'\cong\mathcal{R}^\tau$ for some $\tau\in G_{\mathbb{Q}}$ fixing (E'',ι''_0). We say that an m-dimensional compatible system $\mathcal{R}$ is Schur unobstructed if for all ℓ and ι, $r_{\ell,\iota}$ stabilizes an m-dimensional E_λ-vector space – where $\lambda|\ell$ is the prime ideal of E associated to ι – and hence can be defined over E_λ (and not just over $\overline{E_\lambda}=\overline{\mathbb{Q}_\ell}$).

Remark. Note that $\mathcal{R}$ coefficient-wise descending to E'' is only a statement about the field of definition of the Frobenius characteristic polynomials. Each representation $r_{\ell,\iota''}$ will still typically only be defined over $\overline{\mathbb{Q}_\ell}$. In particular, $\mathcal{R}$ coefficient-wise descending to E'' does not necessarily mean that $\mathrm{Des}_{E''}\mathcal{R}$ is Schur unobstructed.

We say that $\mathcal{R}$ is of symplectic (resp. orthogonal) type if every $r_{\ell,\iota}$ is essentially self-dual and preserves a symplectic (resp. orthogonal) pairing up to similitude. If the base field of $\mathcal{R}$ is totally real and $\mathcal{R}$ is symplectic or orthogonal type, we write that $\mathcal{R}$ is odd type if its similitude character is totally odd. Suppose that E is CM, with totally real subfield E^+ and complex conjugation automorphism c. We say that $r_{\ell,\iota}$ is essentially conjugate self-dual if $r_{\ell,\iota}^\vee\cong {}^c r_{\ell,\iota\circ c}\otimes\chi_\lambda$ for some character χ_λ We say that $\mathcal{R}$ is of unitary type if every $r_{\ell,\iota}$ is essentially conjugate self-dual.

2.5 Representation theoretic lemmas

For ξ a representation of G, $\mathrm{Res}_H\,\xi\cong(\mathrm{Res}_H\,\xi)_\sigma$ for all $\sigma\in G$. The following lemma captures the extent to which the converse (or at least, some projective version thereof) holds.

Lemma 2.9. *Let G be a group, $H \subseteq G$ be a finite-index normal subgroup, and V a vector space over an algebraically closed field $\bar{k}$. Let $\rho\colon H \to \mathrm{GL}(V)$ be an irreducible representation such that $\rho_\sigma \cong \rho$ for all $\sigma \in G$.*

We can associate a cocycle $[c] \in H^2(G/H, \bar{k}^\times)$ to (G, ρ), and there exists a representation $\tilde{\rho}\colon G \to \mathrm{GL}(V)$ such that $\mathrm{Res}_H \tilde{\rho} \cong \rho$ (an extension of ρ to G) if and only if $[c]$ vanishes.

Example 2.10. The nontrivial character of $\{\pm 1\} \subset Q_8$ is stable under conjugation, but is not the restriction of any character of Q_8. I learned this example from Chengyang Bao. Indeed, $Q_8/\{\pm 1\} \cong (\mathbb{Z}/2\mathbb{Z})^2$ and one can check that $H^2((\mathbb{Z}/2\mathbb{Z})^2, \mathbb{C}^\times) \cong \mathbb{Z}/2\mathbb{Z}$.

Proof. This is a standard result in Clifford theory, and a proof can also be found in e.g. [Isa06]. Choose representatives $\tilde{\sigma} \in G$ for each $\sigma \in G/H$. To specify an extension $\tilde{\rho}$ to G, it is sufficient to specify the values of $T_{\tilde{\sigma}} := \tilde{\rho}(\tilde{\sigma})$ for all $\sigma \in G/H$. We can then define $\tilde{\rho}(\tilde{\sigma}h) := T_{\tilde{\sigma}}\rho(h)$. Let $h_{\sigma,\sigma'} := \widetilde{\sigma\sigma'}^{-1}\tilde{\sigma}\tilde{\sigma}'$. The map $\tilde{\rho}$ is a group homomorphism if and only if

$$T_{\tilde{\sigma}}\rho(h)T_{\tilde{\sigma}'}\rho(h') = T_{\widetilde{\sigma\sigma'}}\rho(h_{\sigma,\sigma'})\rho(\tilde{\sigma}'^{-1}h\tilde{\sigma}'h')$$

for all $\sigma, \sigma' \in G/H$ and $h, h' \in H$. Equivalently, we require that

$$T_{\tilde{\sigma}}T_{\tilde{\sigma}'} = T_{\widetilde{\sigma\sigma'}}\rho(h_{\sigma,\sigma'}) \qquad \text{and} \qquad \rho(\tilde{\sigma}'^{-1}h\tilde{\sigma}') = T_{\tilde{\sigma}'}^{-1}\rho(h)T_{\tilde{\sigma}'} \tag{3}$$

for all $\sigma, \sigma' \in G/H$ and $h, h' \in H$. The second condition in Equation (3) means that each $T_{\tilde{\sigma}}$ must intertwine ρ and $\rho_{\tilde{\sigma}}$ – such intertwiners exist by our assumption that $\rho \cong \rho_{\tilde{\sigma}}$, and are well-defined up to scaling by Schur's lemma. By definition of $h_{\sigma,\sigma'}$,

$$\rho(x) = \rho((h_{\sigma,\sigma'}^{-1}\widetilde{\sigma\sigma'}^{-1}\tilde{\sigma}\tilde{\sigma}')x(h_{\sigma,\sigma'}^{-1}\widetilde{\sigma\sigma'}^{-1}\tilde{\sigma}\tilde{\sigma}')^{-1}) = (\rho(h_{\sigma,\sigma'}^{-1})T_{\widetilde{\sigma\sigma'}}^{-1}T_{\tilde{\sigma}}T_{\tilde{\sigma}'})\rho(x)(\rho(h_{\sigma,\sigma'}^{-1})T_{\widetilde{\sigma\sigma'}}^{-1}T_{\tilde{\sigma}}T_{\tilde{\sigma}'})^{-1}.$$

Again by Schur's lemma, $\rho(h_{\sigma,\sigma'}^{-1})T_{\widetilde{\sigma\sigma'}}^{-1}T_{\tilde{\sigma}}T_{\tilde{\sigma}'}$ is a scalar, so the first condition in Equation (3) always holds up to scaling. Let $c(\sigma,\sigma') := \rho(h_{\sigma,\sigma'})^{-1}T_{\widetilde{\sigma\sigma'}}^{-1}T_{\tilde{\sigma}}T_{\tilde{\sigma}'}$. Furthermore, one can check that c is a 2-cocycle for G/H, and that the associated class $[c] \in H^2(G/H, \bar{k}^\times)$ is well-defined independent of the choices of lifts $\{\tilde{\sigma}\}$ and intertwiners $\{T_{\tilde{\sigma}}\}$. In particular, $[c]$ vanishes if and only if there exist intertwiners $\{T_{\tilde{\sigma}}\}$ satisfying the conditions of Equation (3). □

Remark. Suppose we want to work with projective representations instead of genuine representations. Suppose that $\mathrm{Proj}\,\rho_\sigma \cong \mathrm{Proj}\,\rho$ for all $\sigma \in G$. If we are only interested in finding projective extensions of ρ, then we only need Equation (3) to hold up to scaling, and setting $\tilde{\rho}(\tilde{\sigma}) := T_{\tilde{\sigma}}$ (for any choice of intertwiner) for all σ is good enough for this. Therefore, there always exists a projective representation $\tilde{\rho}$ such that $\mathrm{Res}_H \tilde{\rho} \cong \rho$ (with no cocycle condition).

In general, we will not be able to assume that $H^2(G/H, \bar{k}^\times) = 0$ (see Example 2.10), although this is true when G/H is cyclic. In our application however, G is the absolute Galois group of a number field, and we can use Tate's theorem to say something stronger.

Corollary 2.11. *Let F'/F be a finite Galois extension of number fields, $V/\bar{k}$ be a finite-dimensional vector space over an algebraically closed characteristic 0 field, and $\rho\colon G_{F'} \to \mathrm{GL}(V)$ be a strongly irreducible representation such that $\rho_\sigma \cong \rho$ for all $\sigma \in \mathrm{Gal}(F'/F)$. Then, there exists a representation $\xi\colon G_F \to \mathrm{GL}(V)$ and a finite-order character $\psi\colon G_{F'} \to \bar{k}^\times$ such that $\mathrm{Res}_{G_{F'}} \xi \cong \rho \otimes \psi$ (and hence $\mathrm{Res}_{\ker\psi}\xi \cong \mathrm{Res}_{\ker\psi}\rho$).*

Proof. Tate proved (see [Pat19, Theorem 2.11]) that $H^2(G_F, \mu_\infty(\bar{k})) = 0$, where we take continuous cohomology and impose the discrete topology on $\mu_\infty(\bar{k})$. Since $\operatorname{coker}(\mu_\infty(\bar{k}) \hookrightarrow \bar{k}^\times)$ is uniquely divisible, $H^2(G_F, \operatorname{coker}(\mu_\infty(\bar{k}) \hookrightarrow \bar{k}^\times)) = 0$. Thus, $H^2(G_F, \bar{k}^\times) = 0$. Because

$$H^2(G_F, \bar{k}^\times) = H^2(\varprojlim \operatorname{Gal}(\tilde{F}/F), \bar{k}^\times) = \varinjlim_{\tilde{F}/F} H^2(\operatorname{Gal}(\tilde{F}/F), \bar{k}^\times),$$

for any nonzero class $[c] \in H^2(\operatorname{Gal}(F'/F), \bar{k}^\times)$ there exists a finite Galois extension $\tilde{F}/F'$ such that the image of $[c]$ in $H^2(\operatorname{Gal}(\tilde{F}/F), \bar{k}^\times)$ is 0. By Lemma 2.9, it follows that $\operatorname{Res}_{G_{\tilde{F}}} \rho$ admits an extension ξ to G_F. Because ρ is irreducible, it follows from Frobenius reciprocity there is a finite-order character ψ of $\operatorname{Gal}(\tilde{F}/F')$ such that $\operatorname{Res}_{G_{F'}} \xi \cong \rho \otimes \psi$. □

Lemma 2.12. *Let H be a finite-index normal subgroup of G. Suppose that ρ is an absolutely irreducible representation of H and $\tilde{\rho}$ is a semisimple representation of G extending ρ. If ξ is a semisimple representation of G such that $\langle \operatorname{Res}_H \xi, \rho \rangle = r$, then there exists an r-dimensional representation ϱ of G/H such that $\tilde{\rho} \otimes \varrho$ is a subrepresentation of ξ.*

Proof. Because ρ has an extension to G,

$$\operatorname{Ind}_H^G \rho = \bigoplus_{\varrho'} (\tilde{\rho} \otimes \varrho')^{\oplus \dim \varrho'},$$

where ϱ' ranges over the irreducible representations of G/H. By Frobenius reciprocity,

$$r = \langle \operatorname{Res}_H \xi, \rho \rangle = \langle \xi, \bigoplus (\tilde{\rho} \otimes \varrho')^{\oplus \dim \varrho'} \rangle.$$

As such, $\tilde{\rho} \otimes \left(\bigoplus_{\varrho' \in S} \varrho' \right)$ is a subrepresentation of ξ for some multiset S of ϱ' such that $\sum_{\varrho' \in S} \dim \varrho = r$, and we define $\varrho := \bigoplus_{\varrho' \in S} \varrho'$. □

3 A structure theorem for Galois representations of abelian varieties

In this subsection, we prove Theorem 1.3. We note again that, while we were unaware of it when writing this paper, Corollary 1.4, the same structural result on the level of ℓ-adic Galois representations, first appears in [FG22, Theorem 1.1].

3.1 Proving the structure theorem

Let A/F be a g-dimensional geometrically simple abelian variety over a number field F. Let $D := \operatorname{End}_{\bar{F}}^0(A)$ (the geometric endomorphism algebra of A), and let E be the center of D. E is a number field, and we define $e := [E : \mathbb{Q}]$ and $r := \sqrt{\dim_E D}$. By [Sil92, Proposition 2.2], there exists a unique minimal extension L/F such that all the endomorphisms in D are defined over L, and this L/F is Galois. Because any automorphism of D preserves its center E, the Galois action of $\operatorname{Gal}(L/F)$ on D restricts to an action on E. We define K/F as the field fixed by the kernel of this action – it is the minimal extension over which A has endomorphism algebra containing E, and it is automatically Galois.

Let $\mathcal{V} := \mathcal{V}(A) = (F; \mathbb{Q}, S, \{Q_v(X)\}_v, \{\rho_{A,\ell}\}_\ell)$ be the compatible system of Galois representations associated to A, where $V_\ell(A) := H^1_{\text{ét}}(A, \mathbb{Q}_\ell)^\vee$, $\rho_{A,\ell}$ is the Galois representation on $V_\ell(A)$, and S is the set of primes of bad reduction for A.

By Faltings' theorem [Fal83], $\mathrm{End}_{G_{F'}}(V_\ell(A)) \cong \mathrm{End}_{F'}(A) \otimes \mathbb{Q}_\ell$ for any finite extension F'/F. In particular, taking $F' = L$, $\mathrm{End}^0_L(A) \otimes \overline{\mathbb{Q}_\ell} = D \otimes \overline{\mathbb{Q}_\ell} \cong M_r(\overline{\mathbb{Q}_\ell})^e$, and so

$$\mathrm{Res}_{G_L} V_\ell(A) \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{i \in [e]} \rho_i^{\oplus r} \tag{4}$$

for irreducible $\frac{2g}{er}$-dimensional representations $\{\rho_i\}_{i\in[e]}$.

By the Albert classification, $\mathrm{End}^0_{F'}(A)$ for any F'/F is either a totally real field ($A_{F'}$ is type I), a totally indefinite quaternion algebra over a totally real field ($A_{F'}$ is type II), a totally definite quaternion algebra over a totally real field ($A_{F'}$ is type III), or a division ring whose center is a CM field ($A_{F'}$ is type IV). We say that A is geometrically type X if $A_{\overline{F}}$ has Albert type X. Note that the geometric Albert type and the Albert type often differ, so the distinction is important.

Theorem 1.3. *Let A be a geometrically simple abelian variety of dimension g, let L/F be the unique minimal extension over which all the geometric endomorphisms of A are defined, and let K/F be the minimal extension over which $E \hookrightarrow \mathrm{End}^0_K(A)$. Then, there exist a compatible system $\mathcal{R}$ of strongly irreducible $\frac{2g}{er}$-dimensional G_K-representations and a compatible system $\mathcal{A}$ of r-dimensional Artin G_K-representations, both well-defined up to twist by a finite-order character, such that*

1. *The projective kernel of $\mathcal{A}$ is G_L;*
2. *$\mathcal{R}\otimes\mathcal{A}$ is a Schur unobstructed (see Section 2.4) compatible system that coefficient-wise descends to E;*
3. *$\mathrm{Ind}_K^F \mathrm{Des}_E(\mathcal{R} \otimes \mathcal{A})$ is a Schur unobstructed compatible system that coefficient-wise descends to E_0; and*
4. $$\mathcal{V}(A) \cong \mathrm{CRes}_{E_0/\mathbb{Q}} \left(\mathrm{Des}_{E_0} \left(\mathrm{Ind}_K^F \mathrm{Des}_E(\mathcal{R} \otimes \mathcal{A})\right)\right).$$

Furthermore, $\mathcal{R}$ is of symplectic type if A is geometrically (Albert) type I or II, orthogonal type if A is geometrically type III, and unitary type if A is geometrically type IV.

Suppose that $E' \hookrightarrow \mathrm{End}^0_{F'}(A)$. Then, setting $e' := [E' : \mathbb{Q}]$, we can write

$$\mathrm{Res}_{G_{F'}} \rho_A \colon G_{F'} \longrightarrow \mathrm{GL}_{\frac{2g}{e'}}(E' \otimes \mathbb{Q}_\ell) \cong \prod_{\lambda'|\ell} \mathrm{GL}_{\frac{2g}{e'}}(E'_{\lambda'}), \tag{5}$$

and we define $\rho'_{\lambda'} \colon G_{F'} \to \mathrm{GL}_{\frac{2g}{e'}}(E'_{\lambda'})$ to be the projection of $\mathrm{Res}_{G_{F'}} \rho_A$ onto the factor corresponding to λ'. Let $V := V_\ell(A)$. Because E' is commutative, the action of E' on $V \otimes \overline{\mathbb{Q}_\ell}$ decomposes as a sum of characters $E' \to \overline{\mathbb{Q}_\ell}$. The possible characters are indexed by $\{\jmath_\ell \circ \iota' \colon \iota' \in \Sigma_{E'}\}$, and we call the subspace where E' acts by $\jmath_\ell \circ \iota'$ the ι'-eigenspace of $V \otimes \overline{\mathbb{Q}_\ell}$ and denote it by V'_ι. Since the E'-action commutes with the $G_{F'}$-action, V'_ι is a $G_{F'}$-subrepresentation for all $\iota' \in \Sigma_{E'}$. Viewed in this light, the subrepresentation $\rho'_{\lambda'}$ acts on subspace where the action of E' factors through the map $E' \hookrightarrow E'_{\lambda'}$, and, over $\overline{\mathbb{Q}_\ell}$, this subspace is the direct sum of the eigenspaces $V_{\iota'}$ associated to the embeddings $\jmath_\ell \circ \iota'$ extending $E' \hookrightarrow E'_{\lambda'}$. With S as above, let $S_{F'}$ denote the primes of F' dividing the primes of S.

Proposition 3.1 (see [Rib76, Theorem 2.1.2])**.** *The representations ρ'_λ appearing below Equation (5) fit into a $\frac{2g}{e'}$-dimensional Schur unobstructed compatible system $\mathcal{W}' := \left(F'; E', S_{F'}, \{Q_v(X)\}, \{\rho'_{\lambda'}\}\right)$ such that $\mathrm{Res}_{G_{F'}} \mathcal{V} \cong \mathrm{CRes}_{E'/\mathbb{Q}} \mathcal{W}'$.*

For any ℓ, we have $D \otimes_{\mathbb{Q}} \overline{\mathbb{Q}_\ell} \cong \prod_{\iota \in \Sigma_E} M_r(\overline{\mathbb{Q}_\ell})$. In the ιth factor of this decomposition, an endomorphism $e \in E$ is represented by the scalar $(\jmath \circ \iota)(e) \in \overline{\mathbb{Q}_\ell}$. Applying Proposition 3.1 with $F' = L$ and $E' = E$, there is a compatible system $\mathcal{U}' := (L; E, S_L, \{Q_w(X)\}_w, \{u'_\lambda\}_\lambda)$, w ranging over primes of L not in S_L and λ ranging over primes of E, such that $\mathrm{Res}_{G_L} \mathcal{V} \cong \mathrm{CRes}_{E/\mathbb{Q}} \mathcal{U}'$. For our choice of ℓ,

$$\mathrm{Res}_{G_L} \rho_A \cong \bigoplus_{\lambda | \ell} u'_\lambda, \tag{6}$$

and there exist representations u_λ such that $u'_\lambda \otimes_{E_\lambda} \overline{\mathbb{Q}_\ell} \cong u_\lambda^{\oplus r}$, where in the tensor product we choose an arbitrary embedding of E_λ into $\overline{\mathbb{Q}_\ell}$. Note that when λ is split in D, we do not need to tensor with $\overline{\mathbb{Q}_\ell}$ in order to define u_λ. The representations u_λ are strongly irreducible as otherwise, by Faltings' theorem, the geometric endomorphism algebra of A would be larger than D. Let w be a prime of L not in S_L. Because the characteristic polynomial $u'_\lambda(\mathrm{Frob}_w)$ is monic, has coefficients in E, and is independent of λ, the same is true for the characteristic polynomial of $u_\lambda(\mathrm{Frob}_w)$. Therefore, $\mathcal{U} := (L; E, S_L, \{Q_w(X)^{1/r}\}, \{u_\lambda\}_\lambda)$ also defines a (Schur obstructed, when $r > 1$ and D has nonarchimedean ramification) weakly compatible system with coefficients in E, and $\mathcal{U}' \cong \mathcal{U}^{\oplus r}$. Because the $(u_\lambda)_\lambda$ are strongly irreducible, $\mathcal{U}$ is strongly irreducible.

Decomposing Equation (6) even further, we have

$$\mathrm{Res}_{G_L} \rho_A \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{\iota \in \Sigma_E} \xi_\iota^{\oplus r}, \tag{7}$$

where $\xi_\iota^{\oplus r}$ acts on the subspace V_ι defined prior to Proposition 3.1. Consider the inner conjugation action of $\sigma \in G_F$ on the direct summands in Equation (7). We have

$$\bigoplus_{\iota \in \Sigma_E} (\xi_\iota)_\sigma^{\oplus r} \cong \mathrm{Res}_{G_L}(\rho_A)_\sigma \cong \mathrm{Res}_{G_L} \rho_A \cong \bigoplus_{\iota \in \Sigma_E} \xi_\iota^{\oplus r}, \tag{8}$$

where in the middle isomorphism we use that ρ_A is a representation of G_F (and not just of G_L). By Equation (8), G_F acts on the set $\{\xi_\iota\}$ by inner conjugation. Because $E \hookrightarrow \mathrm{End}^0_K(A)$, any $\sigma \in G_K$ must respect the E-eigenspace decomposition of $V_\ell(A)$, and hence conjugation by any $\sigma \in G_K$ must stabilize every ξ_ι. (As we will see later, G_K is in fact the shared stabilizer of all of the ξ_ι under the inner conjugation action).

Applying Proposition 3.1 with $F' = K$ and $E' = E$, there exists a compatible system of G_K-representations $\mathcal{W}$ with coefficients in E such that $\mathrm{Res}_{G_K} \mathcal{V} \cong \mathrm{CRes}_{E/\mathbb{Q}} \mathcal{W}$ and $\mathrm{Res}_{G_L} \mathcal{W} \cong \mathcal{U}^{\oplus r}$. By definition, $\mathcal{U}_\sigma \cong \mathcal{U}$ for every $\sigma \in G_K$. We claim that there are compatible systems $\mathcal{R}$ and $\mathcal{A}$ of G_K-representations, as in the statement of Theorem 1.3, with coefficients in some extension of E, such that the compatible system $\mathcal{R} \otimes \mathcal{A}$ descends to E and $\mathcal{W} \cong \mathrm{Des}_E(\mathcal{R} \otimes \mathcal{A})$.

Fix a prime $\lambda_0 | \ell_0$ of E. We have $(u_{\lambda_0})_\sigma \cong u_{\lambda_0}$ for every $\sigma \in G_K$, so by Corollary 2.11, there exists a G_K-representation r_0 and a finite Galois extension M_0/K such that $\mathrm{Res}_{G_{M_0}} r_0 \cong \mathrm{Res}_{G_{M_0}} u_{\lambda_0}$. By Lemma 2.12, there then exists an Artin representation $a_0 \colon G_K \twoheadrightarrow \mathrm{Gal}(M_0/K) \to \mathrm{GL}_r(\overline{\mathbb{Q}_{\ell_0}})$ such that $w_{\lambda_0} \cong r_0 \otimes a_0$. In particular, $r_0 \otimes a_0$ is a subrepresentation of $\mathrm{Res}_{G_K} \rho_A$. Because $\mathrm{End}_{G_L} V_\ell(A) \cong D \otimes \overline{\mathbb{Q}_\ell}$, the projective kernel of a_0 is G_L.

Consider the abelian variety $A' := \mathrm{Res}_{M_0/K} A_{M_0}$, the Weil restriction to K of the base change to M_0 of A. Let ρ_A be the ℓ_0-adic representation of associated to A. The ℓ_0-adic Galois representation $\rho_{A'}$ associated to A' is $\mathrm{Ind}_{G_{M_0}}^{G_K}(\mathrm{Res}_{G_{M_0}} \rho_A) \cong (\mathrm{Res}_{G_K} \rho_A) \otimes \mathbb{Q}_{\ell_0}[\mathrm{Gal}(M_0/K)]$. Because $a_0^\vee$ is a subrepresentation of the regular representation $\mathbb{Q}_{\ell_0}[\mathrm{Gal}(M_0/K)]$, $a_0 \otimes a_0^\vee$ has a trivial representation summand, and r_0 is a direct summand of $\rho_{A'}$. By [Gu25, Lemma 1.6], it follows that r_0 is a member of a compatible system of Galois representations $\mathcal{R}$ over K with coefficients in some extension

E'/E. Concretely, this means that $r_0 = r_{\lambda_0'}$ for some prime $\lambda_0'|\ell_0$ of E'. Up to replacing $\mathcal{R}$ with a coefficient-wise Galois conjugate, we can assume that $\lambda_0'|\lambda_0$ and that $\iota_{\lambda_0'}$ extends ι_λ. The compatible system $\mathcal{R}$ is strongly irreducible because $\mathrm{Res}_{G_{M_0}} \mathcal{R} \cong \mathrm{Res}_{G_{M_0}} \mathcal{U}$ and $\mathcal{U}$ is strongly irreducible.

Assume without loss of generality that E' is large enough so that a_0 can be defined over E'. Then, the Artin representation a_0 fits into a compatible system $\mathcal{A} = \left((\jmath_\ell \circ \sigma_{\lambda'} \circ \jmath_{\ell_0}^{-1})(a_0)\right)_{\lambda'}$, where $\lambda'|\ell$ ranges over the primes of E' and $\sigma_{\lambda'} \in G_{\mathbb{Q}}$ is an element taking $\iota_{\lambda_0'}$ to $\iota_{\lambda'}$ (recall that these are both elements of $\Sigma_{E'}$, see Section 2.1). In particular, $a_0 = a_{\lambda_0'}$.

It follows that $\mathcal{W}' := \mathcal{R} \otimes \mathcal{A}$ is a compatible system over K with coefficients in E'. We want to show that $\mathcal{W}'$ descends to E and that $\mathcal{W} \cong \mathrm{Des}_E \mathcal{W}'$. Fixing a prime λ' of E', it suffices to show that the Frobenius characteristic polynomials of $w'_{\lambda'} = r_{\lambda'} \otimes a_{\lambda'}$ lie in $((\jmath_\ell \circ \iota_{\lambda'})(E))[X]$. Because $\mathcal{W}'$ is a compatible system, these characteristic polynomials are equal to $j_\ell \circ \iota_{\lambda'} \circ \iota_{\lambda_0'}^{-1} \circ \jmath_{\ell_0}^{-1}$ applied to the Frobenius characteristic polynomial of $r_{\lambda_0'} \otimes a_{\lambda_0'} \cong w_{\lambda_0}$. As w_{λ_0} is a member of a compatible system $\mathcal{W}$ with coefficients in E and $\lambda_0'|\lambda_0|\ell$, its Frobenius characteristic polynomials lie in $((\jmath_{\ell_0} \circ \iota_{\lambda_0})(E))[X]$. Because $\iota_{\lambda_0}(E) = \iota_{\lambda_0'}(E)$, it follows that $\mathcal{W}'$ descends to E and that $\mathcal{W} \cong \mathrm{Des}_E \mathcal{W}' \cong \mathrm{Des}_E(\mathcal{R} \otimes \mathcal{A})$ as desired.

So far, we have shown that

$$\mathrm{Res}_{G_K} \mathcal{V} \cong \mathrm{CRes}_{E/\mathbb{Q}} \mathcal{W} \cong \mathrm{CRes}_{E/\mathbb{Q}} \mathrm{Des}_E(\mathcal{R} \otimes \mathcal{A}). \tag{9}$$

It remains to understand $\mathcal{V}$ itself.

Earlier, we fixed an ℓ and considered the inner conjugation action of $\sigma \in G_F$ on $\bigoplus_{\iota \in \Sigma_E} \xi_\iota^{\oplus r}$, and observed that G_K is contained in the intersection of the stabilizers of the ξ_ι. Now, we can do better. By Equation (9), we have

$$(\mathrm{Res}_{G_K} \mathcal{V})_{E^{\mathrm{gal}}} \cong \bigoplus_{\tau \in \mathcal{T}} {}^\tau \mathcal{W}.$$

For any $\sigma \in G_F$, $(\mathrm{Res}_{G_K} \mathcal{V})_\sigma \cong \mathrm{Res}_{G_K} \mathcal{V}$. Suppose that σ stabilizes ${}^\tau \mathcal{W}$ for some $\tau \in \mathcal{T}$. Because inner and outer Galois conjugation commute, it follows that σ stabilizes ${}^\tau \mathcal{W}$ for every $\tau \in \mathcal{T}$. But then, looking at any r_λ, we see that σ stabilizes the E-eigenspace decomposition, and hence that it lies in the kernel of the map $G_F \to \mathrm{Aut}_{\mathbb{Q}}(E)$. In particular, any $\sigma \in G_F$ fixing any of the ${}^\tau \mathcal{W}$ fixes all of them and hence lies in G_K. Therefore, $\mathrm{Gal}(K/F)$ acts freely on the $\{{}^\tau \mathcal{W}\}_{\tau \in \mathcal{T}}$, and hence – picking an ℓ, on the $\{w_\iota\}_{\iota \in \Sigma_E}$, where $w_\iota = \xi_\iota \otimes \varrho_\iota$. Let $S \subset \Sigma_E$ be a set of orbit representatives for the $\mathrm{Gal}(K/F)$ action. Because $\mathrm{Res}_{G_K} \rho_A \cong \bigoplus_{\iota \in \Sigma_E} w_\iota$ and the $\{w_\iota\}$ are pairwise nonisomorphic, Frobenius reciprocity implies that

$$\rho_A \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{\iota \in S} \mathrm{Ind}_{G_K}^{G_F} w_\iota \cong \bigoplus_{\iota \in S} \mathrm{Ind}_{G_K}^{G_F}(\xi_\iota \otimes \varrho_\iota),$$

proving Corollary 1.4. The $\mathrm{Ind}_{G_K}^{G_F} w_\iota$ are not guaranteed to be irreducible (because each w_ι need not be, as ϱ_ι can be reducible) but each $\mathrm{Ind}_{G_K}^{G_F} w_\iota$ is a minimal direct sum of E-eigenspaces which are stable under the G_F-action (on $E \hookrightarrow \mathrm{End}_F^0(A)$). By construction, each is necessarily an E_0-eigenspace as $E_0 = E^{\mathrm{Gal}(K/F)}$. Applying Proposition 3.1 with $E' = E_0$ and $F' = F$, we deduce that that $\mathrm{Ind}_K^F \mathcal{W}$ is Schur unobstructed and descends to E_0, and furthermore that

$$\mathcal{V} \cong \mathrm{CRes}_{E_0/\mathbb{Q}} \mathrm{Des}_{E_0} \left(\mathrm{Ind}_K^F \mathcal{W}\right) \cong \mathrm{CRes}_{E_0/\mathbb{Q}} \mathrm{Des}_{E_0} \mathrm{Ind}_K^F \mathrm{Des}_E \left(\mathcal{R} \otimes \mathcal{A}\right),$$

concluding the proof of Theorem 1.3.

3.2 Potential automorphy

The following is our main tool to prove potential automorphy.

Corollary 3.2. *An abelian variety A/F is potentially automorphic if L/F is solvable and the compatible system $\mathcal{R}$ associated to A by Theorem 1.3 is potentially automorphic via Theorem 2.3 or Theorem 2.4.*

Proof. It suffices to show that $\rho_{A,\ell}$ is potentially automorphic. Because $\mathcal{R}$ is potentially automorphic, we may assume that all of the $\{\xi_{\iota_0}\}_{\iota_0}$ appearing in Corollary 1.4 are potentially automorphic.

Let M/K be the kernel of the Artin compatible system $\mathcal{A}$ appearing in Theorem 1.3. First, M/K is solvable (it is a cyclic extension of L/K), so $\mathrm{Res}_{G_M} \xi_j$ is potentially automorphic by Lemma 2.2. Therefore, $\mathrm{Res}_{G_M}(\xi_j \otimes \varrho_j) \cong (\mathrm{Res}_{G_M} \xi_j)^{\oplus r}$ is potentially automorphic, meaning that $\xi_j \otimes \varrho_j$ itself is. Second, because K/F is solvable, applying Lemma 2.2 implies that $\mathrm{Ind}_{G_K}^{G_F}(\xi_j \otimes \varrho_j)$ is potentially automorphic. By Lemma 2.2, the direct sum of these potentially automorphic representations is potentially automorphic, so the result follows from Corollary 1.4. □

Combining Corollary 3.2, class field theory, Theorem 2.3, and noting that a two-dimensional semisimple Galois representation without big image is (after restricting to a finite extension) a direct sum of characters, we deduce the following.

Corollary 3.3. *Let F be a totally real or CM field and A/F an abelian variety. Let the extension K/F and the compatible system $\mathcal{R}$ be as in Theorem 1.3. If K is totally real or CM, L/F is solvable, $\mathcal{R}$ is one- or two-dimensional, and (if $\dim \mathcal{R} = 2$) $\mathcal{R}$ satisfies the hypothesis of Theorem 2.3, then A is potentially automorphic.*

We will use Corollary 3.3 to prove potential modularity of some atypical abelian fourfolds.

3.3 Pairings

Throughout this subsection, we fix a rational prime ℓ. As noted in the proof of Theorem 1.3, $E \hookrightarrow \mathrm{End}_K^0(A)$, and

$$\mathrm{Res}_{G_K} V_\ell(A) \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{\iota \in \Sigma_E} (\xi_\iota \otimes \varrho_\iota). \tag{10}$$

Lemma 3.4.

1. *If A is not type IV, the compatible system $\mathrm{Des}_E(\mathcal{R} \otimes \mathcal{A})$ associated to A_K by Theorem 1.3 is of symplectic type with cyclotomic similitude character.*
2. *If A is type IV, the compatible system $\mathrm{Des}_E(\mathcal{R} \otimes \mathcal{A})$ associated to A_K by Theorem 1.3 is of unitary type with cyclotomic similitude character.*

Proof. Since $V_\ell(A)^\vee \cong V_\ell(A) \otimes \varepsilon^{-1}$, for any ι we have $(\xi_\iota \otimes \varrho_\iota)^\vee \cong (\xi_{\iota'} \otimes \varrho_{\iota'}) \otimes \varepsilon^{-1}$ for some ι'. Let $\phi\colon V_\ell(A) \to V_\ell(A)^\vee \otimes \varepsilon$ be induced by the polarization on A. Because the Rosati involution is the adjoint involution for the polarization, for any $d \in \mathrm{End}^0(A) \hookrightarrow \mathrm{End}_{G_F} V_\ell(A)$ we have $d^\vee \circ \phi = \phi \circ d^\dagger$. If $e \in E \subset D$ is self-adjoint with respect to the Rosati involution, then distinct E-eigenspaces (the $\{\xi_\iota \otimes \varrho_\iota\}$) are orthogonal with respect to the pairing ψ on $V_\ell(A)$ induced by ϕ. If A is not geometrically type IV, then this applies to every $e \in E$. The pairing on $V_\ell(A)$ is then the sum of the pairings on the E-eigenspaces, so each E-eigenspace must preserve a symplectic pairing and the first part of the result follows. If E is geometrically type IV, then we can apply the same reasoning to the eigenspaces of the totally real subfield E^+ (the subfield of E fixed by $\dagger$)

of E. Each E^+-eigenspace is a direct sum of two complex-conjugate E-eigenspaces corresponding to embeddings $\iota, \bar{\iota} \in \mathrm{Hom}_{E^+}(E, \overline{\mathbb{Q}})$. Without loss of generality, suppose that x and y are in the E-eigenspace corresponding to ι. Then, $\iota(e)\psi(x,y) = \psi(ex,y) = \psi(x,e^\dagger y) = \iota(e^\dagger)\psi(x,y)$. Taking $e \in E \setminus E^+$, $\iota(e) \neq \iota(e^\dagger)$ and $\psi(x,y) = 0$. By nondegeneracy of the Weil pairing, we deduce that the two E-eigenspaces are paired against one another and that $\xi_\iota \otimes \varrho_\iota$ is essentially conjugate self-dual. Because the pairing on the entire E^+-eigenspace is alternating with cyclotomic similitude character, the pairing on each E-eigenspace is skew-Hermitian and with cyclotomic similitude character. □

As before, let M/K be the extension through which the $\{\varrho_\iota\}$ (or equivalently, the compatible system $\mathcal{A}$) factor(s).

Lemma 3.5. *When A is not type IV, the $\{\xi_\iota\}$ and $\{\varrho_\iota\}$ appearing in Equation (10)) (and hence, the $\{\xi_{\iota_0}\}$ and $\{\varrho_{\iota_0}\}$ appearing in Corollary 1.4) are self-dual up to similitude. Furthermore, the Weil pairing on $V_\ell(A)$ is the sum of the products of the pairings on the $\{\xi_\iota\}$ and $\{\varrho_\iota\}$.*

Proof. Fix an $\iota_\ell := \jmath_\ell \circ \iota \colon E \hookrightarrow \overline{\mathbb{Q}_\ell}$. By Lemma 3.4, $\xi_\iota \otimes \varrho_\iota$ preserves a symplectic pairing up to cyclotomic similitude. Therefore,

$$\wedge^2(\xi_\iota \otimes \varrho_\iota) \cong \mathrm{Sym}^2 \xi_\iota \otimes \wedge^2 \varrho_\iota \oplus \wedge^2 \xi_\iota \otimes \mathrm{Sym}^2 \varrho_\iota \tag{11}$$

must have a one-dimensional irreducible constituent. Restricting Equation (11) to G_M, at least one of $\mathrm{Sym}^2(\mathrm{Res}_{G_M} \xi_\iota)$ and $\wedge^2(\mathrm{Res}_{G_M} \xi_\iota)$ has a one-dimensional irreducible constituent, and it follows that $\mathrm{Res}_{G_M} \xi_\iota$ is self-dual up to similitude. By Frobenius reciprocity and strong irreducibility of ξ_ι, ξ_ι itself is self-dual up to similitude. It follows that at least one of $\wedge^2 \varrho_\iota$ or $\mathrm{Sym}^2 \varrho_\iota$ has a one-dimensional irreducible constituent, so ϱ_ι is also self-dual up to similitude.

Because ξ_ι is strongly irreducible, the dimension of the space of pairings preserved (up to similitude) by ξ_ι is at most one-dimensional – Schur's lemma ensures that there is at most a one-dimensional space of isomorphisms from $\xi_\iota^\vee$ to $\xi_\iota \otimes \nu$ for any ν, and if $\xi_\iota^\vee \cong \xi_\iota \otimes \nu \cong \xi_\iota \otimes \nu'$ for different similitude characters ν and ν', then $\xi_\iota \cong \xi_\iota \otimes \nu'\nu^{-1}$, contradicting Frobenius reciprocity. By Equation (11), the symplectic pairing on $\xi_\iota \otimes \varrho_\iota$ is then the product of the unique pairing on ξ_ι and a complementary pairing on ϱ_ι – note that there may be multiple choices of pairing on ϱ as it need not be irreducible – and the (cyclotomic) similitude character of the pairing on $\xi_\iota \otimes \varrho_\iota$ is the product of the respective similitude characters. □

The $\{\xi_\iota\}$ do not need to preserve symplectic pairings when $r > 1$ – as we will see, either summand in Equation (11) can have the one-dimensional subrepresentation, depending on the geometric Albert type of the abelian variety. Suppose A is of geometric Albert type II or III, so $D := \mathrm{End}_F^0(A)$ is a quaternion algebra and $r = 2$. Let $\mathfrak{hg}_\ell$ be the traceless Lie algebra of the Zariski closure over $\mathbb{Q}_\ell$ of the identity component of image of the ℓ-adic Galois representation.

Let $W_\iota := \mathrm{Res}_M \xi_\iota$. Restricting both sides of Equation (10) to G_M, we have

$$\mathrm{Res}_{G_M} V_\ell(A) \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{\iota \colon E \hookrightarrow \overline{\mathbb{Q}_\ell}} W_\iota^{\oplus 2}$$

as $\mathfrak{hg}_\ell$-representations. The following result is well-known.

Lemma 3.6. *Suppose that A is geometrically type II or III. For all $\iota \colon E \hookrightarrow \overline{\mathbb{Q}_\ell}$, the representation of $\mathfrak{hg}_\ell$ on W_ι is self-dual. This representation preserves a symplectic pairing if D is indefinite, and an orthogonal pairing if D is definite.*

Proof. Because $V_\iota := W_\iota^{\oplus 2} \cong \mathrm{Res}_M(\xi_\iota \otimes \varrho_\iota) \cong (\mathrm{Res}_M \xi_\iota)^{\oplus 2}$, self-duality of W_ι follows from the self-duality (up to similitude) of ξ_ι given by Lemma 3.5. Henceforth, we write $V := V_\iota$ and $W := W_\iota$. Because W is irreducible, the pairing is unique by Schur's lemma. Therefore, we can write the matrix of the pairing on V as $\Omega' \otimes \Omega_W$ where Ω_W is the matrix of the pairing on W and $\Omega' = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\overline{\mathbb{Q}_\ell})$. If $(\Omega')^T = \Omega'$, then the pairing on W is alternating, and if $(\Omega')^T = -\Omega'$, then the pairing on W is symmetric. In the former case, we may change basis so that $\Omega' = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, and in the latter, $\Omega' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. This lets us express the pairing $\langle \cdot, \cdot \rangle_V$ on V in terms of that on W.

Since we are working with a single ι-eigenspace, we pick a corresponding embedding of D into $M_2(\overline{\mathbb{Q}_\ell})$. Because the endomorphisms in D commute with the action of $\mathfrak{hg}_\ell$, D acts via this 2×2 matrix on the decomposition $V \cong W \oplus W$. The Rosati involution respects the pairing, so writing $x, y \in V$ as $(x_1, x_2), (y_1, y_2) \in W \oplus W$, we have

$$\langle d(x_1, x_2), (y_1, y_2) \rangle_V = \langle dx, y \rangle_V = \langle x, d^\dagger y \rangle_V = \langle (x_1, x_2), d^\dagger (y_1, y_2) \rangle.$$

By direct computation, we can work out the matrix of $d^\dagger \in M_2(\overline{\mathbb{Q}_\ell})$ should be in terms of that of d, and we find that $\dagger$ is the standard involution on $M_2(\overline{\mathbb{Q}_\ell})$ when Ω_W is symmetric, and the transpose when Ω_W is alternating. The standard involution (which is invariant under base change and change of basis) is a positive involution of D if D is definite, and is not a positive involution if D is indefinite. Since the Rosati involution is a positive involution, the result follows. □

Remark. When A is a "Weil type" type IV abelian fourfold (see e.g. [LPS13, Section 2]), $\mathrm{End}^0_{\overline{\mathbb{Q}}}(A)$ is an imaginary quadratic field. One can check in this case that $\mathrm{Res}_{G_L} \rho_A \otimes \overline{\mathbb{Q}_\ell} \cong \xi \oplus \xi^\tau$ and that $\xi^\vee \cong \xi^\tau \otimes \varepsilon^{-1}$ – in this case, ξ preserves a unitary pairing, rather than a symplectic or orthogonal one.

All told, each ξ_j in Theorem 1.3 preserves – up to similitude – a symplectic pairing when A is type I or II, an orthogonal pairing when A is type III, and a unitary pairing when A is type IV.

The following corollary addresses a question of [BCGP21], by showing that their counterexamples to the original version of the paramodular conjecture of Brumer–Kramer (see e.g. [Mar18]) cannot be abelian varieties of type III.

Corollary 3.7. *Let $A/\mathbb{Q}$ be a geometrically simple abelian fourfold. Suppose that $\rho_A \otimes \overline{\mathbb{Q}_\ell} \cong \rho_\ell \oplus \rho_\ell$ for some strongly irreducible $\rho_\ell \colon G_\mathbb{Q} \to \mathrm{GSp}_4(\overline{\mathbb{Q}_\ell})$. Then, A is of type* II(1).

Proof. Because A is geometrically simple and $\mathrm{End}_{G_\mathbb{Q}} V_\ell(A) \otimes \overline{\mathbb{Q}_\ell} \cong M_2(\overline{\mathbb{Q}_\ell})$, $\mathrm{End}^0(A)$ is a quaternion algebra $D/\mathbb{Q}$. Suppose that D is definite. Let W_ℓ be the vector space underlying ρ_ℓ. By the Mumford–Tate conjecture (known for fourfolds with geometric quaternionic multiplication by [MZ95]), Lemma 3.6 implies that the representation of $\mathfrak{hg}_\ell$ on W_ℓ preserves an orthogonal pairing. Because ρ_ℓ preserves a symplectic pairing up to similitude, the representation of $\mathfrak{hg}_\ell$ on V_ℓ also preserves an symplectic pairing. Because ρ_ℓ is strongly irreducible, the representation of $\mathfrak{hg}_\ell$ is irreducible, and so by Schur's lemma there is a unique nonzero $\mathfrak{hg}_\ell$-invariant pairing on V_ℓ up to scaling. This pairing cannot be both orthogonal and symplectic, so we have a contradiction. It follows that D is type II(1). □

3.4 Fields of definition of endomorphisms of abelian varieties

Theorem 1.3 lets us reinterpret some facts about the field of definition of endomorphisms of abelian varieties in the language of Galois representations.

The compatible systems $\mathcal{R}$ and $\mathcal{A}$ appearing in Theorem 1.3 are only canonical up to twist. Nonetheless, it makes sense to talk about the projective kernel of $\mathcal{A}$, or equivalently, the shared projective kernel of the $\{\operatorname{Proj} \varrho_{\iota_0}\}_{\iota_0}$ appearing in Corollary 1.4. The field fixed by this projective kernel is exactly the field L over which $\operatorname{End}_{G_L} V_\ell(A) \otimes \overline{\mathbb{Q}_\ell} \cong M_r(\overline{\mathbb{Q}_\ell})^e$, i.e. the field over which the geometric endomorphisms of A are defined.

As in Theorem 1.3, let K the minimal field of definition of the endomorphisms in E. By Skolem-Noether, $\operatorname{Gal}(L/K)$ embeds into $D^\times/E^\times$. Picking an embedding of E into $\mathbb{C}$ gives us an embedding of $D^\times/E^\times$ into $\mathrm{PGL}_2(\mathbb{C})$. When A is type II, choosing an embedding of E into $\mathbb{R}$ induces an embedding of $D^\times/E^\times \hookrightarrow \mathrm{PGL}_2(\mathbb{R})$, and the only finite subgroup of $\mathrm{PGL}_2(\mathbb{R})$ are cyclic or dihedral It follows that any finite subgroup of $D^\times/E^\times$ is necessarily cyclic or dihedral when D is indefinite, and hence $\operatorname{Gal}(L/K)$ is cyclic or dihedral for geometrically type II abelian varieties (this is part of a much stronger result proved in [CF00, Lemma 2.8]). Lemma 4.1 gives an alternative proof of this fact – it reflects the Artin compatible system $\mathcal{A}$ being of orthogonal type, which in turns follows from $\mathcal{R}$ being of symplectic type.

We can make the relationship between $\mathcal{A}$ and the Galois action on D more concrete. For any embedding $\jmath_\ell \circ \iota \colon E \hookrightarrow \overline{\mathbb{Q}_\ell}$, we obtain as above a map ${}^{\mathbb{P}}\varrho' \colon \operatorname{Gal}(L/K) \to \mathrm{PGL}_2(\overline{\mathbb{Q}_\ell})$ sending σ to a matrix γ_σ realizing the Galois conjugation action on $(\jmath_\ell \circ \iota)(D)$ as the inner conjugation $(\jmath_\ell \circ \iota)(d) \mapsto \gamma_\sigma(\jmath_\ell \circ \iota)(d)\gamma_\sigma^{-1}$. (That such a γ_σ exists follows from the Skolem–Noether theorem). As discussed in the proof of Corollary 2.11, Tate's lifting theorem (see [Pat19, Theorem 2.11]) implies the existence of genuine lifts $\varrho'_\iota \colon G_K \to \mathrm{GL}_2(\overline{\mathbb{Q}_\ell})$ of ${}^{\mathbb{P}}\varrho'$.

Lemma 3.8. *Let ϱ_ι be as in Corollary 1.4 and ϱ'_ι be as above. Then, ϱ'_ι and ϱ_ι are isomorphic up to twist.*

Proof. For simplicity, assume that $E = \mathbb{Q}$, so $\rho_{A,\ell} \cong \xi_\ell \otimes \varrho_\ell$. Abusing notation and writing d for both an element of D and its image in $D \otimes \overline{\mathbb{Q}_\ell}$, because ${}^\sigma(dx) = {}^\sigma d\, {}^\sigma x$ for any $x \in V_\ell(A)$, the action of G_K on $D \otimes \overline{\mathbb{Q}_\ell}$ is given exactly by conjugation by $\rho_D(\sigma)$. For any K'/K, Faltings' theorem and the strong irreducibility of ξ_ℓ imply that

$$M_r(\overline{\mathbb{Q}_\ell}) \cong D \otimes \overline{\mathbb{Q}_\ell} \cong \operatorname{End}_{G_{K'}} \xi_\ell \otimes \varrho_\ell \cong \operatorname{End}_{G_{K'}} \varrho_\ell.$$

Working through this isomorphism, the image of $\gamma \in M_r(\overline{\mathbb{Q}_\ell})$ is the block matrix $I \otimes \gamma \in \operatorname{End} V_\ell(A)$. Therefore, conjugation by a matrix $\xi_\ell(\sigma) \otimes \varrho_\ell(\sigma)$ acts by sending $\gamma \mapsto \varrho_\ell(\sigma)\gamma\varrho_\ell(\sigma)^{-1}$, and the result follows. For general E, the same argument works, replacing $M_r(\overline{\mathbb{Q}_\ell})$ with $M_r(\overline{\mathbb{Q}_\ell})^e$ and noting that the subrepresentations corresponding to different ι do not interact. □

Example 3.9. There are abelian varieties where $E \subsetneq \operatorname{End}^0_K(A)$. Let $f \in S_2(\Gamma_0(243))$ be the newform (243.2.a.d)

$$f := q + \beta q^2 + 4q^4 - \beta q^5 + 2q^7 + 2\beta q^8 - 6q^10 + \beta q^11 - q^13 + 2\beta q^14 + \cdots,$$

which has coefficients over $\mathbb{Q}(\sqrt{6})$. Let A be the abelian variety of GL_2-type associated to f, so $\rho_A \otimes \overline{\mathbb{Q}_\ell} \cong \rho_{f,\ell} \oplus \rho_{{}^\tau f,\ell}$. This f admits the inner twist ${}^\tau f = f \otimes \left(\frac{-3}{\cdot}\right)$, so writing $L := \mathbb{Q}(\sqrt{-3})$, $\operatorname{Res}_{G_L} V_\ell(A) \otimes \overline{\mathbb{Q}_\ell} \cong (\operatorname{Res}_{G_L} \rho_f)^{\oplus 2}$. In particular, $\operatorname{End}^0_{\mathbb{Q}}(A) \otimes \overline{\mathbb{Q}} \cong M_2(\overline{\mathbb{Q}})$, so $\operatorname{End}_{G_L}(A)$ is either $M_2(\mathbb{Q})$ or a quaternion algebra D with center $\mathbb{Q}$. Looking at the Fourier coefficients of f, we see that $a_p(f) \in \mathbb{Q}$ whenever p is split in L, so $\operatorname{Res}_L \rho_f$ has traces in $\mathbb{Q}$. If it were defined over $\mathbb{Q}_\ell$ for all ℓ, then $\operatorname{Res}_L \rho_f$ would be the Galois representation associated to an elliptic curve E/F that is isogenous to its Galois conjugate (a so-called $\mathbb{Q}$-curve), A would be geometrically reducible, and $\operatorname{End}^0_F(A)$ would be $M_2(\mathbb{Q})$. However, it turns out that ρ_f is not defined over $\mathbb{Q}_\ell$ for all ℓ, and $\operatorname{End}^0_L(A)$ is in fact the quaternion algebra $D/\mathbb{Q}$ with discriminant 6, In particular, $\operatorname{End}^0_{\mathbb{Q}}(A) \cong \mathbb{Q}(\sqrt{6})$, $E := \mathbb{Q}$, and D is the aforementioned quaternion algebra.

4 Geometrically type II abelian varieties

In this section, we discuss some phenomena specific to the case of geometrically type II abelian varieties, where D/E is an indefinite quaternion algebra and E is a totally real field.

Lemma 4.1. *Let A/F an abelian variety of geometric type II. The $\{\varrho_\iota\}$ appearing in Equation (10) (and in particular the $\{\varrho_{\iota_0}\}$ appearing in Corollary 1.4) preserve orthogonal pairings up to similitude and have either cyclic or dihedral projective image.*

Proof. Let E be the center of the quaternion algebra $D := \mathrm{End}^0_{\overline{F}}(A)$. By Lemma 3.6, the $\{\xi_\iota\}_\iota$ appearing in Equation (10) all preserve symplectic pairings up to similitude. By Lemma 3.5, the $\{\varrho_\iota\}_\iota$ must hence preserve orthogonal pairings up to similitude. Because the $\{\varrho_\iota\}_\iota$ are two-dimensional orthogonal representations, they have cyclic or dihedral projective images. □

We recall the notion of a distinguished quadratic subfield of D from [LSSV24, Subsection 3.4]. By Skolem-Noether, the Rosati involution on $d \in D$ has the form $d \mapsto \mu\overline{d}\mu^{-1}$ for some nontrivial $\mu \in D^\times/E^\times$, where $d \mapsto \overline{d}$ denotes the standard involution. Because D is indefinite, the standard involution on D is not a positive involution, as discussed in the proof of Lemma 3.6, so we must have $\overline{\mu} = -\mu$ and $\mu^2 < 0$ at any place of E. Because the Rosati involution is an involution and the center of D is E, $E' := E(\mu)$ is a well-defined CM extension of E depending only on A and its polarization.

Lemma 4.2. *Suppose that A/F is such that $\mathrm{End}_{\overline{F}}(A)$ contains an indefinite quaternion algebra D/E for some totally real field E. As in Theorem 1.3, let K/F be the minimal field over which $E \hookrightarrow \mathrm{End}^0_K(A)$. Then, there exists a quadratic extension K'/K such that $\mathrm{End}^0_{K'}(A)$ contains the distinguished CM subfield $E' := E(\mu)$ of D.*

Proof. As usual, write ρ_A for the representation on $V_\ell(A)$, and let L/F be the minimal extension over which $\mathrm{End}^0_L(A) = \mathrm{End}^0_{\overline{F}}(A)$. Recall that G_L is a normal subgroup of G_F. As discussed in Lemma 3.8, an element $d \in D \hookrightarrow \mathrm{End}_{G_L}(V_\ell(A))$ is in (the image of) $\mathrm{End}^0_{F'}(A)$ if and only if it commutes with the image of $G_{F'}$, i.e. if $\rho_A(g')d\rho_A(g')^{-1} = d$ for any $g' \in G_{F'}$.

The Rosati involution (or rather, its $\overline{\mathbb{Q}_\ell}$-linear extension to $D \otimes \overline{\mathbb{Q}_\ell}$) commutes with the G_F-action on D – for any $v, v' \in V_\ell(A)$, letting $\psi(\cdot,\cdot)$ denote the Weil pairing on $V_\ell(A)$, we have

$$\psi(\rho_A(g)d\rho_A(g)^{-1}v, v') = \varepsilon(g)\psi(d\rho_A(g)^{-1}v, \rho_A(g)^{-1}v') = \varepsilon(g)\psi(\rho_A(g)^{-1}v, d^\dagger\rho_A(g)^{-1}v') = \psi(v, \rho_A(g)d^\dagger\rho_A(g)^{-1}v').$$

However, we also know that $d^\dagger = \mu\overline{d}\mu^{-1}$ for some $\mu \in D \hookrightarrow M_{2g}(\overline{\mathbb{Q}_\ell})$. As such,

$$(\rho_A(g)d\rho_A(g)^{\iota})^\dagger = \mu\overline{\rho_A(g)d\rho_A(g)^{-1}}\mu^{-1} = \mu\rho_A(g)\overline{d}\rho_A(g)^{-1}\mu^{-1},$$

where in the second equality use that because D is a quaternion algebra, after embedding into $M_2(\overline{\mathbb{Q}_\ell})^e$, the standard involution is given by sending a matrix to its adjugate.

This is equal to $\rho_A(g)d^\dagger\rho_A(g)^{-1}$ for all $d \in D$ if and only if $\rho_A(g)^{-1}\mu\rho_A(g) = \omega(g)\mu$ for some E-valued function ω. Restricting to G_K, we see that $\rho_A(g)$ commutes with μ and so ω is a character. Because E is totally real, ω must be quadratic. As such, taking K' to be the fixed field of the kernel of ω, $\rho_A(g)^{-1}\mu\rho_A(g) = \mu$ for all $g \in G_{K'}$ and hence $\mu \in \mathrm{End}^0_{K'}(A)$. □

Lemma 4.2 is not true without allowing an extension K'/K – consider for instance GL_2-type abelian surfaces over $\mathbb{Q}$, which have endomorphisms by a real quadratic field over $\mathbb{Q}$ but acquire quaternionic multiplication over a quadratic extension.

Lemma 4.3. *Suppose that* $\mathrm{Res}_{G_{F'}} V_\ell(A) \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{\iota \in \Sigma_E} \xi'_\iota \otimes (1 \oplus \eta_\iota)$*, where* $E' := \mathrm{End}^0_{F'}(A)$ *and the* ξ'_ι *are strongly irreducible and self-dual up to similitude. Then, the pairing induced by the Weil pairing on* $(1 \oplus \eta_\iota)$ *(given by Lemma 3.5) is*

1. $e_1 \otimes e_1 + e_2 \otimes e_2$ *if* E'/E *is totally real;*
2. $e_1 \otimes e_2 + e_2 \otimes e_1$ *if* E'/E *is CM,*

where e_1 *and* e_2 *are basis vectors for the subrepresentations* 1 *and* η_ι*, respectively.*

For example, in the context of Corollary 1.4, Lemma 4.3 applies to abelian varieties A/F with fields F'/F such that $\mathrm{End}^0_{F'}(A) = E' \supseteq E$, in which case $\mathrm{Res}_{G_{F'}} \rho_\iota \cong 1 \oplus \eta_\iota$ with $\eta_\iota \neq 1$.

Proof. The subspaces ξ_ι and $\xi_\iota \otimes \eta$ are the eigenspaces of the E'-action on $V_\ell(A)$ with respect to the two embeddings $\tilde{\iota}, \tilde{\iota}' \in \Sigma_{E'}$ extending $\iota \in \Sigma_E$. By Lemma 3.4, the Weil pairing restricts to a pairing on $\xi_\iota \otimes (1 \oplus \eta_\iota)$. The Rosati involution $\dagger \colon E' \to E'$ – which is trivial if E' is totally real and is complex conjugation if E' is CM – is the adjoint operation for $e \in E'$ with respect to the Weil pairing. Let $\tilde{\iota}_\ell := \jmath_\ell \circ \tilde{\iota}_\ell$ and define $\tilde{\iota}'_\ell$ analogously. Writing $\psi(\cdot, \cdot)$ for the Weil pairing this means that for any $e \in E'$, $x \in \xi_\iota$, and $y \in \xi_\iota \otimes \eta$,

$$\tilde{\iota}_\ell(e)\psi(x, y) = \psi(ex, y) = \psi(x, e^\dagger y) = \tilde{\iota}'_\ell(e^\dagger)\psi(x, y).$$

In particular, if E' is totally real, $\dagger$ is trivial, and $\psi(x, y) = 0$. Similarly, taking $x, y \in \xi_\iota$,

$$\tilde{\iota}_\ell(e)\psi(x, y) = \psi(ex, y) = \psi(x, e^\dagger y) = \tilde{\iota}_\ell(e^\dagger)\psi(x, y).$$

In particular, if E'/E is CM, $\dagger$ is nontrivial, and $\psi(x, y) = 0$. We can do the same computation for $x, y \in \xi_\iota \otimes \eta$ and $x, y \in \xi_\iota$ and deduce the result. □

Known automorphy theorems that we might want to apply to the compatible system $\mathcal{R}$ over a totally real field typically impose a parity condition on the representations in $\mathcal{R}$. For example, the automorphy lifting theorem of Boxer–Calegari–Gee–Pilloni (Theorem 2.4) for four-dimensional symplectic Galois representations, which we wish to apply to prove potential automorphy of type I(2) and II(1) abelian fourfolds, requires $\mathcal{R}$ to be odd – in this case, the associated automorphic forms are Siegel modular forms. However, there are in fact abelian varieties whose associated compatible systems $\mathcal{R}$ are even (Section 7). As such, we will content ourself with the following weaker result.

Lemma 4.4. *Suppose that* A/F *is geometrically type II. Assume that the field* K *in Theorem 1.3 is totally real, and there that is no field* K'/K *with a real embedding for which* $\mathrm{End}_{K'}(A)$ *is a CM extension of* E*. Then, the* $\{\xi_\iota\}$ *appearing in Equation (10) (and hence, the compatible system* $\mathcal{R}$ *appearing in Theorem 1.3) are (is) odd.*

Note that because the $\{\xi_\iota\}_\iota$ are representations of G_K, the question of oddness only makes sense when K is totally real. The third condition of Lemma 4.4 may seem unnatural, but Main Theorem 2 demonstrates that a condition of this sort is necessary.

Proof. Fix an embedding $\iota \colon E \hookrightarrow \overline{\mathbb{Q}_\ell}$, and let $\xi := \xi_\iota$ and $\varrho := \varrho_\iota$. As discussed earlier, ϱ preserves an orthogonal pairing (up to similitude). The similitude character of ϱ is then a one-dimensional summand in $\mathrm{Sym}^2 \varrho$.

First suppose that ϱ has cyclic projective image, so – up to possibly replacing ξ by a twist – $\varrho \cong 1 \oplus \eta$. Then,

$$\mathrm{Sym}^2 \varrho \cong 1 \oplus \eta^2 \oplus \eta. \tag{12}$$

The first two direct summands in Equation (12) are automatically even. The similitude character can only be η when the direct summands of ϱ corresponding to 1 and η are paired against each other by the Weil pairing, which by Lemma 4.3 can happen only if $\mathrm{End}^0_K(A)$ is a CM extension of E. Under our assumptions, any possible similitude character in Equation (12) is thus even.

Suppose instead that ϱ has dihedral projective image, so $\varrho \cong \mathrm{Ind}_{G_{K'}}^{G_K} \eta$ for K'/K a quadratic extension, and $\mathrm{End}^0_K(A) = E$. Let M/K' be the field cut out by η. Considering the restriction to $G_{K'}$ and the determinant, one can show that

$$\mathrm{Sym}^2 \mathrm{Ind}_{G_{K'}}^{G_K} \eta \cong \mathrm{Ind}_{G_{K'}}^{G_K} \eta^2 \oplus (\eta \circ \mathrm{Ver}), \tag{13}$$

where $\mathrm{Ver}\colon G_K^{\mathrm{ab}} \to G_{K'}^{\mathrm{ab}}$ is the transfer map. Here, $\eta \circ \mathrm{Ver}$ is the similitude character of a pairing that is given (in an appropriate basis) by the matrix $\begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$. If $\eta^2 \ncong \eta_\sigma^2$ and $\mathrm{Ind}_{G_{K'}}^{G_K} \eta^2$ is irreducible, the similitude character of the orthogonal pairing on ϱ is necessarily $\eta \circ \mathrm{Ver}$ as it is the only one-dimensional summand in Equation (13). We have

$$(\eta \circ \mathrm{Ver})(c) = \begin{cases} \eta(c)\eta(\sigma c \sigma^{-1}) & c \in G_{K'} \\ \eta(c^2) & c \notin G_{K'}. \end{cases}$$

In the second case, $\eta(c^2) = \eta(1) = 1$. In the first case, K' has a real embedding and as $\mathrm{Res}_{G_{K'}} \varrho$ has cyclic projective image, ξ is odd by our assumptions and our discussion of the cyclic projective image case at the real place.

If $\eta^2 \cong \eta_\sigma^2$, then the projective image of ϱ is D_2, and additionally η^2 extends to a character $\widetilde{\eta^2}$ of G_K. In this case, ϱ can be induced from three different index 2 subgroups of G_K, corresponding to the three index 2 subgroups of D_2. Any extension of K over which ϱ becomes reducible must correspond to one of these three extensions, and so by Lemma 4.2, one of the corresponding extensions K'/K is such that $\mathrm{End}^0_{K'}(A)$ is a CM extension of E. We have

$$\mathrm{Sym}^2 \mathrm{Ind}_{G_{K'}}^{G_K} \eta \cong \widetilde{\eta^2} \oplus \left(\widetilde{\eta^2} \otimes \omega_{K'/K}\right) \oplus (\eta \circ \mathrm{Ver}). \tag{14}$$

Because $\mathrm{End}^0_{K'}(A)$ is a CM field, the similitude character of the orthogonal pairing must be $\eta \circ \mathrm{Ver}$ by Lemma 4.3. The result then follows from the earlier discussion. □

5 Twisted moduli spaces of abelian schemes

Let D a division ring with center E of degree e equipped with a positive involution $\dagger$, and let $\mathcal{O}$ be an order of D stable under $\dagger$.

Let V be a $2g$-dimensional vector space over $\mathbb{Q}$ and $\Lambda \subset V$ a lattice defining a weight 1 $\mathcal{O}$-Hodge structure with principal polarization $\psi\colon \Lambda \times \Lambda \to \mathbb{Z}$ such that $\psi(dx, y) = \psi(x, d^\dagger y)$ for $d \in \mathcal{O}$ and $x, y \in \Lambda$. Let h be the Hodge cocharacter associated to Λ.

Define $G := \mathrm{CSp}_D(V, \psi)$ and let X be a conjugacy class X of homomorphisms $\mathbb{S} := \mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m \to G_{\mathbb{R}}$ such that (G, X) is a PEL Shimura datum. Let F_0 be the reflex field of (G, X). Fix an extension F'/F_0, a finite Galois extension L'/F', a finite set of rational primes S. Choose homomorphisms

$$\rho_D\colon \mathrm{Gal}(L'/F') \longrightarrow \mathrm{Aut}_{\mathbb{Q}}(D) \qquad \text{and} \qquad \overline{\rho}_\ell\colon \mathrm{Gal}(L'/F') \longrightarrow \mathrm{GL}(\Lambda/\ell\Lambda) \text{ for all } \ell \in S$$

such that the action of $\mathrm{im}\,\rho_D$ preserves $\mathcal{O}$ and commutes with $\dagger$. Let n be the product of the primes in S. We can combine the data of the $\overline{\rho}_\ell$ into a single group homomorphism $\overline{\rho}_n\colon \mathrm{Gal}(L'/F') \to$

$\mathrm{GL}(\Lambda/n\Lambda)$. Intuitively, ρ_D and $\overline{\rho}_n$ will specify a Galois action on the endomorphism algebra and on the mod-n principal level structure. Let G'_n denote the étale group scheme over F' (of order n^{2g}) associated to $\overline{\rho}_n$. Let $U \subset G(\mathbb{A}_f)$ be the compact open subgroup associated to the principal level-n structure.

The following definitions are largely motivated by [Lan13, Definition 1.4.1.2]. Consider the moduli functor $\mathcal{M}_{F'} \colon \mathbf{Sch}_{/F'} \to \mathbf{Set}$ sending a scheme S to the set of quadruples $(A, \lambda, \iota, \eta)$ (to be defined momentarily) up to isomorphism (also to be defined momentarily).

For any scheme S/F', we define $S_{L'} := S \times_{\operatorname{Spec} F'} \operatorname{Spec} L'$. In each quadruple:

- A is an abelian scheme over S of relative dimension g;
- $\lambda \colon A \to A^\vee$ is a $\mathbb{Z}^\times$-polarization (in the sense of [Lan13, Definition 1.3.2.19]) of A
- $\iota \colon \mathcal{O} \to \mathrm{End}(A \times_S S_{L'})$ defines an $\mathcal{O}$-structure of $(A \times_S S_{L'}, \lambda)$ (in the sense of [Lan13, Definition 1.3.3.1]) such that the action of $\mathrm{Gal}(L'/F')$ on $\iota(D) \subseteq \mathrm{End}(A \times_S S_{L'})$ is given by ρ_D;
- η consists of an $\mathcal{O}$-equivariant group scheme isomorphism $\alpha_n \colon (G'_n)_S \xrightarrow{\sim} A[n]$ and a group scheme isomorphism $\nu_n \colon (\mathbb{Z}/n\mathbb{Z})_S(1) \xrightarrow{\sim} \mu_{n,S}$ such that the diagram

$$\begin{array}{ccc} G'_n \times G'_n & \xrightarrow{\psi(\cdot,\cdot)} & (\mathbb{Z}/n\mathbb{Z})_S(1) \\ \downarrow{\scriptstyle \alpha_n \times \alpha_n} & & \downarrow{\scriptstyle \nu_n} \\ A[n] \times A[n] & \longrightarrow & \mu_{n,S} \end{array} \tag{15}$$

  commutes, where the bottom edge is the Weil pairing.
- $\mathrm{Lie}_{A \times_S S_{L'}}$ satisfies the Kottwitz determinant condition described in [Lan13, Definition 1.3.4.1] given by $(\Lambda \otimes_{\mathbb{Z}} \mathbb{R}, \psi, h)$.

In the definition of η, $(\mathbb{Z}/n\mathbb{Z})_S(1)$ denotes the Tate twist of the constant group scheme over S associated to the finite group $\mathbb{Z}/n\mathbb{Z}$. In order for ψ to define a Galois-equivariant map from $G'_n \times G'_n$ to $(\mathbb{Z}/n\mathbb{Z})_S(1)$, $\overline{\rho}_n$ must have cyclotomic similitude character with respect to the pairing induced by ψ.

For any $y \in A[n]$, $\sigma \in \mathrm{Gal}(L'/F')$, and $d \in \mathcal{O}$, we must have ${}^\sigma(\iota(d)y)) = {}^\sigma \iota(d)^\sigma y$. Because α_n is $\mathcal{O}$-equivariant, we can pull this back to the following linear algebraic relation on the $\mathcal{O}$-module $\Lambda/n\Lambda$:

$$\overline{\rho}_n(\sigma) d = \rho_D(\sigma)(d) \overline{\rho}_n(\sigma), \tag{16}$$

where we view $d \in \mathcal{O}$ as an element of $\mathrm{GL}(\Lambda/n\Lambda)$ via the $\mathcal{O}$-module structure of Λ. Equation (16) must hold in order for $\mathcal{M}_{F'}$ to be nontrivial (i.e. for there to be any quadruples satisfying the aforementioned conditions).

Remark. The definition of η is a generalization of [Lan13, Definition 1.3.6.2]. However, because the polarization we choose on Λ is principal, we may elide the symplectic-liftability condition. However, our notation differs slightly from [Lan13]. There, $(\mathbb{Z}/n\mathbb{Z}(1))_S$ denotes the group $\mathbb{Z}/n\mathbb{Z}$ viewed as a constant group scheme over S, with trivial Galois action. Here, we omit the inner (1), and $(\mathbb{Z}/n\mathbb{Z})_S(1)$ denotes the Tate twist of the constant group scheme $(\mathbb{Z}/n\mathbb{Z})_S$, with cyclotomic Galois action.

An isomorphism $(A, \lambda, \iota, \eta) \to (A', \lambda', \iota', \eta')$ is given by an isomorphism $f \colon A \xrightarrow{\sim} A'$ of abelian schemes such that

1. $\lambda = f^\vee \circ \lambda' \circ f$
2. For any $d \in \mathcal{O}$, the map

$$\begin{aligned} f_*\colon \operatorname{End}^0(A \times_S S_{L'}) &\longrightarrow \operatorname{End}^0(A' \times_S S_{L'}) \\ \iota(d) &\longmapsto f \circ \iota(d) \circ f^{-1} \end{aligned}$$

   is an isomorphism of $\mathrm{Gal}(L'/F')$-modules that sends $\iota(d)$ to $\iota'(d)$.
3. $f|_{A[n]} \circ \eta = \eta'$.

Theorem 5.1. *Fix $(D, \mathcal{O}, \dagger)$, a weight 1 polarized $\mathcal{O}$-Hodge structure (V, Λ, ψ), a Galois extension L'/F', and actions ρ_D and $\overline{\rho}$ of $\mathrm{Gal}(L'/F')$ on D and $\Lambda/n\Lambda$, satisfying the definitions and conditions described above. Let $\mathcal{M}_{F'}$ be the moduli functor described above. Assume that $n \geq 3$. Then, there is a smooth scheme Y'/F' representing $\mathcal{M}_{F'}$ such that Y' is a twist of $\mathrm{Sh}_U(G,X)_{F'}$ – concretely, $Y'_{L'}$ and $\mathrm{Sh}_U(G,X)_{L'}$ are isomorphic.*

Proof. Let $Y := \mathrm{Sh}_U(G,X)_{F'}$. Because $n \geq 3$, Y/F' is a smooth scheme representing the usual PEL moduli problem. We prove Theorem 5.1 in two steps. First, we define a variety Y'/F' via Galois descent from $Y_{L'}$, and then argue that Y' represents the moduli problem $\mathcal{M}_{F'}$. Throughout, we follow the conventions of [Poo17, 4.4]. We let $G := \mathrm{Gal}(L'/F')$ act on the left on L', which defines a right action on $\operatorname{Spec} L'$. The morphism on $\operatorname{Spec} L'$ associated to $\sigma \in G$ is denoted σ^*. Given an L'-scheme S, we write ${}^\sigma S$ for the pullback of S along σ^*.

By [Poo17, Proposition 4.4.2], giving a Galois descent datum is equivalent to providing isomorphisms $\{\tilde{\sigma}_Y\}_{\sigma \in G}$ such that the square

$$\begin{array}{ccc} Y_{L'} & \xrightarrow{\tilde{\sigma}_Y} & Y_{L'} \\ \downarrow & & \downarrow \\ \operatorname{Spec} L' & \xrightarrow{\sigma^*} & \operatorname{Spec} L' \end{array}$$

commutes for all $\sigma \in G$ and such that the "cocycle condition" $\widetilde{(\sigma\tau)}_Y = \tilde{\tau}_Y \circ \tilde{\sigma}_Y$ holds.

By the Yoneda lemma, we can define $\tilde{\sigma}_Y$ by specifying for every L'-scheme S a bijection of sets $\tilde{\sigma}_Y(S)\colon Y_{L'}({}^\sigma S) \to Y_{L'}(S)$ such that $\widetilde{(\sigma\tau)}_Y(S) = \tilde{\tau}_Y(S) \circ \tilde{\sigma}_Y({}^\tau S)$ and such that given a morphism $T \to S$ of L'-schemes, the following square commutes:

$$\begin{array}{ccccc} S & & Y({}^\sigma S) & \xrightarrow{\tilde{\sigma}_Y(S)} & Y(S) \\ \uparrow & & \downarrow & & \downarrow \\ T & & Y({}^\sigma T) & \xrightarrow{\tilde{\sigma}_Y(T)} & Y(T). \end{array} \tag{17}$$

The simplest descent datum on $Y_{L'}$ (which, as we will see later, gives us descent to Y itself) is given by

$$\begin{aligned} \tilde{\sigma}_Y(S)\colon \quad Y_{L'}({}^\sigma S) &\longrightarrow Y_{L'}(S) \\ ({}^\sigma A, {}^\sigma\lambda, {}^\sigma\iota, {}^\sigma\eta) &\longmapsto (A, \lambda, \iota, \eta), \end{aligned}$$

where ${}^\sigma\lambda$ is the pullback of λ, ${}^\sigma\iota$ sends $\iota(d) \in \mathrm{End}(A)$ to its pullback in $\mathrm{End}({}^\sigma A)$, and given $\eta\colon (\Lambda/n\Lambda)_S \to A[n]$, ${}^\sigma\eta\colon (\Lambda/n\Lambda)_{{}^\sigma S} \to {}^\sigma A[n]$ is the pullback of η. The maps $\{\tilde{\sigma}_Y(S)\}_{\sigma,S}$ satisfy the cocycle condition and make the square in Equation (17) commute, so they define a descent datum on Y'_L.

For our purposes, we want to work with a more general descent datum that captures our prescribed Galois action on D and on $A[n]$. Given representations ρ_D and $\overline{\rho}_n$, we can define the maps

$$\begin{aligned} h_\sigma(S)\colon \quad Y(S) &\longrightarrow Y(S) \\ (A,\lambda,\iota,\eta) &\longmapsto (A,\lambda,\iota\circ\rho_D(\sigma),(\alpha_n\circ\overline{\rho}_n(\sigma),\nu_n\circ\varepsilon(\sigma))). \end{aligned}$$

For any given $\sigma \in G$, the maps $\{h_\sigma(S)\}_S$ are compatible for any morphism $T \to S$ of L-schemes, and hence define a morphism h_σ of L-schemes. This map can be thought of as relabelling the endomorphism and level structure on each abelian scheme parameterized by $Y_{L'}$ according to $\rho_D(\sigma)$ and $\rho_n(\sigma)$.

Since ρ_D and ρ_n are homomorphisms, the $(h_\sigma)_\sigma$ satisfy the condition $h_{\sigma\tau} = h_\tau h_\sigma$. Because $\tilde{\tau}_Y(S)\circ h_\sigma({}^\tau S) = h_\sigma(S)\circ\tilde{\tau}_Y(S)$ (as precomposition commutes with pullback), the maps $\tilde{\sigma}'_Y := h_\sigma\circ\tilde{\sigma}_Y$ define a new Galois descent datum on $Y_{L'}$. We write $Y'/\operatorname{Spec} F'$ for the resulting descended variety over F'.

It remains to verify that Y' solves the moduli problem $\mathcal{M}_{F'}$. For this, we will show that for any scheme $S/\operatorname{Spec} F$,

$$Y'(S) = Y'(S_{L'})^G = \mathcal{M}_{L'}(S_{L'})^G = \mathcal{M}_{F'}(S) \tag{18}$$

for appropriately defined Galois actions on $Y'(S_{L'})$ and $\mathcal{M}_{L'}(S_{L'})$. On $Y'(S_{L'}) = \mathrm{Hom}_{L'}(S_{L'}, Y')$, we define the Galois action

$$h \longmapsto (\tilde{\sigma}'_Y)^{-1}\circ h\circ\tilde{\sigma}_S =: \sigma\cdot h.$$

On $\mathcal{M}_{L'}(S_{L'})$, we define the Galois action

$$(A,\lambda,\iota,\eta) \longmapsto \left(\tilde{\sigma}_S^*A,\ \tilde{\sigma}_S^*\lambda,\ \tilde{\sigma}_S^*(\iota\circ\rho_D(\sigma)^{-1}),\ \tilde{\sigma}_S^*(\eta\circ\overline{\rho}_n(\sigma)^{-1})\right)$$

The first equality of Equation (18) is essentially Galois descent for morphisms, and follows from full faithfulness of fpqc descent.

Next, we consider the second equality of Equation (18). Let $(A^\diamond,\lambda^\diamond,\iota^\diamond,\eta^\diamond)$ be the universal abelian scheme on $Y_{L'}$, and $h \in \mathrm{Hom}_{L'}(S_{L'}, Y_{L'})$ the morphism such that $h^*(A^\diamond,\lambda^\diamond,\iota^\diamond,\eta^\diamond) = (A,\lambda,\iota,\eta)$. We have

$$\begin{aligned} \sigma\cdot(A,\lambda,\iota,\eta) &= \left(\tilde{\sigma}_S^*A,\tilde{\sigma}_S^*\lambda,\tilde{\sigma}_S^*(\iota\circ\rho_D(\sigma)^{-1}),\tilde{\sigma}_S^*(\eta\circ\overline{\rho}_n(\sigma)^{-1})\right) \\ &= \left(\tilde{\sigma}_S^*h^*A^\diamond,\tilde{\sigma}_S^*h^*\lambda^\diamond,\tilde{\sigma}_S^*(\iota^\diamond\circ\rho_D(\sigma)^{-1}),\tilde{\sigma}_S^*(\eta^\diamond\circ\overline{\rho}_n(\sigma)^{-1})\right) \\ &= \left((\tilde{\sigma}'_Y\circ(\sigma\cdot h))^*A^\diamond,(\tilde{\sigma}'_Y\circ(\sigma\cdot h))^*\lambda^\diamond,(\tilde{\sigma}'_Y\circ(\sigma\cdot h))^*(\iota^\diamond\circ\rho_D(\sigma)^{-1}),(\tilde{\sigma}'_Y\circ(\sigma\cdot h))^*(\eta^\diamond\circ\overline{\rho}_n(\sigma)^{-1})\right) \\ &= (\sigma\cdot h)^*\left((\tilde{\sigma}'_Y)^*A^\diamond,(\tilde{\sigma}'_Y)^*\lambda^\diamond,(\tilde{\sigma}'_Y)^*(\iota^\diamond\circ\rho_D(\sigma)^{-1}),(\tilde{\sigma}'_Y)^*(\eta^\diamond\circ\overline{\rho}_n(\sigma)^{-1})\right) \\ &= (\sigma\cdot h)^*\tilde{\sigma}_Y^*\left(h_\sigma^*A^\diamond,h_\sigma^*\lambda^\diamond,h_\sigma^*(\iota^\diamond\circ\rho_D(\sigma)^{-1}),h_\sigma^*(\eta^\diamond\circ\overline{\rho}_n(\sigma)^{-1})\right). \end{aligned}$$

By construction, h_σ sends $(A^\diamond, \lambda^\diamond, \iota^\diamond, \eta^\diamond)$ to $(A^\diamond, \lambda^\diamond, \iota^\diamond \circ \rho_D(\sigma), \eta^\diamond \circ \overline{\rho}_n(\sigma))$. By the diagram

$$\begin{array}{ccc} h^*(A^\diamond, \lambda^\diamond, \iota^\diamond, \eta^\diamond) & \longrightarrow & (A^\diamond, \lambda^\diamond, \iota^\diamond, \eta^\diamond) \\ \downarrow{\scriptstyle h^*\pi} & & \downarrow{\scriptstyle \pi} \\ Y & \xrightarrow[h]{} & Y \end{array}$$

we deduce that $\left(h_\sigma^* A^\diamond, h_\sigma^* \lambda^\diamond, h_\sigma^*(\iota^\diamond \circ \rho_D(\sigma)^{-1}), h_\sigma^*(\eta^\diamond \circ \overline{\rho}_n(\sigma)^{-1})\right) \cong (A^\diamond, \lambda^\diamond, \iota^\diamond, \eta^\diamond)$. Because $(A^\diamond, \lambda^\diamond, \iota^\diamond, \eta^\diamond)$ is the base change to $Y_{L'}$ of the universal abelian scheme on Y, $\tilde{\sigma}_Y^*(A^\diamond, \lambda^\diamond, \iota^\diamond, \eta^\diamond) \cong (A^\diamond, \lambda^\diamond, \iota^\diamond, \eta^\diamond)$.

Lastly, we consider the third equality of Equation (18). Suppose that for some quadruple $(A, \lambda, \iota, \eta)$ over $S_{L'}$, and all $\sigma \in \mathrm{Gal}(L'/F')$, we have isomorphisms

$$\left(\tilde{\sigma}_S^* A, \tilde{\sigma}_S^* \lambda,\ \tilde{\sigma}_S^*(\iota \circ \rho_D(\sigma)^{-1}),\ \tilde{\sigma}_S^*(\eta \circ \overline{\rho}_n(\sigma)^{-1})\right) \cong (A, \lambda, \iota, \eta).$$

In particular, over $\operatorname{Spec} L$, there are isomorphisms $f_\sigma \colon {}^\sigma A \cong A$ for all $\sigma \in G$. A priori, the existence of the $\{f_\sigma\}$ does not immediately make them a descent datum on A/L' because these isomorphisms need not satisfy the cocycle condition $f_{\sigma\tau} = f_\sigma \circ {}^\sigma f_\tau$ – we only know that $f_{\sigma\tau}(f_\sigma \circ {}^\sigma f_\tau)^{-1}$ is an automorphism of $(A, \lambda, \iota, \eta)$. However, because $n \geq 3$, $(A, \lambda, \iota, \eta)$ has no nontrivial automorphisms, so the cocycle condition is indeed satisfied and A/L' descends to $\operatorname{Spec} F'$. Descending $S_{L'}$ to S using the descent datum $\{\tilde{\sigma}_S$, Galois descent for morphisms implies that $A/S_{L'}$ descends to S. Further applications of Galois descent for morphisms then show that λ descends, and that ι and λ descend to maps satisfying the conditions of $\mathcal{M}_{F'}$ (where the action of Galois is given by ρ_D and $\overline{\rho}_n$). Therefore, the third equality is proved. □

While we constructed our twisted moduli varieties by starting with PEL Shimura varieties over L' and descending them to F', our descent datum is perfectly well-defined integrally. If $Y_{L'}$ has an integral model $\mathcal{Y}/\mathcal{O}_{L'_w}$ for some place w of L', then the $\{\tilde{\sigma}_Y\}$ can be defined – in exactly the same way – as descent morphisms lying over $\operatorname{Spec} \mathcal{O}_{L'_w} \xrightarrow{\sigma^*} \operatorname{Spec} \mathcal{O}_{L'_w}$. For a place v of F lying below w, the map $\operatorname{Spec} \mathcal{O}_{L'_w} \to \operatorname{Spec} \mathcal{O}_{F'_v}$ is fpqc whenever L'_w/F'_v is unramified [Poo17, Proposition 3.8.2]. Because $\mathcal{Y}/\mathcal{O}_{L'_w}$ is affine (and in particular quasiaffine), descent on $\mathcal{Y}$ to $\mathcal{O}_{F'_v}$ is effective by [Poo17, Theorem 4.3.5].

We will ultimately want to apply the Moret–Bailly theorem to a connected component of a twisted PEL Shimura variety produced by Theorem 5.1. In order to apply Theorem 2.4, we need totally real points on the connected components, so we certainly require the connected components to be defined over a totally real field. However, in the case of (untwisted) PEL Shimura varieties, the field of definition of the connected components need not be totally real even if the reflex field is.

Example 5.2. The connected components of the modular curve $Y(n)$ can only be defined over $\mathbb{Q}(\zeta_n)$. To understand this, consider the square Equation (15) with $G'_n := (\Lambda/n\Lambda)_S$ and $(\mathbb{Z}/n\mathbb{Z})_S$ instead of $(\mathbb{Z}/n\mathbb{Z})_S(1)$ – this is the usual definition of an $\mathcal{O}$-equivariant symplectic isomorphism from [Lan13, Definition 1.3.6.1]. Then, applying Galois conjugation by $\sigma \in \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ changes ν_n. Because ν_n takes one of $(\mathbb{Z}/n\mathbb{Z})^\times$ discrete values, conjugation by σ moves points between components. This reflects the fact that ν_n is a map from a group scheme with trivial Galois action to one with cyclotomic Galois action. By replacing $(\mathbb{Z}/n\mathbb{Z})_S$ by $(\mathbb{Z}/n\mathbb{Z})_S(1)$, we remove this issue, and conjugation by σ no longer changes ν_n. Therefore, we are able to descend each individual component of the twisted modular curve – with $\overline{\rho}_n$ of cyclotomic determinant – to the reflex field $\mathbb{Q}$.

When D is a totally real field or a quaternion algebra over a totally real field, the reflex field is $\mathbb{Q}$, and [Zin83, 3.1] (see also [Mil05, Proposition 8.14]) implies that there exists a unique conjugacy class of homomorphisms X defining a PEL datum. In particular, all of the geometrically connected components of X are associated to the level structure. As in the example, the fact that $\overline{\rho}_n$ has cyclotomic similitude allows us to descend the geometrically connected components of $\mathrm{Sh}_K(G,X)$ all the way to the reflex field, and we obtain the following corollary.

Corollary 5.3. *If D is either a totally real field or a quaternion algebra over a totally real field, then every geometrically connected component of Y' is defined over $\mathbb{Q}$.*

As noted before, we can also obtain integral models for the geometrically connected components at all but finitely many places. This will be used implicitly when we apply [MP12, Proposition 1.13] to varieties produced by Corollary 5.3 later on, as their results apply to moduli schemes which are defined over the ring of integers of a local field.

In these paper, we use these twisted moduli schemes as an input to the Moret–Bailly theorem in Section 6 and **??**. However, we expect them to be of independent interest, as the following theorem demonstrates.

Theorem 5.4.

6 Applications to abelian fourfolds

In this section, we apply Theorem 1.3 and Theorem 5.1 to prove Main Theorem 1.

We first address the most interesting cases: geometric types I(2) and II(1). For ease of exposition, we first give the proof in the case of geometrically type II(1) fourfolds, and afterwards describe the modifications needed to treat geometrically type I(2) fourfolds.

6.1 Potential automorphy of geometrically type II(1) abelian fourfolds

Let A/F be a type II(1) abelian fourfold with $D := \mathrm{End}^0_{\overline{F}}(A)$, and let $\mathcal{V}(A) = (F; \mathbb{Q}, S, \{Q_{\mathcal{V},v}(X)\}_v, \{\rho_{A,\ell}\}_\ell)$ be the compatible system of 8-dimensional representations associated to A. By Theorem 1.3, $E_0 = E = \mathbb{Q}$ and $\mathcal{V}(A) \cong \mathrm{Des}_{\mathbb{Q}}(\mathcal{R}_A \otimes \mathcal{A}_A)$, where $\mathcal{R}_A = (F; E', S, \{Q_{\mathcal{R},v}(X)\}_v, \{r_\lambda\}_\lambda)$ is a compatible system of four-dimensional symplectic representations and $\mathcal{A}_A = (F; E', S, \{Q_{\mathcal{A},v}(X)\}_v, \{a_\lambda\}_\lambda)$ is a compatible system of two-dimensional Artin representations, both (after possibly enlarging coefficients) with coefficients in some $E'/\mathbb{Q}$. Let S_0 be the set of rational primes divisible by primes in S. For each rational prime ℓ, we pick a prime $\lambda|\ell$ in E' and define $\xi_\ell := r_\lambda$ and $\varrho_\ell := a_\lambda$, so that $\rho_{A,\ell} \otimes \overline{\mathbb{Q}_\ell} \cong \xi_\ell \otimes \varrho_\ell$ for all $\ell \notin S_0$ – because $\mathcal{R}_A \otimes \mathcal{A}_A$ descends to $\mathbb{Q}$, the specific choice does not matter. As before, we write M/F for the extension through which $\mathcal{A}_A$ factors. In this subsection, we prove the following theorem.

Theorem 1.5. *Let F be a totally real field and A/F be an odd type abelian fourfold with $\mathrm{End}^0_{\overline{F}}(A) = D$, where D is an indefinite quaternion algebra with center $\mathbb{Q}$. Then, there exists a compatible system $\mathcal{R}$ of four-dimensional strongly irreducible symplectic (odd) representations and a compatible system $\mathcal{A}$ of two-dimensional Artin representations with cyclic or dihedral projective image such that*

1. *$\mathcal{R} \otimes \mathcal{A}$ descends to $\mathbb{Q}$;*
2. *$\mathcal{V}(A) \cong \mathrm{Des}_{\mathbb{Q}}(\mathcal{R} \otimes \mathcal{A})$; and*
3. *A is potentially automorphic, and is automorphic whenever any r_λ in $\mathcal{R}$ is automorphic.*

By Theorem 1.3 and Corollary 3.2, it suffices to prove that $\mathcal{R}$ is potentially automorphic.

Definition 6.1 (Following [BCGP21, Definition 9.2.3])**.** Let p be a rational prime and v a prime of F dividing p. We say that v is good for A/F if

- $(\xi_p)\big|_{G_{F_v}}$ is p-distinguished weight 2 ordinary, as defined in [BCGP21, Definition 7.3.1];
- v splits completely in the extension M/F, where M is the extension through which the Artin compatible system $\mathcal{A}$ factors; and
- The image of $\overline{\xi}_p$ contains $\mathrm{Sp}_4(k)$ for some finite extension $k/\mathbb{F}_p$.

We say that a rational prime p is good for A/F if:

- For all primes $v|p$ in F, v is good;
- p splits completely in $F/\mathbb{Q}$;
- $p \nmid [M : F]$;
- p is split in D; and
- p does not divide the conductor of A.

Lemma 6.2. *Let A be a geometrically type II(1) abelian fourfold. The set of good rational primes for A/F has relative density 1 in the set of rational primes which split completely in $M/\mathbb{Q}$, and in particular, have positive relative density in the set of all rational primes.*

Proof. It suffices to show that a density 1 subset of the primes (ordered by norm) of M are good. We can restrict to the density one subset of primes of M above rational primes which split completely in $M/\mathbb{Q}$. By Theorem 1.3, $\mathrm{Res}_M \mathcal{V} \cong \mathrm{Des}_{\mathbb{Q}}((\mathrm{Res}_M \mathcal{R})^{\oplus 2})$. Let $A' := A_M$ and $\mathcal{V}' := \mathrm{Res}_M \mathcal{V}$. For any finite place v of M, $Q_{\mathcal{V}',v}(X) = Q^2_{\mathrm{Res}_M \mathcal{R},v}$. In particular, because $Q_{\mathcal{V}',v}(X) \in \mathbb{Z}[X]$ and both $Q_{\mathcal{V}',v}(X)$ and $Q_{\mathrm{Res}_M \mathcal{R},v}(X)$ are monic, $Q_{\mathrm{Res}_M \mathcal{R},v}(X) \in \mathbb{Z}[X]$, and we may define $\mathcal{R}' := \mathrm{Des}_{\mathbb{Q}}(\mathrm{Res}_M \mathcal{R})$. Let $\{r'_\ell\}_\ell$ be the ℓ-adic representations associated to $\mathcal{R}'$.

The remainder of the proof is largely the same as the proof of [BCGP21, Lemma 9.2.5]. However, this is the step which we must modify in the case of geometrically type I(2) fourfolds later, so we spell out some of the details. Because $\rho^\vee_{A'} \cong \rho_{A'} \otimes \epsilon^{-1}$, the roots of $Q_{\mathcal{R}',v}(X)$ have the form $\{\alpha_v, \beta_v, p\alpha_v^{-1}, p\beta_v^{-1}\}$, and we can write $Q_{\mathcal{R}',v}(X) := x^4 - a_1(\mathrm{Frob}_v)x^3 + a_2(\mathrm{Frob}_v)x^2 - pa_1(\mathrm{Frob}_v)x + p^2$. The representation $(r'_p)|_{G_{F_v}}$ is (weight 2) ordinary (at v) if and only if $v_p(a_2) = 0$ (equivalent to $v_p(\alpha_v) = v_p(\beta_v) = 0$). The representation $(r'_p)|_{G_{F_v}}$ is p-distinguished if and only if $v_p(a_1(\mathrm{Frob}_v)^2 - 4a_2(\mathrm{Frob}_v)) = 0$ (equivalent to $v_p(\alpha_v - \beta_v) = 0$).

These criteria let us translate p-distinguished weight 2 ordinarity – a condition on $r'_p(\mathrm{Frob}_v)$ – into a condition on $Q_{\mathcal{R}',v}(X)$, independent of the representation r'_p. By the Weil bounds, for any embedding $\jmath\colon \overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$, $|\jmath(\alpha_v)| = |\jmath(\beta_v)| = \sqrt{p}$. Therefore, $|a_1(\mathrm{Frob}_v)| \leq 4\sqrt{p}$ and $|a_2(\mathrm{Frob}_v) - 2p| \leq 4p$. In particular, v fails to be ordinary if and only if $a_2(\mathrm{Frob}_v) = cp$ for some $c \in \mathbb{Z} \cap [-6, 6]$, and fails to be p-distinguished if and only if $a_1(\mathrm{Frob}_v)^2 = 4a_2(\mathrm{Frob}_v) + c'p$ for some $c' \in \mathbb{Z} \cap [-40, 40]$. Now pick an auxiliary prime ℓ split in D. Let G_ℓ be the ℓ-adic algebraic monodromy group of A' – the Zariski closure (over $\mathbb{Q}_\ell$) of the image of $\rho_{A'}$ in $\mathrm{GSp}_{2g}(\mathbb{Q}_\ell)$. The Mumford–Tate group G of a geometrically type II(1) abelian fourfold is a twisted form G of $\mathrm{GSp}_{4,\mathbb{Q}}$ which splits over $\mathbb{Q}_\ell$ whenever ℓ is unramified in D [KR99]. The Mumford–Tate conjecture is known for abelian fourfolds with geometric quaternionic multiplication [MZ95], so $G_\ell \cong G \times_{\mathbb{Q}} \mathbb{Q}_\ell$. In particular, we deduce that $r'_\ell\colon G_F \to G_\ell \cong \mathrm{GSp}_4(\overline{\mathbb{Q}_\ell})$ has Zariski dense image. Consider $\rho := \rho_\ell := (r'^{\otimes 2} \oplus \wedge^2 r') \otimes \varepsilon^{-1}$. This

representation factors through $\rho\colon G_M \to G_\ell \to \mathrm{GL}_{22}(\mathbb{Q}_\ell)$. For $g \in \mathrm{GL}_{22}(\mathbb{Q}_\ell)$, let $\mathrm{tr}_i(g)$, $i \in \{1,2\}$, be the trace of the ith block in ρ_ℓ, Because $a_1^2(\mathrm{Frob}_v) = \mathrm{tr}(r'^{\otimes 2})$ and $a_2(\mathrm{Frob}_v) = \mathrm{tr}(\wedge^2 r')$, v fails to be ordinary or fails to be p-distinguished if and only if $\rho(\mathrm{Frob}_v)$ lands in the union Y of the hypersurfaces cut out by the equations

$$\{\mathrm{tr}_2(g) = c\colon c \in \mathbb{Z} \cap [-6,6]\} \qquad \text{and} \qquad \{\mathrm{tr}_1(g) - 4\,\mathrm{tr}_2(g) = c'\colon c \in \mathbb{Z} \cap [-40,40]\}. \tag{19}$$

Because Y does not contain $\operatorname{im} G_\ell$ (for each of the equations in Equation (19), there is a matrix in $\mathrm{GSp}_4(\mathbb{Q}_\ell)$ whose image in $\mathrm{GL}_{22}(\mathbb{Q}_\ell)$ does not satisfy it) the preimage Z of Y under the aforementioned map $G_\ell \to \mathrm{GL}_{22}(\mathbb{Q}_\ell)$ is a subvariety of positive codimension. The Chebotarev density theorem applied to $\rho\colon G_F \to G_\ell$ then implies that Frob_v avoids Z (and hence that v is ordinary and p-distinguished) for a density 1 subset of primes of M. Lastly, because $\mathcal{R}'$ satisfies the Mumford–Tate conjecture, [Lar95, Theorem 3.17] implies that $\overline{r'_p} \otimes k$ has image containing $\mathrm{Sp}_4(k)$ (for some finite extension $k/\mathbb{F}_p$) for a density 1 subset of primes p. □

Corollary 6.3. *Let A be a geometrically type II(1) abelian fourfold. There exist good rational primes p and q such that for all $v|p$, $\overline{\xi_{A,q}}(\mathrm{Frob}_v)$ has distinct eigenvalues.*

Proof. This is [BCGP21, Lemma 9.2.6]. □

Theorem 6.4. *Let A/F be an abelian variety over a totally real field and p and q a pair of good primes for A given by Corollary 6.3. There exist a totally real field extension $\tilde{F}/F$ and a totally real quadratic extension F_1/F such that*

- *p and q split completely in $\tilde{F}/\mathbb{Q}$ and in $F_1/\mathbb{Q}$; and*
- *$\tilde{F}/F$ and F_1/F are linearly disjoint from each other, from L/F, and from the kernels of the actions of G_F on $A[p]$ and $A[q]$.*

and a Galois representation $\overline{\rho}_q^{\text{mm}}\colon G_{\tilde{F}} \to \mathrm{GSp}_4(\mathbb{F}_q)$ with cyclotomic similitude such that

- *$\overline{\rho}_q^{\text{mm}}$ is automorphic;*
- *For any complex conjugation in $G_{\tilde{F}}$, $\overline{\rho}_q^{\text{mm}}(c) \sim \mathrm{diag}(1,1,-1,-1)$;*
- *$\overline{\rho}_q^{\text{mm}}|_{G_{\tilde{F}_v}} \cong \overline{\xi_q}|_{G_{F_v}}$ for all places $v|p$;*
- *$\overline{\rho}_q^{\text{mm}}|_{G_{\tilde{F}_w}} \cong \left(\overline{\xi_q}|_{G_{\tilde{F}_w}}\right)^{\mathrm{ss}}$ for all places $w|q$; and*
- *$\overline{\rho}_q^{\text{mm}}$ is vast and tidy in the sense of [BCGP21, Definitions 7.5.6 and 7.5.11]*

In particular,

$$\left(\overline{\rho}_q^{\text{mm}}|_{G_{\tilde{F}_v}}\right)^{\oplus 2} \cong A[q]|_{G_{\tilde{F}_v}} \qquad \text{and} \qquad \left(\overline{\rho}_q^{\text{mm}}|_{G_{\tilde{F}_w}}\right)^{\oplus 2} \cong \left(A[q]|_{G_{\tilde{F}_w}}\right)^{\mathrm{ss}}.$$

Proof. This follows from the proof of [BCGP21, Lemma 9.2.7] with their $\overline{\rho}_{A,q}$ (for them A is an abelian surface) replaced with our $\overline{\xi_q}$. Note that because q is good, $(\xi_q)|_{G_{F_w}}$ is q-distinguished weight 2 ordinary. The last statement follows because p and q split completely in M/F and hence the Artin factor in $\overline{\rho}_{A,q} \cong \overline{\xi}_q \otimes \overline{\varrho}_q$ becomes trivial upon restriction to $G_{\tilde{F}_v}$ or $G_{\tilde{F}_w}$. □

Lemma 6.5. *Let p and q be good rational primes. Let w be a prime of F above q. There exists an abelian variety A^{can}/F_w such that for all primes w' of L above w,*

1. $\mathrm{End}^0_{F_w}(A) \hookrightarrow \mathrm{End}^0_{F_w}(A^{\mathrm{can}})$ *as* $\mathrm{Gal}(L_{w'}/F_w)$*-modules.*

2. $A^{\mathrm{can}}[p] \cong \left(A[p]\big|_{G_{F_w}}\right)$

3. $A^{\mathrm{can}}[q] \cong \left(A[q]\big|_{G_{F_w}}\right)^{\mathrm{ss}}$.

Note that because L/F is Galois, $L_{w'}/F_w$ is Galois and independent of w'.

Proof. We take A^{can}/F_w to be the Serre–Tate canonical lift of the special fibre of A_{F_w}. As in the proof of Lemma 6.2, let $\{\alpha_w, \beta_w, q\alpha_w^{-1}, q\beta_w^{-1}\}$ be the eigenvalues (each with multiplicity 2) of $\rho_{A,q}(\mathrm{Frob}_w)$. The defining property of the Serre–Tate canonical lift is that

$$A^{\mathrm{can}}[q] \cong \left(A[q]\big|_{G_{F_w}}\right)^{\mathrm{ss}} \sim \begin{pmatrix} \lambda_{\overline{\alpha}_w} & & & \\ & \lambda_{\overline{\beta}_w} & & \\ & & \varepsilon\lambda_{\overline{\alpha}_w}^{-1} & \\ & & & \varepsilon\lambda_{\overline{\beta}_w}^{-1} \end{pmatrix}^{\oplus 2}.$$

The third property follows immediately.

Because A^{can}/F_w has good reduction and residue field characteristic q, its ℓ-adic Galois representation is unramified for any $\ell \neq q$, and the eigenvalues of Frob_w do not depend on ℓ. Furthermore, they are the same as the Frobenius eigenvalues of A_{F_w}. Taking $\ell = p$, we therefore have

$$A^{\mathrm{can}}[p] \cong \left(A[p]\big|_{G_{F_w}}\right) \sim \begin{pmatrix} \lambda_{\overline{\alpha}_w} & & & \\ & \lambda_{\overline{\beta}_w} & & \\ & & \varepsilon\lambda_{\overline{\alpha}_w}^{-1} & \\ & & & \varepsilon\lambda_{\overline{\beta}_w}^{-1} \end{pmatrix}^{\oplus 2}.$$

The first property follows from [Mes72, Corollary 3.4] and functoriality of the canonical lift. □

We are now ready to prove the third part of Theorem 1.5. We follow the approach (and where possible, the notation) of [BCGP21, Section 9.2]. Let p and q be a pair of good rational primes A/F as given by Corollary 6.3. As usual, we have $\rho_{A,p} \cong \xi_{A,p} \otimes \varrho_{A,p}$ (and similarly for q). We want to apply Theorem 2.4, first to a q-adic representation whose residual representation is $\overline{\rho}_q^{\mathfrak{m}}$ (which is related to $\overline{\xi_{A,q}}$), and then to $\xi_{A,p}$.

Suppose that the similitude character of $\xi_{A,p}$ is $\varepsilon_p\nu_p$ (and that of ϱ_p is ν_p^{-1}) for some finite-order character ν_p (similarly, ν_q), . Because $\mathcal{R}_A$ and $\mathcal{A}_A$ are compatible systems, the kernel of ν_p is the same as that of the kernel of the similitude character of ν_q. Because the GSp_4-automorphy lifting theorem (Theorem 2.4) can only be applied to representations with cyclotomic similitude character, we replace F with the extension fixed by $\ker\nu_p = \ker\nu_q$ – because A is odd type, ν_p and ν_q are even, so this extension is still totally real.

For ease of notation, for the remainder of this proof we replace F with the extension $\tilde{F}/F$ appearing in Theorem 6.4, A with its base change to $\tilde{F}$, L with $L\tilde{F}$, and the extension F_1/F in Theorem 6.4 with $F_1\tilde{F}/\tilde{F}$. Note that $\mathrm{Gal}(L\tilde{F}/\tilde{F}) = \mathrm{Gal}(L/F)$.

The main difference between this proof and the approach of [BCGP21, Section 9.2] is that we replace the moduli of abelian surfaces with a twisted moduli space of geometrically type II(1) abelian fourfolds capturing the desired Galois action on their endomorphism algebras.

Let Y'_A/F be a geometrically connected component of the scheme given by applying Theorem 5.1 with $F' = F$, L' the compositum of L and the splitting fields of $A[p]$, $A[q]$, and $\overline{\rho}_q^{\mathfrak{m}}$, $\rho_D\colon \mathrm{Gal}(L/F) \to$

$\mathrm{Aut}(\mathrm{End}_L^0(A)) \cong \mathrm{Aut}(D)$ the map given by the Galois action on the geometric endomorphisms of A, $\overline{\rho}_p := A[p]$, and $\overline{\rho}_q := \overline{\rho}_q^{\mathrm{mm}} \otimes \overline{\varrho_q}$ (with $\overline{\rho}_q^{\mathrm{mm}}$ as defined in Theorem 6.4). By Lemma 3.8, ϱ_q is a lift of the projective representation $\mathrm{Gal}(L/F) \to D^\times/\mathbb{Q}^\times \hookrightarrow \mathrm{PGL}_2(\overline{\mathbb{Q}_q})$ realizing the Galois conjugation action on $D \otimes_{\mathbb{Q}} \overline{\mathbb{Q}_q}$ as inner conjugation by a matrix in $\mathrm{PGL}_2(\overline{\mathbb{Q}_q})$. Because $\mathcal{O}$ is Galois stable, reducing everything mod n shows that Equation (16) is satisfied.

We want to apply the Moret–Bailly theorem (see e.g. [BLGGT14, Proposition 3.1.1]) to Y'_A to find a totally real field F''/F in which p and q split completely and such that Y'_A has a point over F'' with good distinguished ordinary reduction at p and q. Additionally, we want F''/F to be linearly disjoint from F_1/F, M/F, and the kernels of the G_F-actions on $A[p]$, $A[q]$, and $\overline{\rho}_q^{\mathrm{mm}}$. By [MP12, Proposition 1.13] (this is discussed in greater detail in the proof of Corollary 7.4), for every $v|p$, there exists an open subset $\Omega_{D,v} \subset Y'_A(F_v)$ containing A which is disjoint from the union of the extra geometric endomorphism (beyond D) locus of $Y'_A(F_v)$. To impose the local conditions upon F''/F, we require $\Omega'_{A,u} := Y'_A(F_u)$, $\Omega'_{A,v} := Y'_A(F_v)^{\mathrm{ord}} \cap \Omega_{D,v}$, and $\Omega'_{A,w} := Y'_A(F_w)^{\mathrm{ord}}$ to be nonempty for every real place $u|\infty$, every $v|p$, and every $w|q$. We check these in turn.

Because A is defined over a totally real field, $\rho_D(c_u)$ by definition is exactly the action of complex conjugation on $A \times_{F,u} \mathbb{C}$, $\overline{\rho}_p(c_u) = A[p](c_u)$ (because $\overline{\rho}_p = A[p]$), and $\overline{\rho}_q(c_u) \cong A[q](c_u)$ (by the second property of $\overline{\rho}_q^{\mathrm{mm}}$ in Theorem 6.4), A_{F_u} defines a point in $\Omega'_{A,u}$ for every $u|\infty$. Similarly, because A is defined over a field where p splits completely, has geometric endomorphisms exactly by D, and $\overline{\rho}_q|_{G_{F_v}} \cong A[q]|_{G_{F_v}}$ (by the third property of $\overline{\rho}_q^{\mathrm{mm}}$ in Theorem 6.4), A_{F_v} defines a point in $\Omega'_{A,v}$ for every $v|p$. Lastly, because A is defined over a field where q splits completely, for any $w|q$ the abelian variety A^{can}/F_w provided by Lemma 6.5 defines a point in $\Omega'_{A,w}$ for every $w|q$.

Therefore, by the Moret–Bailly theorem, Y'_A has a point over F'' for some finite extension F''/F satisfying the aforementioned conditions. Such a point corresponds, by Theorem 5.1, to an abelian fourfold B/F'' with ordinary good reduction at p and q such that B has geometric endomorphism algebra exactly D,

$$B[p] \cong A[p]\big|_{G_{F''}} \qquad \text{and} \qquad B[q] \cong \left(\overline{\rho}_q^{\mathrm{mm}} \otimes \overline{\varrho_{A,q}}\right)\big|_{G_{F''}}.$$

Let $\mathcal{R}_B$ and $\mathcal{A}_B$ be the compatible systems of Galois representations associated by Theorem 1.3 to B, and let for each rational prime ℓ let $\{\xi_{B,\ell}\}_\ell$ and $\{\varrho_{B,\ell}\}_\ell$ be the ℓ-adic representations given by Corollary 1.4. Because the ρ_D used in the construction of Y' captures the $\mathrm{Gal}(L/K)$-action on $\mathrm{End}_{\overline{F}}^0(A)$, Lemma 3.8 implies that the Artin representation $\varrho_{B,\ell}$ has the same projectivization as $\varrho_{A,\ell}|_{G_{F''}}$ for all ℓ (recall that F''/F was chosen to be linearly disjoint from M/F). Up to replacing $\mathcal{R}_B$ and $\mathcal{A}_B$ by twists, we may therefore assume that $\mathcal{A}_B \cong \mathrm{Res}_{F''} \mathcal{A}_B$. Henceforth, for any rational prime ℓ, we write $\varrho'_\ell := \varrho_{A,\ell}\big|_{G_{F''}} \cong \varrho_{B,\ell}$. We have

$$\overline{\xi_{B,q}} \otimes \overline{\varrho'_q} \;\cong \overline{\rho_{B,q}} \;\cong \overline{\rho}_q^{\mathrm{mm}}\big|_{G_{F''}} \otimes \overline{\varrho'_q}.$$

Because F''/F is linearly disjoint from the kernel of $\overline{\rho}_q^{\mathrm{mm}}$, $\overline{\rho}_q^{\mathrm{mm}}\big|_{G_{F''}}$ is irreducible, We want to use Frobenius reciprocity to argue that $\overline{\xi_{B,q}}$ is automorphic, but need to be somewhat careful because we are working in characteristic q. Nonetheless,

$$\begin{aligned}\mathrm{Hom}_{\overline{\mathbb{F}}_q[G_{MF''}]}\left(\overline{\xi_{B,q}}, \overline{\rho}_q^{\mathrm{mm}}\big|_{G_{F''}}\right) &\cong \mathrm{Hom}_{\overline{\mathbb{F}}_q[G_{F''}]}\left(\overline{\xi_{B,q}} \otimes_{\overline{\mathbb{F}}_q[G_M]} \overline{\mathbb{F}}_q[G_F], \overline{\rho}_q^{\mathrm{mm}}\big|_{G_{F''}}\right) \\ &\cong \mathrm{Hom}_{\overline{\mathbb{F}}_p[G_F]}\left(\overline{\xi_{B,q}} \otimes_{\overline{\mathbb{F}}_q} \overline{\mathbb{F}}_q[\mathrm{Gal}(M/F)], \overline{\rho}_q^{\mathrm{mm}}\big|_{G_{F''}}\right).\end{aligned}$$

Because q is good for A, $q \nmid [M:F]$, so $\overline{\mathbb{F}}_q[\mathrm{Gal}(M/F)]$ is a direct sum of irreducible representations. Schur's lemma implies that $\overline{\xi_{B,q}}$ is also irreducible and is isomorphic to a twist of $\overline{\rho}_q^{\mathrm{mm}}$ by a character

of $\mathrm{Gal}(MF''/F'') \cong \mathrm{Gal}(M/F)$. In particular, $\overline{\xi_{B,q}}$ is automorphic. Because ϱ_q has trivial similitude, $\overline{\xi_{B,q}}$ has cyclotomic similitude. Additionally, because q is good for A and $\overline{\rho}_q^{\mathfrak{m}}\big|_{G_{F''_w}} \cong \left(\overline{\xi_{A,q}}\big|_{G_{F''_w}}\right)^{\mathrm{ss}}$, $\xi_{B,q}$ is not only ordinary at w (which we already know from $B \in \Omega'_{A,w}$) but q-distinguished ordinary. With the rest of Theorem 6.4 in hand, Theorem 2.4 implies that $\xi_{B,q}\big|_{G_{F''}}$, and hence $\mathrm{Res}_{F''}\mathcal{R}_B$, are automorphic. From this, we deduce the automorphy of $\xi_{B,p}\big|_{G_{F''}}$.

Let $\xi'_{A,p} := \xi_{A,p}\big|_{G_{F''}}$. We have

$$\overline{\xi_{B,p}} \otimes \overline{\varrho_p} \cong B[p] \cong A[p] \cong \overline{\xi'_{A,p}} \otimes \overline{\varrho_p}.$$

By the same argument as above, $\overline{\xi'_{A,p}}$ is automorphic. Again applying Theorem 2.4, we deduce that $\xi'_{A,p}$ is automorphic and hence that $\mathrm{Res}_{F''}\mathcal{R}_A$ is automorphic, as desired.

Remark. Gu [Gu25] proves the potential automorphy of some four-dimensional compatible systems of symplectic type associated to abelian varieties. In particular, [Gu25, Theorem 6.11] implies the potential automorphy of type II(1) abelian fourfolds A/F where F a totally real field – abelian fourfolds for which $\mathrm{End}^0_F(A) = \mathrm{End}^0_{\overline{F}}(A)$ is an indefinite quaternion algebra with center $\mathbb{Q}$. In general, the minimal field over which the geometric endomorphisms of A are defined need not be totally real, and in these cases Theorem 1.5 does not follow from his theorem. However, his methods, like ours, follow the approach of [BCGP21, Section 9.2]. Because he is interested in abelian varieties with quaternionic multiplication defined over a general totally real field, he is able to work with real points on the moduli of such abelian varieties directly without worrying about the issue of Galois actions on the endomorphism algebra.

6.2 Potential modularity of geometrically type I(2) abelian fourfolds

Let A be a geometrically type I(2) abelian fourfold. Its geometric endomorphism algebra is a real quadratic field, which we denote by $E/\mathbb{Q}$. By Theorem 1.3, there exists a compatible system $\mathcal{R}$ of four-dimensional strongly irreducible symplectic representations with coefficients in E such that either

$$\mathcal{V}(A) \cong \mathrm{CRes}_{E/\mathbb{Q}}\mathcal{R} \qquad \text{or} \qquad \mathcal{V}(A) \cong \mathrm{Des}_{\mathbb{Q}}(\mathrm{Ind}_K^F\mathcal{R}). \tag{20}$$

A key distinction from the geometric type II(1) case is that, by Lemma 3.4, $\mathcal{R}$ has cyclotomic similitude, and is in particular odd. Suppose that F is totally real. We call A/F challenging if it is in the second case of Equation (20) and K/F is not totally real.

Theorem 6.6. *Let F be a totally real field and A/F be an abelian fourfold with $\mathrm{End}^0_{\overline{\mathbb{Q}}}(A) = E$, where E is a real quadratic field. Then, there exists a compatible system $\mathcal{R}$ of four-dimensional strongly irreducible symplectic (odd) representations with coefficients in E such that*

1. *$\mathcal{V}(A) \cong \mathrm{CRes}_{E/\mathbb{Q}}\mathcal{R}$ or $\mathcal{V}(A) \cong \mathrm{Des}_{\mathbb{Q}}(\mathrm{Ind}_K^F\mathcal{R})$;*
2. *A is potentially modular unless we are either in the second case above and K/F is not totally real, or $E \subset K^{\mathrm{gal}}$; and*
3. *A is modular whenever any r_λ in $\mathcal{R}$ is modular.*

The proof of Theorem 6.6 is largely the same as the approach used in Section 6.1 to prove Theorem 1.5, but with a few deviations.

- Because the compatible system $\mathcal{R}$ has coefficients in a real quadratic field E rather than in $\mathbb{Q}$, we need a slightly different argument (Lemma 6.7, below) to prove that there are enough good primes.
- Because there is no Artin compatible system $\mathcal{A}$, $\mathcal{R}$ is guaranteed to be odd, so we do not need an "odd type" condition in Theorem 6.6.
- If A is challenging, then because $\mathcal{R}$ is not over a totally real field, we cannot apply Theorem 2.4. When A is not challenging, then by replacing A with A_K we may assume that $\mathcal{V}(A) \cong \mathrm{CRes}_{E/\mathbb{Q}} \mathcal{R}$ for the purposes of proving a potential modularity theorem.
- For a type I(2) abelian fourfold, we have $\rho_{A,\ell} \otimes \overline{\mathbb{Q}_\ell} \cong \xi_{1,\ell} \oplus \xi_{2,\ell}$, where the Frobenius characteristic polynomials of $\xi_{1,\ell}$ and $\xi_{2,\ell}$ in $E[X]$ are Galois conjugate to one another. Instead of having a single $\overline{\rho}_q^{\mathfrak{m}}$, we use the proof of [BCGP21, Theorem 9.2.7] to build $\overline{\rho}_{1,q}^{\mathfrak{m}}$ and $\overline{\rho}_{2,q}^{\mathfrak{m}}$, associated to $\xi_{1,q}$ and $\xi_{2,q}$ respectively, such that

$$(\overline{\rho}_{1,q}^{\mathfrak{m}} \oplus \overline{\rho}_{2,q}^{\mathfrak{m}})\big|_{G_{\tilde{F}_v}} \cong (\overline{\xi_{1,q}} \oplus \overline{\xi_{2,q}})\big|_{G_{\tilde{F}_v}} \cong A[q]\big|_{G_{\tilde{F}_v}}$$

  and

$$(\overline{\rho}_{1,q}^{\mathfrak{m}} \oplus \overline{\rho}_{2,q}^{\mathfrak{m}})\big|_{G_{\tilde{F}_w}} \cong \left((\overline{\xi_{1,q}} \oplus \overline{\xi_{2,q}})\big|_{G_{\tilde{F}_w}}\right)^{\mathrm{ss}} \cong \left(A[q]\big|_{G_{\tilde{F}_w}}\right)^{\mathrm{ss}}.$$

  Because good primes are inert in $E/\mathbb{Q}$, the representations $\xi_{1,\ell}$ and $\xi_{2,\ell}$ are coefficient-wise Galois conjugate, and we must take $\overline{\rho}_{1,q}^{\mathfrak{m}}$ and $\overline{\rho}_{2,q}^{\mathfrak{m}}$ to be coefficient-wise Galois conjugate as well to ensure the appropriate E-action everywhere.
- We can set up our moduli problem to capture abelian fourfolds B for which $B[p] \cong A[p]$ and $B[q] \cong \overline{\rho}_{1,q}^{\mathfrak{m}} \oplus \overline{\rho}_{2,q}^{\mathfrak{m}}$. Previously, in order for A to define a totally real point on our moduli space Y_A', we needed to prescribe a nontrivial Galois module structure on the geometric endomorphisms of abelian varieties parameterized by Y_A'. Here, because A already has RM by E, we do not need to define a twisted Galois action on the endomorphism algebra when defining Y_A' (i.e. we may take $\rho_D = 1$ when applying Theorem 5.1).

Thus, given the following lemma to ensure that we have enough good primes, the proof goes through in the same way as the proof of Theorem 1.5.

Lemma 6.7. *Let A/F be a geometrically type I(2) abelian fourfold for which $E \not\subseteq K^{\mathrm{gal}}$. The set of good rational primes for A has relative density 1 in the set of rational primes which are inert in $E/\mathbb{Q}$ and split completely in $K/\mathbb{Q}$, and in particular, have positive relative density in the set of all rational primes.*

Proof. This proof has the same structure as the proof of Lemma 6.2. By Theorem 1.3, $\mathcal{R} = (r_\lambda)_\lambda$ is Schur unobstructed and has coefficients in E. Again, taking v split completely in $K/\mathbb{Q}$, we have $Q_{\mathcal{R},v}(X) := x^4 - a_1(\mathrm{Frob}_v)x^3 + a_2(\mathrm{Frob}_v)x^2 - pa_1(\mathrm{Frob}_v)x + p^2$, and a prime $v|p$ of K fails to be ordinary or p-distinguished if and only if $v_p(a_2(\mathrm{Frob}_v)) > 0$ or $v_p(a_1(\mathrm{Frob}_v)^2 - 4a_2(\mathrm{Frob}_v)) > 0$. Ultimately, we want to express these conditions using polynomials with rational coefficients, so taking norms, v fails to be p-distinguished ordinary if and only if $p^{f_p} \mid \mathrm{Nm}(a_2(\mathrm{Frob}_v))$ or $p^{f_p} \mid \mathrm{Nm}(a_1(\mathrm{Frob}_v)^2 - 4a_2(\mathrm{Frob}_v))$, where f_p is the inertia degree of p in $E/\mathbb{Q}$. Again by the Weil bounds, for any embedding $\jmath\colon \overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$, $|\jmath(a_1(\mathrm{Frob}_v))| \leq 4\sqrt{p}$ and $|\jmath(a_2(\mathrm{Frob}_v) - 2p)| \leq 4p$. Taking norms and rearranging, we obtain the bounds

$$|\mathrm{Nm}(a_2(\mathrm{Frob}_v))| \leq 36p^2 \qquad \text{and} \qquad |\mathrm{Nm}(a_1(\mathrm{Frob}_v)^2 - 4a_2(\mathrm{Frob}_v))| \leq 1600p^2 \tag{21}$$

The Mumford–Tate group of a geometrically type I(2) abelian fourfold is $G := \{g \in \mathrm{Res}_{E/\mathbb{Q}} \mathrm{GSp}_{4,E} \colon \nu g \in \mathbb{G}_{m,\mathbb{Q}}\}$. Again we have the Mumford–Tate conjecture in this setting by [MZ95]. Choosing any rational prime ℓ inert in $E/\mathbb{Q}$, we define the representation $\rho := \rho_\ell := (r^{\otimes 2} \oplus \wedge^2 r) \otimes \varepsilon^{-1}$, which we think of as valued in $\mathrm{Res}_{E_\lambda/\mathbb{Q}_\ell} \mathrm{GL}_{22}(E_\lambda)$. Given a polynomial $P(g)$ in the entries of $\mathrm{GSp}_{22}(E_\lambda)$ – and in particular, the tr_1 and tr_2 from Equation (19)) – $\mathrm{Nm}\, P(g)$ defines a polynomial with rational coefficients in $\mathrm{Res}_{E_\lambda/\mathbb{Q}} \mathrm{GL}_{22}(E_\lambda)$. If p is inert in $E/\mathbb{Q}$, a prime $v|p$ of K fails the conditions if and only if $\rho(\mathrm{Frob}_v)$ lands in the union Y of the hypersurfaces cut out by the equations

$$\{\mathrm{Nm}(\mathrm{tr}_2(g)) = c \colon c \in \mathbb{Z} \cap [-36, 36]\} \qquad \text{and} \qquad \{\mathrm{Nm}(\mathrm{tr}_1(g) - 4\,\mathrm{tr}_2(g)) = c' \colon c \in \mathbb{Z} \cap [-1600, 1600]\}. \tag{22}$$

The image of the monodromy group is not contained in Y, so as before, $\rho(\mathrm{Frob}_v)$ avoids Y for a density 1 subset of the primes of K. It is important that $E \cap K^{\mathrm{gal}} = \mathbb{Q}$, as otherwise it would not be possible for a prime p to split completely in K and to be inert in $\mathbb{Q}$. The remainder of the argument is exactly the same, as we again know the Mumford–Tate conjecture for such fourfolds [MZ95]. □

Remark. The strategy of Lemma 6.7 relies crucially on the existence of infinitely many rational primes which are totally inert in $E/\mathbb{Q}$ – by restricting to these primes, we can describe our ordinarity and distinguished ordinarity conditions in Equation (22) using only finitely many equations. This is related to the problem of proving that there are infinitely many ordinary primes for a weight 2 modular form with coefficients in an extension field $E/\mathbb{Q}$, which is also open for general E.

Remark. In [BCGP21], a challenging abelian surface is defined to be any surface A which is either geometrically type I(1) (typical, $\mathcal{V}(A) = \mathcal{R}$) or geometrically type I(2) with $\mathcal{V}(A) \cong \mathrm{Ind}_K^F \mathcal{R}$ for a two-dimensional compatible system of G_K-representations $\mathcal{R}$. The former are challenging because $\mathcal{R}$ is irregular, and the latter are challenging because K need not be totally real. Our notion of a challenging type I(2) abelian fourfold is a generalization of the latter. [BCGP21] deal with challenging type I(2) abelian surfaces by treating them alongside the case of typical abelian surfaces – they Moret–Bailly to find a point B on an appropriately twisted Siegel threefold with p-torsion related to A and automorphic q-torsion. We cannot do the same in our case because B would generically be a typical abelian fourfold, and we do not have a modularity lifting theorem for 8-dimensional symplectic representations.

6.3 Abelian fourfolds of geometric types I(4), II(2), and IV(1,4)

Recall that we call an abelian fourfold A of geometric types I(4) or II(2) challenging if $\mathcal{V}(A) \cong \mathrm{Ind}_K^F \mathcal{R} \otimes \mathcal{A}$ for K a field that is neither totally real nor CM. In these cases, note that $\dim \mathcal{R} = 2$ and $\dim \mathcal{A} = r$.

Lemma 6.8. *If A/F is an abelian fourfold such that*

1. *A is of geometric type I(4) and not challenging;*
2. *A is of geometric type II(2) and not challenging; or*
3. *A is of geometric type IV(1,4),*

then A is potentially automorphic.

Proof. If A/F is geometric type I(4), then by Theorem 1.3, $\mathcal{V}(A) \cong \mathrm{CRes}_{E_0/\mathbb{Q}} \mathrm{Ind}_K^F \mathcal{R}$, where $\mathcal{R}$ is 2-dimensional and $[E_0 : \mathbb{Q}][K : F] = [E : \mathbb{Q}] = 4$. In this case, $L = K$ because $r = 1$. By assumption, K is totally real or CM, so the result follows from Corollary 3.3.

If A/F is geometric type II(2), then $\mathcal{V}(A) \cong \mathrm{CRes}_{E_0/\mathbb{Q}} \mathrm{Ind}_K^F \mathcal{R} \otimes \mathcal{A}$, where $\mathcal{R}$ and $\mathcal{A}$ are both 2-dimensional and $[E_0 : \mathbb{Q}][K : F] = [E : \mathbb{Q}] = 2$. By self-duality of $\mathcal{R}$, its associated Hodge–Tate weights are $(0, 1)$ at every place of F, and L/K is solvable because ϱ has cyclic or dihedral projective image by Lemma 4.1, so potential automorphy follows from Theorem 2.3.

If A/F is geometric type IV(1,4), then $\mathcal{R}$ is 1-dimensional and its automorphy foLlows from class field theory. Every order 8 group is solvable, so we are again done by Corollary 3.3 □

Recall from the proof of Theorem 1.3 that K/F is Galois because Galois conjugation preserves the center of $\mathrm{End}^0_{\overline{\mathbb{Q}}}(A)$ and G_K is the kernel of the induced morphism $G_F \to \mathrm{Aut}_{E_0}(E)$. Therefore, if $F = \mathbb{Q}$, then A is not challenging for geometrically I(4) or geometrically I(2) varieties; this is because K/F is Galois quartic or quadratic and hence abelian, and any abelian extension of $\mathbb{Q}$ is automatically either totally real or CM. Therefore, we deduce the following corollary.

Corollary 6.9. *If an abelian fourfold $A/\mathbb{Q}$ is of geometric type I(4) or II(2), then A is potentially automorphic.*

One might expect the proof of Lemma 6.8 to also give a proof of potential modularity for abelian fourfolds of geometric type IV(1,2). To understand the problem, let A/F be an abelian variety with $E := \mathrm{End}^0(A) = \mathrm{End}^0_{\overline{F}}(A)$ a CM field. Let c denote complex conjugation in E and let E^+ be the totally real subfield of E. Then, we have

$$\rho_{A,\ell} \otimes \overline{\mathbb{Q}_\ell} \cong \bigoplus_{\iota \in \Sigma_{E^+}} \xi_{\tilde{\iota}} \oplus \xi_{\tilde{\iota}'},$$

where $\tilde{\iota}$ and $\tilde{\iota}' := \tilde{\iota} \circ c$ are the two embeddings of E extending E^+. By Lemma 3.4 and the Weil pairing on $\rho_{A,\ell}$, $\xi_{\tilde{\iota}} \oplus \xi_{\tilde{\iota}'}$ is self-dual with cyclotomic similitude for all ι, and we have $\xi_{\tilde{\iota}}^\vee \cong \xi_{\tilde{\iota}'} \otimes \varepsilon^{-1}$. Again by self-duality and cyclotomic similitude, the Hodge–Tate weights of $\xi_{\tilde{\iota}} \oplus \xi_{\tilde{\iota}'}$ are $0^{(\frac{2g}{e})}, 1^{(\frac{2g}{e})}$. A key feature in the unitary setting is that $\xi_{\tilde{\iota}}|_{G_{F_v}}$ and $\xi_{\tilde{\iota}'}|_{G_{F_v}}$ can have *different* Hodge–Tate weights. In particular, if the Hodge–Tate weights of $\xi_{\tilde{\iota}}|_{G_{F_v}}$ are $(0^{(a)}, 1^{(b)})$ for $a + b = \frac{2g}{e}$, we need not have $a = b$. This phenomenon appears already in the setting of CM elliptic curves – if A/F is an elliptic curve with has CM by E, then $\rho_{A,\ell}$ has Hodge–Tate weights $(0, 1)$ and is a direct sum of characters $\xi_{\tilde{\iota}}$ and $\xi_{\tilde{\iota}'}$ of G_F. Therefore, (WLOG) $\xi_{\tilde{\iota}}$ has Hodge–Tate weight 1 and $\xi_{\tilde{\iota}'}$ has Hodge–Tate weight 0. Returning to this general case, and picking an embedding $F \hookrightarrow \mathbb{C}$, this discrepancy is related to the E-action on $\mathrm{Lie}(A_{\mathbb{C}})$. Assume now that ℓ is totally split in E. We can view $\mathrm{Lie}(A_{\mathbb{C}})$ as a module over $E \otimes_{\mathbb{Q}} \mathbb{C} \cong \bigoplus_{\tilde{\iota} \in \Sigma_E} E \otimes_{E, \jmath \circ \tilde{\iota}} \mathbb{C}$, Let $n_{\tilde{\iota}}$ be the dimension of the subspace of $\mathrm{Lie}(A_{\mathbb{C}})$ where E acts through the embedding $\jmath \circ \tilde{\iota}$. Because the action of E on $H^1(A(\mathbb{C}), \mathbb{Q})$ is defined over $\mathbb{Q}$, the dimension of the $\tilde{\iota}$-eigenspace of the E-action on $H^1(A(\mathbb{C}), \overline{\mathbb{Q}})$ is $\frac{2g}{e}$. The E-action commutes with the almost complex structure J. Because the E-action is defined over $\mathbb{Q}$, the eigenvalues of the E-action on $\mathrm{Lie}(A(\mathbb{C})) = H^{1,0}(A(\mathbb{C}))^\vee$ are complex conjugates (this is well defined because E is CM) of the eigenvalues of the E-action on $H^{0,1}(A(\mathbb{C}))^\vee$. Therefore, $n_{\tilde{\iota}} + n_{\tilde{\iota}'} = \frac{2g}{e}$ for all $\tilde{\iota}$. And again, we need not have $n_{\tilde{\iota}} = n_{\tilde{\iota}'}$, as is apparent already when $g = 1$, $e = 2$, and A is a CM elliptic curve.

By work of Tate and Raynaud (see e.g. [**?**, §3.1, Théorème 2]), without loss of generality $\{n_{\tilde{\iota}}, n_{\tilde{\iota}'}\} = \{a, b\}$. where a and b are the multiplicities of the Hodge–Tate weights 0 and 1 in $\xi_{\tilde{\iota}}$. Theorem 2.3 only gives us potential automorphy of two-dimensional representations ξ with Hodge–Tate weights $(0, 1)$ for every $w|\ell$.

For an abelian fourfold of geometric type IV(1,2), $E/\mathbb{Q}$ is a CM quartic field. As noted in [MZ95, 7.5], writing $\Sigma_E = \{\tilde{\iota}_1, \tilde{\iota}_1', \tilde{\iota}_2, \tilde{\iota}_2'\}$, $\{n_{\tilde{\iota}_1}, n_{\tilde{\iota}_1'}\} = \{2, 0\}$ and $\{n_{\tilde{\iota}_2}, n_{\tilde{\iota}_2'}\} = \{1, 1\}$. In particular, $\xi_{\tilde{\iota}_1}$ does not satisfy the hypotheses of Theorem 2.3, and therefore we cannot deduce its automorphy.

6.4 Modularity of abelian fourfolds beyond Main Theorem 1

If A is of geometric type III, $\mathrm{End}_{\overline{F}}(A)$ is a quaternion algebra with center $\mathbb{Q}$ [MZ95]. By Theorem 1.3, $\mathcal{V} \cong \mathcal{R} \otimes \mathcal{A}$, where $\mathcal{R}$ is of orthogonal type. Potential modularity of four-dimensional geometric orthogonal representations with Hodge–Tate weights $(0,0,1,1)$ is open, so we cannot immediately deduce potential modularity of geometrically type III abelian fourfolds. This will be addressed in forthcoming work.

If A is of geometric type IV(1,1) or IV(1,2), progress requires further understanding modularity theorems for compatible systems of unitary type.

7 Even symplectic representations appearing in geometry

To the best of this author's knowledge, all known modularity theorems for symplectic (resp. orthogonal) Galois representations assume that the representations in question are odd (resp. even). However, not every symplectic (resp. orthogonal) compatible system appearing in geometry is odd (resp. even). It is not hard – for example by tensoring an elliptic curve by an odd Artin representation – to produce even symplectic compatible systems "appearing in" geometry. However, the methods of this paper suggest a more refined question – is the strongly irreducible system $\mathcal{R}$ appearing in Theorem 1.3 odd if and only if it is symplectic (resp. even if and only if it is orthogonal)? For $g = 1$ and $g = 2$, $\mathcal{R}$ is always symplectic or unitary, and in the symplectic cases it follows from Lemma 4.4 and Theorem 1.3 that $\mathcal{R}$ is odd. When $g = 4$ and A is geometrically type III (i.e. A has geometric endomorphisms by a definite quaternion algebra over $\mathbb{Q}$), one can deduce (as we learned from Frank Calegari) that the geometric endomorphisms of A cannot be defined over the reals. Equivalently, writing $\mathcal{V} \cong \mathrm{Des}_{\mathbb{Q}}(\mathcal{R} \otimes \mathcal{A})$ as in Theorem 1.3, the two-dimensional symplectic Artin compatible system $\mathcal{A}$ is odd, and hence the four-dimensional (orthogonal, by Lemma 3.6) compatible system $\mathcal{R}$ is even. Despite all of this, we will show that there exist abelian fourfolds such that $\mathcal{R}$ is symplectic and even! To prove a modularity result for such fourfolds (the "not odd type" case of Table 1), we need some way to attach a Siegel Maass form to such representations, which seems far out of reach at present.

The following theorem and its proof, the key idea of which was suggested to the author by F. Calegari, were the motivation for the general twisted moduli construction given in **??** We are grateful to him for allowing us to include his idea in this article.

Lemma 7.1. *Let $D/\mathbb{Q}$ be an indefinite quaternion division algebra equipped with a positive involution $\dagger$. There exists a g-dimensional abelian variety A over the real numbers such that $D \hookrightarrow \mathrm{End}^0_{\mathbb{C}}(A)$ and $E' \hookrightarrow \mathrm{End}^0(A)$, where E' is the distinguished quadratic subfield of $(D,\dagger)$ as defined in Lemma 4.2, if and only if $g \equiv 0 \pmod 4$.*

Remark. If $\mathrm{End}^0(A) = E'$ for some E'/E, and E'/E is CM, then $\dagger$ must restrict to complex conjugation on E'/E by the Albert classification. Because D/E is indefinite, the subspace of D on which $\dagger$ acts with eigenvalue -1 is 1-dimensional over E, so E' must be the distinguished quadratic subfield of $(D,\dagger)$.

Before proving Lemma 7.1, we prove a preparatory lemma.

Lemma 7.2. *Let J be a linear operator on a real vector space U such that $J^2 = -I$. Fix a real number $\lambda > 0$. There exists a matrix M such that $M^2 = -\lambda I$ and $MJ = -JM$ if and only if $\dim U \equiv 0 \pmod 4$.*

Proof. First, $\dim U$ cannot be odd because the $+i$ and $-i$ eigenvalues of J come in conjugate pairs. Fix a basis in which J is the block matrix $\begin{pmatrix} & I \\ -I & \end{pmatrix}$. If $M = \begin{pmatrix} W & X \\ Y & Z \end{pmatrix}$, then the conditions

$$MJ = -JM \qquad \text{and} \qquad M^2 = \lambda I$$

translate into the conditions $W = -Z$, $X = Y$, $WX = XW$, and $W^2 + X^2 = \lambda I$. Because W is a matrix on an odd-dimensional real vector space, it has a real eigenvector v with eigenvalue μ. Because X commutes with W, the subspace $\langle X\rangle v := \{v, Xv, X^2v, \dots\}$ is contained within the μ-eigenspace of W. If $\langle X\rangle v$ is odd-dimensional, then X has a real eigenvector $w \in \langle X\rangle v$ as before, with eigenvalue μ'. But then, $(W^2 + X^2)w = (\mu^2 + \mu'^2)w$, so $\lambda > 0$. If $\langle X\rangle v$ is even-dimensional, then we induct on the complement of $\langle X\rangle v$. □

Corollary 7.3. *The square of a $\mathbb{C}$-antilinear operator on an odd-dimensional complex vector space cannot be a negative scalar.*

Proof of Lemma 7.1. Choose $i \in D$ such that $E' = \mathbb{Q}(i) \subset D$, and let $a := -i^2 > 0$. Choose $j \in D$ such that $ij = -ji$ and $b := j^2 > 0$ – such a j exists because D is indefinite.

Suppose that $A/\mathbb{R}$ has an endomorphism algebra satisfying the properties in the lemma statement. Because $H^1(A(\mathbb{C}), \mathbb{Q})$ is a free D-module, 4 must divide $2g$, so g must be even (this does not use the fact that A is defined over $\mathbb{R}$). The Lie algebra $\mathrm{Lie}(A)$ is a g-dimensional real vector space, and $\mathrm{Lie}(A_{\mathbb{C}}) = \mathrm{Lie}(A) \otimes_{\mathbb{R}} \mathbb{C}$. Let γ denote the map on $\mathrm{Lie}(A_{\mathbb{C}})$ induced by the complex conjugation morphism on $A(\mathbb{C})$. It is a $\mathbb{C}$-antilinear involution on $\mathrm{Lie}(A_{\mathbb{C}})$. Let $\iota\colon D \hookrightarrow \mathrm{End}(A_{\mathbb{C}})$ be the embedding of D into the endomorphism algebra. Because $\iota(i)$ is defined over $\mathbb{R}$ and $\iota(j)$ is not (and using the fact that $j^2 = b$ is defined over $\mathbb{R}$), we have the relations

$$\iota(i)^2 = -a, \iota(j)^2 = b, \gamma^2 = 1, \iota(i)\iota(j) = -\iota(j)\iota(i), \iota(i)\gamma = \gamma\iota(i), \text{ and } \gamma\iota(j) = -\iota(j)\gamma. \tag{23}$$

Because $\iota(i)$ is defined over $\mathbb{R}$, the endomorphism $\iota(i)$ acts $\mathbb{R}$-linearly on $\mathrm{Lie}(A)$ – in particular, it gives a decomposition of $\mathrm{Lie}(A_{\mathbb{C}})$ into $\pm\sqrt{-a}$-eigenspaces. Let $W \subset \mathrm{Lie}(A_{\mathbb{C}})$ be the $\sqrt{-a}$-eigenspace of $\iota(i)$. Because $\iota(j)$ and γ anticommute with $\iota(i)$, the product $\iota(j)\gamma$ acts on the $\frac{g}{2}$-dimensional complex vector space W. By Equation (23), $(\iota(j)\gamma)^2 = -b$. However, this product is $\mathbb{C}$-antilinear, so by Corollary 7.3, $(\iota(j)\gamma)^2$ cannot be a negative scalar if $g \equiv 2 \pmod 4$.

Now suppose that $g \equiv 0 \pmod 4$. Because $\mathbb{Q}(i)$ is the distinguished quadratic subfield, $i^\dagger = -i$ and $j^\dagger = j$, and we can define the $\dagger$-stable order $\mathcal{O} := \mathbb{Z}[\frac{1}{N}]\langle i, j\rangle \subset D$, where N is an integer that will be chosen later. Define $\Lambda' := \mathcal{O}^{\oplus \frac{g}{4}}$, $\Lambda := \Lambda' \oplus \Lambda'$, and $V := \Lambda \otimes_{\mathbb{Z}} \mathbb{Q}$ Let $\phi\colon \Lambda \times \Lambda \to \mathcal{O}$ be the alternating $\mathcal{O}$-skew Hermitian pairing given (as a block matrix on $\Lambda' \oplus \Lambda'$) by $J = \begin{pmatrix} & I \\ -I & \end{pmatrix} \in M_{\frac{g}{2}}(\mathcal{O})$. Then, defining $h\colon \mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m \to \mathrm{GL}(\Lambda \otimes_{\mathbb{Z}} \mathbb{R})$ by $h(i) = J$, Λ is a polarized weight 1 $\mathcal{O}$-Hodge structure with polarization given by $\mathrm{tr}_{\mathcal{O}/\mathbb{Z}}\, \phi$, and hence corresponds to an complex abelian variety $A/\mathbb{C}$ with endomorphism ring containing $\mathcal{O}$. A descent datum on $A/\mathbb{C}$ corresponds to an involution $\delta\colon H^1(A(\mathbb{C}), \mathbb{Z}) \to H^1(A(\mathbb{C}), \mathbb{Z})$ such that $\delta^2 = 1$ and $\delta J = -J\delta$. By the double centralizer theorem [**?**], we can write $\mathrm{End}_{\mathbb{Q}}(V) \cong D \otimes_{\mathbb{Q}} \mathrm{End}_D(V)$, where we think of the left D as the endomorphism algebra of the abelian variety. We have the inclusions

$$1 \otimes_{\mathbb{Q}} \mathrm{End}_D(V) \hookrightarrow \mathrm{End}_E(V) \cong \mathbb{Q}(i) \otimes_{\mathbb{Q}} \mathrm{End}_D(V) \hookrightarrow D \otimes_{\mathbb{Q}} \mathrm{End}_D(V) \cong \mathrm{End}_Q(V).$$

Our goal will be to choose an integral element $\delta \in \mathrm{End}_E(V) \setminus \mathrm{End}_D(V)$ so that δ commutes with i but not with all of D. Because δ is an involution, this is equivalent to requiring that

$\delta i = i\delta, \delta j = -j\delta$, where the equations are evaluated in $\mathrm{End}_{\mathbb{Q}}(V)$. (Indeed, because $\delta j = -\delta j$, it follows that $\delta = \beta i \otimes \delta_1 \in \mathbb{Q}(i) \otimes_{\mathbb{Q}} \mathrm{End}_D(V)$ for some $\beta \in \mathbb{Q}$.)

Because E' splits D, b is a norm in E' ([Voi21, Theorem 5.4.4]), so there exist $\alpha, \beta \in \mathbb{Q}$ such that $\alpha^2 - (\beta\sqrt{-a})^2 = b$. We choose N to contain the denominators of α and β. Let $\mathcal{O}_{E'} := \mathcal{O} \cap E'$, and fix an $\mathcal{O}_{E'}$-linear embedding $\mathcal{O} \hookrightarrow M_2(\mathcal{O}_{E'})$ given by

$$1 \longmapsto \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}, i \longmapsto \begin{pmatrix} & i \\ i & \end{pmatrix} \; j \longmapsto \begin{pmatrix} \alpha & \beta\sqrt{-a} \\ -\beta\sqrt{-a} & -\alpha \end{pmatrix}, ij \longmapsto \begin{pmatrix} -i\beta\sqrt{-a} & -i\alpha \\ i\alpha & i\beta\sqrt{-a} \end{pmatrix}. \tag{24}$$

We will construct δ as a matrix in $\mathrm{End}_{\mathcal{O}_{E'}}(\Lambda) \setminus \mathrm{End}_{\mathcal{O}}(\Lambda)$, or equivalently, in $M_g(\mathcal{O}_{E'}) \setminus M_{\frac{g}{2}}(\mathcal{O})$. As noted earlier, we require with $\delta J = -J\delta$, $\delta^2 = 1$ so that δ defines a descent datum on A. We must also have $\delta^T J \delta = -J$ so that the polarization descends to the real variety.

To satisfy all of these conditions, we take

$$\delta = \begin{pmatrix} & & 1 \\ & \iddots & \\ 1 & & \end{pmatrix} \in M_g(\mathcal{O}_{E'}).$$

Note that any symmetric matrix in $M_{\frac{g}{2}}(\mathcal{O}) = \mathrm{End}_{\mathcal{O}}(\Lambda)$ in the image of Equation (24) has totally imaginary antidiagonal, so $\delta \in \mathrm{End}_{\mathcal{O}_{E'}}(\Lambda) \setminus \mathrm{End}_{\mathcal{O}}(\Lambda)$ as desired. □

Corollary 7.4. *With the notation of Lemma 7.1, there exists a g-dimensional abelian variety A over a totally real field F such that $\mathrm{End}^0_F(A) = \mathrm{End}^0_{\mathbb{R}} E'$ (for any embedding of F into $\mathbb{R}$) and $\mathrm{End}^0_{\overline{F}}(A) = D$ if and only if $g \equiv 0 \pmod 4$.*

Proof. Because D is indefinite with center $\mathbb{Q}$, the PEL Shimura variety Y parameterizing g-dimensional abelian varieties with endomorphisms by D (and any neat level structure) has reflex field $\mathbb{Q}$. We want to apply Theorem 5.1 to construct a moduli scheme on which the abelian variety of Lemma 7.1 is a real point.

Pick a prime $p \geq 3$ such that $p \equiv 3 \pmod 4$, p is split in D, and a is a quadratic residue modulo p. Let $L := \mathbb{Q}(\zeta_p)$ and $K := \mathbb{Q}(\sqrt{-p})$ the unique imaginary quadratic subextension. The idea is what we really care about is $\mathrm{Gal}(K/\mathbb{Q})$, as it controls the action of complex conjugation, but in order to ensure that $\overline{\rho}_p$ has cyclotomic similitude (so we can isolate a single geometrically connected component) we work with all of $\mathrm{Gal}(L/\mathbb{Q})$.

Define $\rho_D \colon \mathrm{Gal}(L/\mathbb{Q}) \twoheadrightarrow \mathrm{Gal}(K/\mathbb{Q}) \to \mathrm{Aut}_{\mathbb{Q}}(D)$ to be the map sending the complex conjugation $c \in \mathrm{Gal}(K/\mathbb{Q})$ to to the automorphism $(d \mapsto idi^{-1})$. Let $\mathcal{O}$, Λ, and Λ' be as in the proof of Lemma 7.1. Then, $\mathrm{End}_{\mathcal{O}} \Lambda \otimes_{\mathbb{Z}} \mathcal{O} \hookrightarrow \mathrm{End}_{\mathbb{Z}} \Lambda$. We write $\overline{\mathcal{O}_p} := \mathcal{O} \otimes \mathbb{F}_p \cong M_2(\mathbb{F}_p)$. Let $\delta \in \mathcal{O} \otimes_{\mathbb{Z}} \mathrm{End}_{\mathcal{O}} \Lambda$ be the descent datum chosen in the proof of Lemma 7.1, and write $\overline{\delta}$ for its reduction mod p. We define $\overline{\rho}_p \colon \mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/(\frac{p-1}{2})\mathbb{Z} \to \overline{\mathcal{O}_p} \otimes_{\mathbb{Z}_p} \mathrm{End}_{\overline{\mathcal{O}_p}}(\Lambda)$ by sending the $\mathbb{Z}/2\mathbb{Z}$ factor to $\overline{\delta}$ and the other factor to any representation of valued in $\mathrm{End}_{\overline{\mathcal{O}_p}}(\Lambda) \cong M_{\frac{g}{2}}(\mathcal{O}_p) \cong M_g(\mathbb{Z}_p)$ with the appropriate similitude character (for example, writing ϵ_0 for the restriction of ϵ to $Z/(\frac{p-1}{2})\mathbb{Z}$, $\mathrm{diag}(1, \cdots, 1, \epsilon_0, \cdots \epsilon_0)$ in the basis where the symplectic pairing is given by pairing e_i against e_{g+i} in the usual way). Let $c \in \mathrm{Gal}(L/\mathbb{Q})$ denote complex conjugation. Because $\overline{\rho}_n(c) = \overline{\delta}$, and $\delta^2 = 1$, and the image of the rest of the representation commutes with δ, **??** is satisfied. Applying Theorem 5.1, we obtain a moduli scheme $Y'/\mathbb{Q}$, and by construction, the abelian variety $A/\mathbb{R}$ produced by Lemma 7.1 corresponds to a real point on $Y'/\mathbb{Q}$.

It remains to upgrade our real point to a point defined over a totally real field. Because $Y'(\mathbb{R})$ is nonempty, the Moret–Bailly theorem implies that Y' has a totally real point. However, this does not

force the totally real point to have geometric endomorphism algebra exactly D. We might naively hope that the subset of $Y'(\mathbb{R})$ without extra endomorphisms is open, but we should not expect this to be the case – already on the modular curve, the CM points (which correspond to the images of quadratic irrationalities of the upper half-plane under the j-invariant map) are dense in the complex analytic topology. However, because the j-invariant of a CM point is an algebraic integer, the CM points of the modular curve over $\mathbb{Q}_p$ are definitely not dense. Indeed, much more is true – for any prime p, it follows from [MP12, Proposition 1.13] applied to the universal abelian scheme over Y' that the extra endomorphism locus in $Y'(\mathbb{C}_p)$ is *nowhere* dense. In particular, there exists a finite Galois extension $K_p/\mathbb{Q}_p$ and an open subset $\Omega_p \subset Y'(K_p)$ such that Ω_p is nonempty. Applying the Moret–Bailly theorem with $\Omega_{\mathbb{R}} = Y'(\mathbb{R})$ and $\Omega_p \subset Y'(K_p)$ as just constructed, we deduce that Y' has a totally real point with no extra geometric endomorphisms beyond D. □

Main Theorem 2. *Let $F/\mathbb{Q}$ be a totally real number field. There exist type II abelian fourfolds A/F for which the four-dimensional compatible system $\mathcal{R}$ given by Theorem 1.3 is even type and symplectic type. In particular, there exists a representation $\xi\colon G_F \to \mathrm{GSp}_4(\overline{\mathbb{Q}_\ell})$ such that ξ is geometric, has Zariski dense image, and is totally even.*

Proof. Let A/F be an abelian fourfold produced by applying Corollary 7.4 with $g = 4$. By Corollary 1.4, we can write $\rho_A \cong \xi \otimes \varrho$, where ϱ has cyclic projective image. Up to replacing ξ and ϱ by twists, we can assume that $\varrho \cong 1 \oplus \eta$ for a character η. By Equation (12) and Lemma 4.3, because $\mathrm{End}_F^0(A)$ is a CM field, the similitude character of ϱ is η. Because the geometric endomorphisms of A are not defined over $\mathbb{R}$, $\ker \eta$ is G_L for some totally imaginary field L/F. In particular, $\eta(c) = -1$ for every complex conjugation c, and ϱ is odd. Therefore, ξ must be even. Because A is geometrically type II(1), ξ has big image in GSp_4, and the result follows. □

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